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TECHNICAL SPECIFICATION

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**5G;**
**Telecommunication management;**
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**Integration Reference Point (IRP);**
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# Foreword

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# Introduction

The present document is part of a TS-family covering the 3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; Telecommunication management; as identified below:

28.621 Generic Network Resource Model (NRM) Integration Reference Point (IRP); Requirements;

**28.622 Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS) ;**

28.623 Generic Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions.

The interface Itf-N, defined in 3GPP TS 32.102 [2], is built up by a number of Integration Reference Points (IRPs) and a related Name Convention, which realise the functional capabilities over this interface. The basic structure of the IRPs is defined in 3GPP TS 32.150 [4].

The present document is part of a set that has been developed for converged management solutions.

The present document is part of a set that is used for management and orchestration of 5G networks and network slicing.

# 1 Scope

The present document specifies the Generic network resource information that can be communicated between an MnS producer and MnS consumer for telecommunication network management purposes, including management of converged networks and networks that include virtualized network functions.

This document specifies the semantics of information object class attributes and relations visible across the reference point in a protocol and technology neutral way. It does not define their syntax and encoding.

This document supports the Federated Network Information Model (FNIM) concept described in TS 32.107 [8] in that the relevant Information Object Class (IOC)s defined in this specification are directly or indirectly inherited from those specified in the Umbrella Information Model (UIM) of TS 28.620 [9].

Note that the present document is applicable to deployment scenarios using the Service Based Management Architecture (SBMA) as defined in TS 28.533 [32]. For deployment scenarios using the IRP framework as defined in TS 32.102 [2] the latest Rel-14 version of TS 28.622 is applicable.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
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[1] 3GPP TS 32.101: "Telecommunication management; Principles and high level requirements".

[2] 3GPP TS 32.102: "Telecommunication management; Architecture".

[3] 3GPP TS 32.302: "Telecommunication management; Configuration Management (CM); Notification Integration Reference Point (IRP): Information Service (IS)".

[4] 3GPP TS 32.150: "Telecommunication management; Integration Reference Point (IRP) Concept and Definitions".

[5] 3GPP TS 23.003: "Technical Specification Group Core Network and Terminals; Numbering, addressing and identification"

[6] Void

[7] ITU-T Recommendation X.710 (1991): "Common Management Information Service Definition for CCITT Applications".

[8] TS 32.107: "Telecommunication management; Fixed Mobile Convergence (FMC) Federated Network Information Model (FNIM)"

[9] TS 28.620: "Telecommunication management; Fixed Mobile Convergence (FMC) Federated Network Information Model (FNIM) Umbrella Information Model (UIM)"

[10] TS 32.156: "Telecommunication management; Fixed Mobile Convergence (FMC) Model Repertoire"

[11] Void

[12] Void

[13] 3GPP TS 32.300: "Telecommunication management; Configuration Management (CM); Name convention for Managed Objects".

[14] 3GPP TS 32.600: "Telecommunication management; Configuration Management (CM); Concept and high-level requirements".

[15] ETSI GS NFV 003 V1.1.1: "Network Functions Virtualisation (NFV); Terminology for Main Concepts in NFV".

[16] ETSI GS NFV-IFA 008 V3.5.1 (2021-11): "Network Functions Virtualisation (NFV) Release 3; Management and Orchestration; Ve-Vnfm reference point - Interface and Information Model Specification".

[17] Void.

[18] ETSI ES 202 336-12 V1.1.1: "Environmental Engineering (EE); Monitoring and control interface for infrastructure equipment (power, cooling and building environment systems used in telecommunication networks); Part 12: ICT equipment power, energy and environmental parameters monitoring information model".

[19] ITU-T Recommendation X.731: "Information technology - Open Systems Interconnection - Systems Management: State management function".

[20] 3GPP TS 28.552: "Management and orchestration; 5G performance measurements".

[21] 3GPP TS 28.625: "State Management Data Definition Integration Reference Point (IRP); Information Service (IS) ".

[22] 3GPP TS 23.501: "System Architecture for the 5G System".

[23] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[24] IETF RFC 791: "Internet Protocol".

[25] IETF RFC 2373: "IP Version 6 Addressing Architecture".

[26] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[27] 3GPP TS 28.532: "Management and orchestration; Generic management services".

[28] 3GPP TS 28.554: "Management and orchestration; 5G end to end Key Performance Indicators (KPI)".

[29] 3GPP TS 32.421: "Telecommunication management; Subscriber and equipment trace; Trace concepts and requirements".

[30] 3GPP TS 32.422: "Telecommunication management; Subscriber and equipment trace; Trace control and configuration management".

[31] Void

[32] 3GPP TS 28.533: "Management and orchestration; Architecture framework".

[33] 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description; Stage 2".

[34] 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)".

[35] 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception".

[36] 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification".

[37] 3GPP TS 36.321: "Evolved Universal Terrestrial Radio Access (E-UTRA); Medium Access Control (MAC) protocol specification".

[38] 3GPP TS 38.331: "NR; Radio Resource Control (RRC); Protocol specification".

[39] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification".

[40] 3GPP TS 25.321: "Medium Access Control (MAC) protocol specification".

[41] 3GPP TS 25.331: "Radio Resource Control (RRC); Protocol specification".

[42] 3GPP TS 38.304: "NR; User Equipment (UE) procedures in Idle mode and RRC Inactive state".

[43] 3GPP TS 37.320: "Universal Terrestrial Radio Access (UTRA) and Evolved Universal Terrestrial Radio Access (E-UTRA); Radio measurement collection for Minimization of Drive Tests (MDT); Overall description; Stage 2".

[44] 3GPP TS 28.705: "Telecommunication management; IP Multimedia Subsystem (IMS) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[45] 3GPP TS 28.702: "Telecommunication management; Core Network (CN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[46] 3GPP TS 28.652: "Telecommunication management; Universal Terrestrial Radio Access Network (UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[47] 3GPP TS 28.708: "Telecommunication management; Evolved Packet Core (EPC) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[48] 3GPP TS 28.541: " Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3".

[49] Void

[50] 3GPP TS 28.405: "Telecommunication management; Quality of Experience (QoE) measurement collection; Control and configuration".

[51] 3GPP TS 26.247: "Transparent end-to-end Packet-switched Streaming Service (PSS); Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH)".

[52] 3GPP TS 26.114: "IP Multimedia Subsystem (IMS); Multimedia Telephony; Media handling and interaction".

[53] 3GPP TS 26.118: "Virtual Reality (VR) profiles for streaming applications".

[54] IETF RFC 3339: "Date and Time on the Internet: Timestamps".

[55] Void

[56] 3GPP TS 28.658: "Telecommunication management; Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[57] 3GPP TS 28.558: "Management and orchestration; UE level measurements for 5G system".

[58] 3GPP TS 28.111: "Fault management"

[59] Void

[60] IETF RFC 4648: "The Base16, Base32, and Base64 Data Encodings"

# 3 Definitions and abbreviations

## 3.1 Definitions

For the purposes of the present document, the following terms and definitions apply. For terms and definitions not found here, please refer to 3GPP TS 32.101 [1], 3GPP TS 32.102 [2], 3GPP TS 32.150 [4] and 3GPP TS 32.600 [14].

**Association**: In general, it is used to model relationships between Managed Objects. Associations can be implemented in several ways, such as:

1) name bindings,

2) reference attributes, and

3) association objects.

This IRP stipulates that name containment associations shall be expressed through name bindings, but it does not stipulate the implementation for other types of associations as a general rule. These are specified as separate entities in the object models (UML diagrams). Currently however, all (non-containment) associations are modelled by means of reference attributes of the participating MOs.

**Information Object Class (IOC):** An IOC represents the management aspect of a network resource. It describes the information that can be passed/used in management interfaces. Their representations are technology agnostic software objects. IOC has attributes that represents the various properties of the class of objects. See the term "attribute" defined in TS 32.156 [10]. Furthermore, IOC can support operations providing network management services invocable on demand for that class of objects. An IOC may support notifications that report event occurrences relevant for that class of objects. It is modelled using the stereotype "Class" in the UML meta-model. See TS 32.156 [10] for additional information on IOC.

**Managed Object (MO)**: A MO is an instance of a Managed Object Class (MOC) representing the management aspects of a network resource. Its representation is a technology specific software object. It is sometimes called MO instance (MOI). The MOC is a class of such technology specific software objects. An MOC is the same as an IOC except that the former is defined in technology specific terms and the latter is defined in technology agnostic terms. MOCs are used/defined in SS level specifications. IOCs are used/defined in IS level specifications.

**Management Information Base (MIB)**: A MIB is an instance of an NRM and has some values on the defined attributes and associations specific for that instance. In the context of the present document, an MIB consists of:

1) a Name space (describing the MO containment hierarchy in the MIB through Distinguished Names),

2) a number of Managed Objects with their attributes and

3) a number of Associations between these MOs. Also note that TMN (ITU-T Recommendation X.710 [7]) defines a concept of a Management Information Tree (also known as a Naming Tree) that corresponds to the name space (containment hierarchy) portion of this MIB definition. Figure 3.1 depicts the relationships between a Name space and a number of participating MOs (the shown association is of a non-containment type)

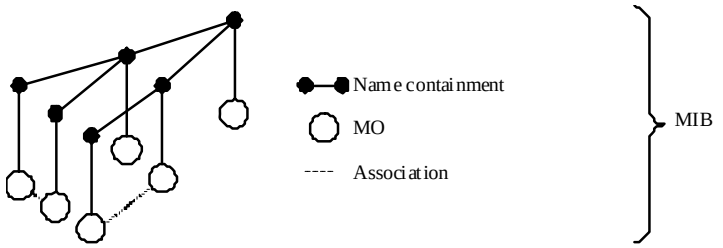



**Figure 3.1: Relationships between a Name space and a number of participating MOs**

**Name space**: A name space is a collection of names. The IRP name convention (see 3GPP TS 32.300 [13]) restricts the name space to a hierarchical containment structure, including its simplest form - the one-level, flat name space.
All Managed Objects in a MIB are included in the corresponding name space and the MIB/name space shall only support a strict hierarchical containment structure (with one root object). A Managed Object that contains another is

said to be the superior (parent); the contained Managed Object is referred to as the subordinate (child). The parent of all MOs in a single name space is called a Local Root. The ultimate parent of all MOs of all managed systems is called the Global Root.

**Network resource:** discrete entity represented by an Information Object Class (IOC) for the purpose of network and service management.

NOTE: A network resource may represent intelligence, information, hardware and software of a telecommunication network.

**Network Resource Model (NRM)**: A collection of IOCs, inclusive of their associations, attributes and operations, representing a set of network resources under management.

**Data node**: This term is defined in TS 32.156 [10].

## 3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [26] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [26].

| | |
|---|---|
| CAG | Closed Access Group |
| DN | Distinguished Name (see 3GPP TS 32.300 [13]) |
| IOC | Information Object Class |
| MN | Master Node |
| MO | Managed Object |
| MOC | Managed Object Class |
| MOI | Managed Object Instance |
| NID | Network ID |
| NFVI | Network Functions Virtualisation Infrastructure (NFVI): Defined in ETSI GS NFV 003 [15]. |
| NPN | Non-Public Network |
| PNI-NPN | Public Network Integrated Non-Public Network |
| RDN | Relative Distinguished Name (see 3GPP TS 32.300 [13]) |
| SN | Secondary Node |
| SNPN | Standalone Non-Public Network |
| SS | Solution Set |
| VNF | Virtualised Network Function |

# 4 Model

## 4.1 Imported information entities and local labels

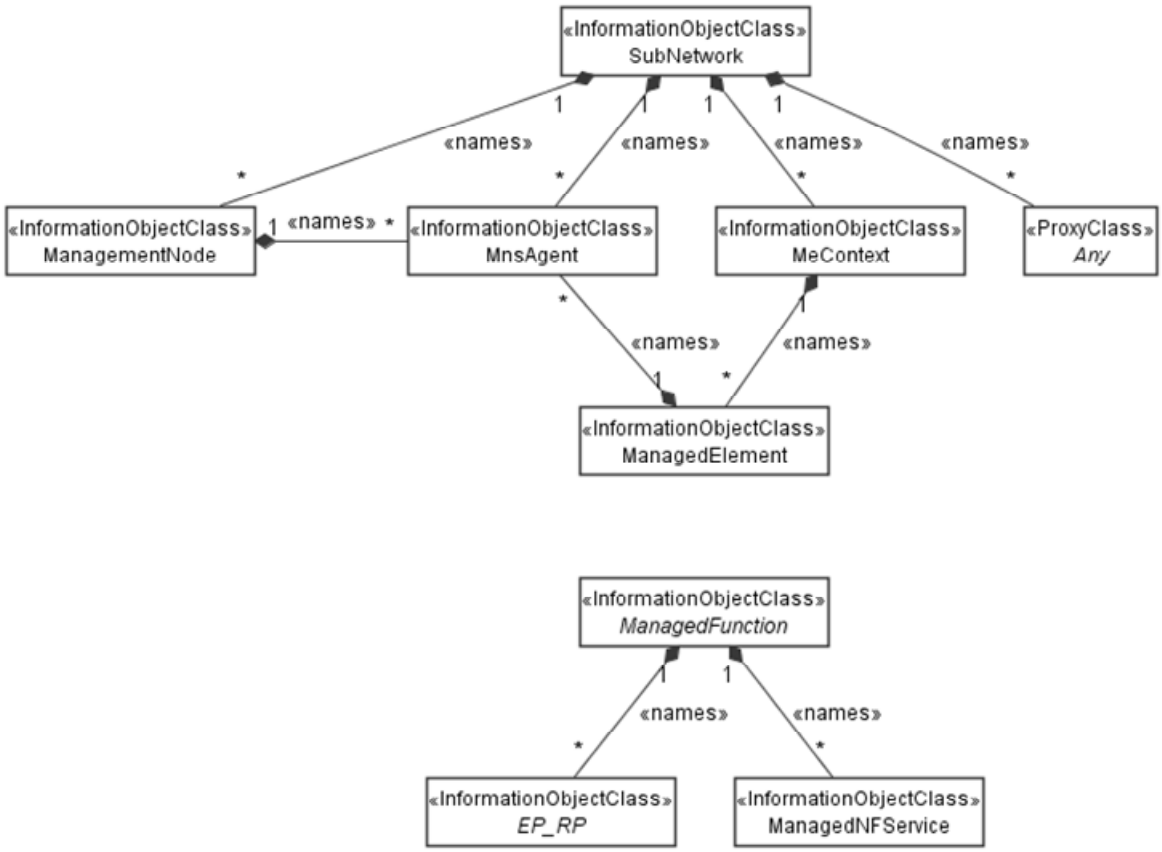
| Label reference | Local label |
|---|---|
| 3GPP TS 28.532 [27], notification, notifyMOICreation | notifyMOICreation |
| 3GPP TS 28.532 [27], notification, notifyMOIDeletion | notifyMOIDeletion |
| 3GPP TS 28.532 [27], notification, notifyMOIAttributeValueChanges | notifyMOIAttributeValueChanges |
| 3GPP TS 28.532 [27], notification, notifyMOIChanges | notifyMOIChanges |
| 3GPP TS 28.111 [58], notification, notifyNewAlarm | notifyNewAlarm |
| 3GPP TS 28.111 [58], notification, notifyClearedAlarm | notifyClearedAlarm |
| 3GPP TS 28.111 [58], notification, notifyChangedAlarm | notifyChangedAlarm |
| 3GPP TS 28.111 [58], notification, notifyChangedAlarmGeneral | notifyChangedAlarmGeneral |
| 3GPP TS 28.111 [58], notification, notifyCorrelatedNotificationChanged | notifyCorrelatedNotificationChanged |
| 3GPP TS 28.111 [58], notification, notifyAckStateChanged | notifyAckStateChanged |
| 3GPP TS 28.111 [58], notification, notifyComments | notifyComments |
| 3GPP TS 28.111 [58], notification, notifyPotentialFaultyAlarmlist | notifyPotentialFaultyAlarmList |
| 3GPP TS 28.111 [58], notification, notifyAlarmlistRebuilt | notifyAlarmListRebuilt |
| 3GPP TS 28.532 [27], notification, notifyFileReady | notifyFileReady |
| 3GPP TS 28.532 [27], notification, notifyFilePreparationError | notifyFilePreparationError |
| 3GPP TS 28.620 [9], IOC, *Domain_* | *Domain_* |
| 3GPP TS 28.620 [9], IOC, *ManagedElement_* | *ManagedElement_* |
| 3GPP TS 28.620 [9], IOC, *Function_* | *Function_* |
| 3GPP TS 28.620 [9], IOC, *ManagementSystem_* | *ManagementSystem_* |
| 3GPP TS 28.620 [9], IOC, *TopologicalLink_* | *TopologicalLink_* |
| 3GPP TS 28.620 [9], IOC, *Top_* | *Top_* |
| 3GPP TS 28.541 [48], dataType, *S-NSSAI* | *S-NSSAI* |
| 3GPP TS 28.658 [56], dataType, PLMNId | *PLMNId* |

## 4.2 Class diagrams

### 4.2.1 Relationships

This clause depicts the set of classes (e.g. IOCs) that encapsulates the information relevant for this IRP. This clause provides the overview of the relationships of relevant classes in UML. Subsequent clauses provide more detailed specification of various aspects of these classes.

The following figure shows the containment/naming hierarchy and the associations of the classes defined in the present document. See Annex A of a class diagram that combines this figure with Figure 1 of TS 32.102 [2], the class diagram of UIM.




NOTE 1: `ManagedElement` may be contained either
-in a `SubNetwork` (since *SubNetwork* inherits from *Domain_* and *ManagedElement* inherits from *ManagedElement_* and *Domain_* name-contained *ManagedElement_* as observed in the figure of Annex A) or
-in a `MeContext` instance as observed by the above figure or in the figure of Annex A.
This either-or relation cannot be shown by using an {xor} constraint in the above figure.
`ManagedElement` may also have no parent instance at all.

NOTE 2: Void

NOTE 3: If the configuration contains several instances of `SubNetwork`, exactly one `SubNetwork` instance shall directly or indirectly contain all the other `SubNetwork` instances.

NOTE 4: The `SubNetwork` instance not contained in any other instance of `SubNetwork` is referred to as "the root `SubNetwork` instance".

NOTE 5: `ManagementNode` shall be contained in the root `SubNetwork` instance.

NOTE 6: If contained in a `SubNetwork` instance, `MnsAgent` shall be contained in the root `SubNetwork` instance.

NOTE 7: Void

NOTE 8: Void

**Figure 4.2.1-1: NRM fragment**

Each Managed Object is identified with a Distinguished Name (DN) according to 3GPP TS 32.300 [13] that expresses its containment hierarchy. As an example, the DN of a `ManagedElement` instance could have a format like:

`SubNetwork`=Sweden,`MeContext`=MEC-Gbg-1,`ManagedElement`=RNC-Gbg-1.




NOTE 8: Void
NOTE 9: Void

**Figure 4.2.1-2: Vendor specific data container NRM fragment**

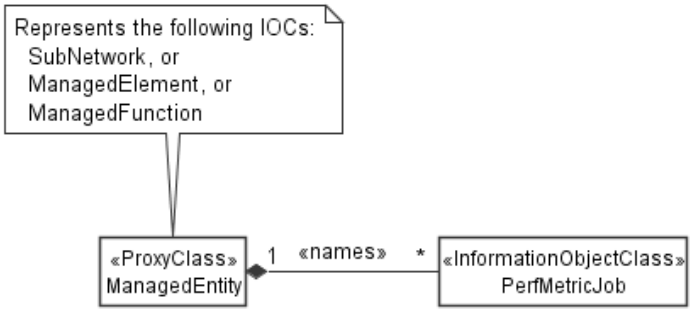



**Figure 4.2.1-3: PM control NRM fragment**

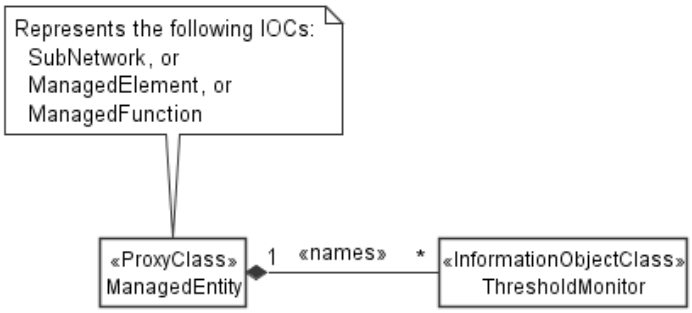



**Figure 4.2.1-4: Threshold monitoring control NRM fragment**




**Figure 4.2.1-5: Notification subscription and heartbeat notification control NRM fragment**

**Figure 4.2.1-6: Void**

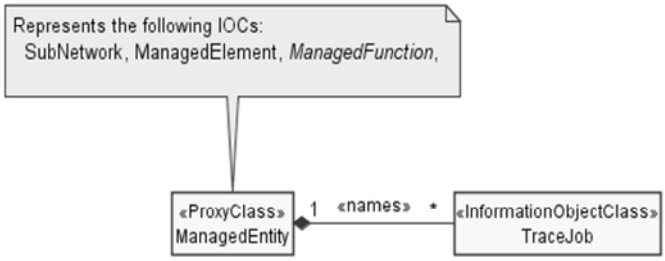



**Figure 4.2.1-7: Trace control NRM fragment**

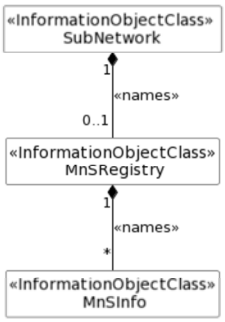



**Figure 4.2.1-8: MnS Registry NRM fragment**

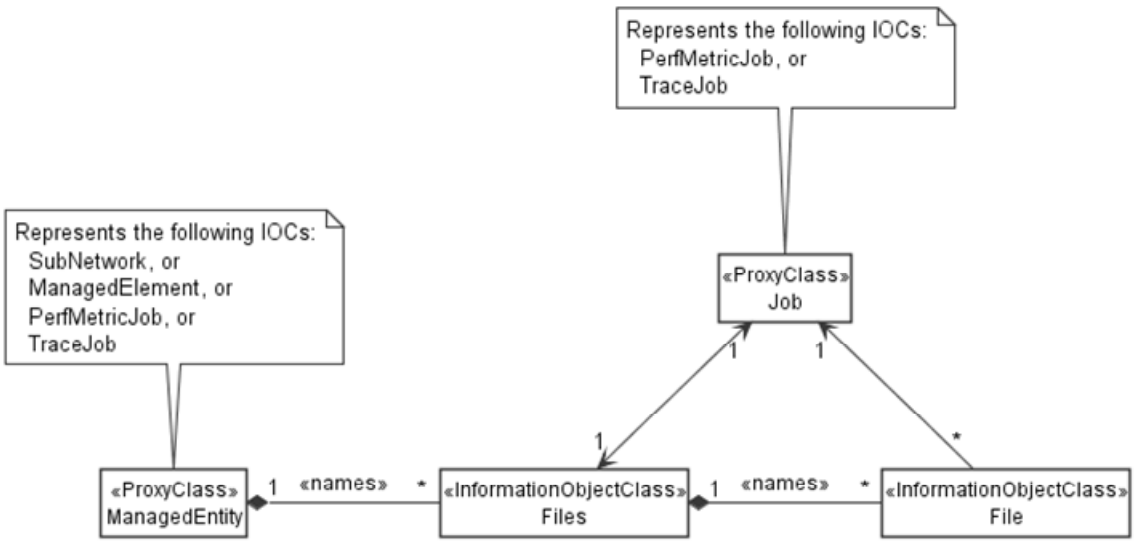



**Figure 4.2.1-9: File retrieval NRM fragment**

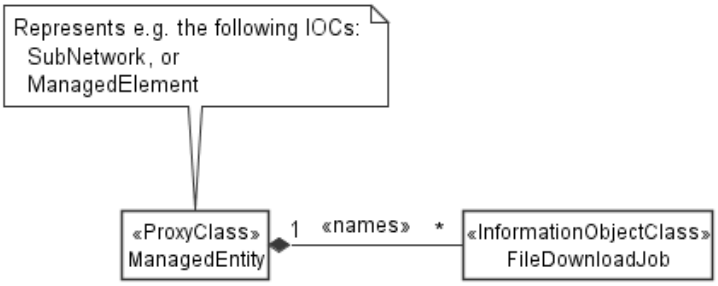



**Figure 4.2.1-10: File download NRM fragment**

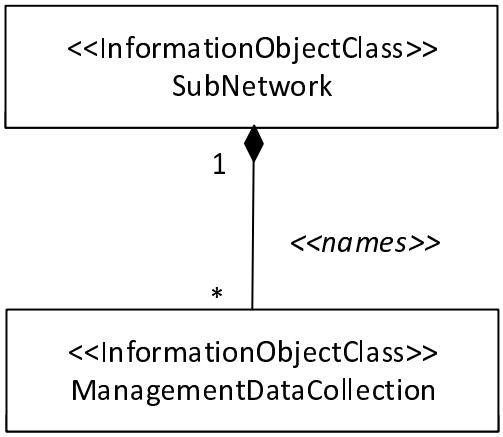



**Figure 4.2.1-11: Management data collection NRM fragment**

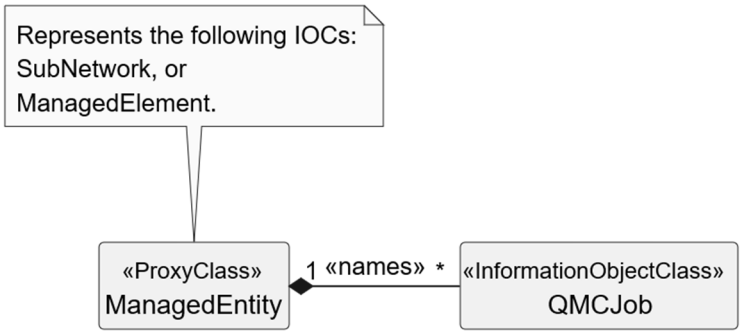



**Figure 4.2.1-12: QoE Measurement Collection NRM fragment**

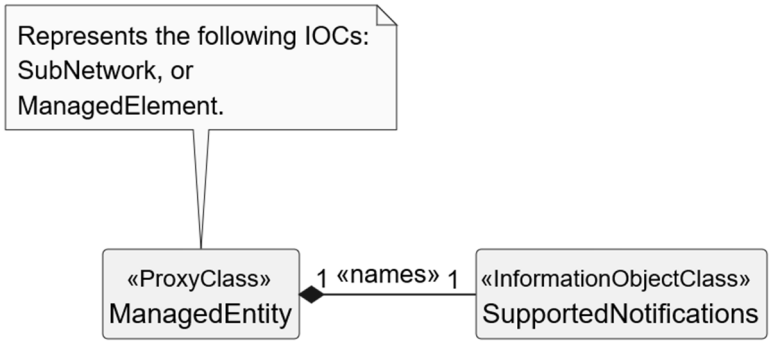



**Figure 4.2.1-13: SupportedNotifications NRM fragment**

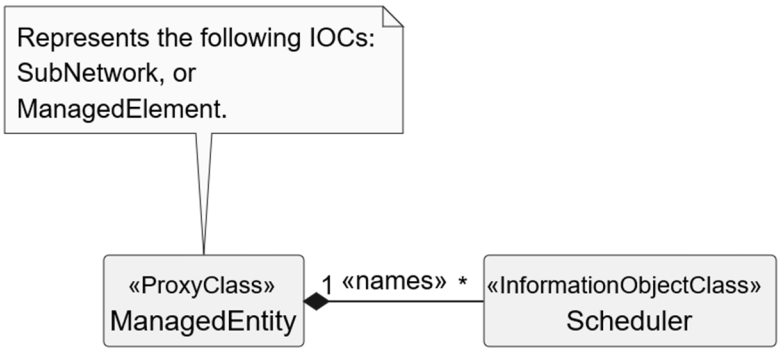



**Figure 4.2.1-14: Scheduler NRM fragment**

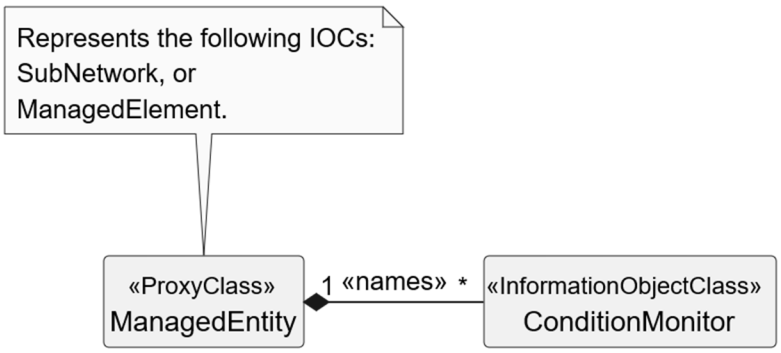



**Figure 4.2.1-15: Condition monitor NRM fragment**

## 4.2.2 Inheritance

This clause depicts the inheritance relationships.

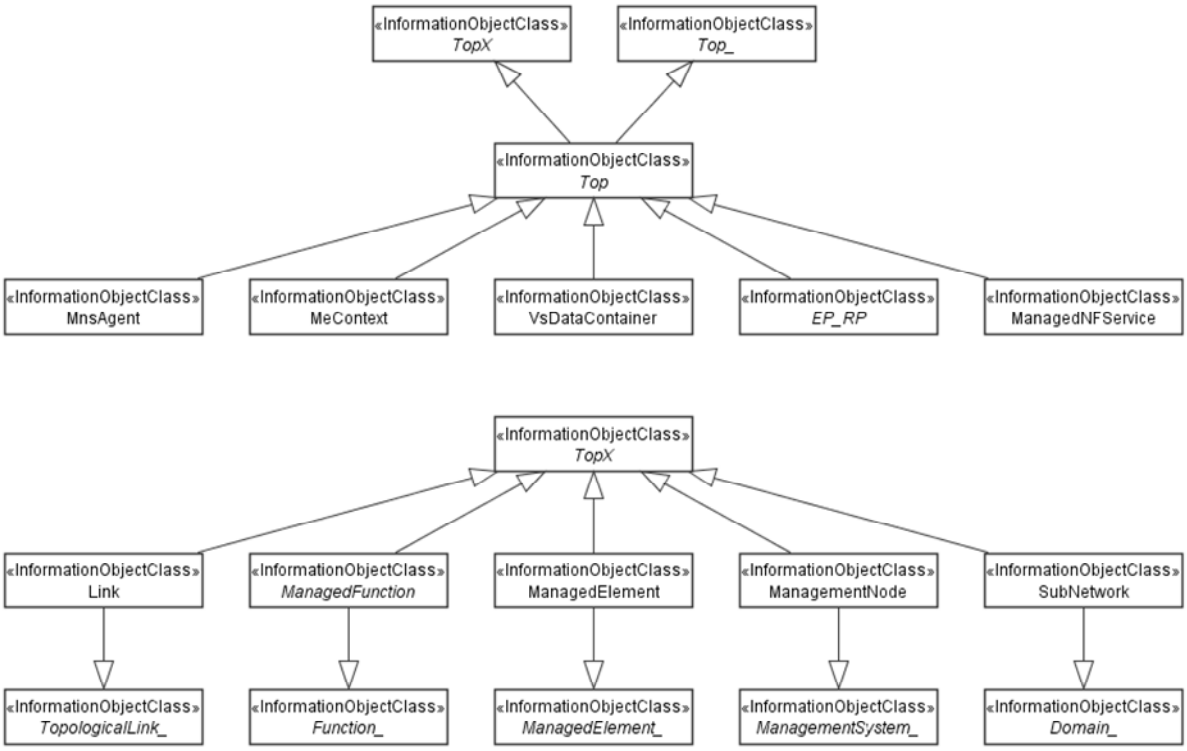



**Figure 4.2.2-1: NRM fragment**

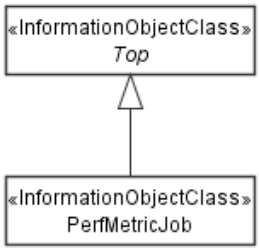



**Figure 4.2.2-2: PM control NRM fragment**

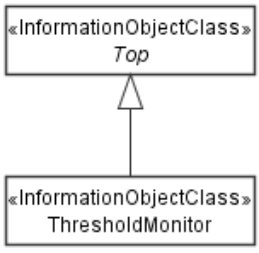



**Figure 4.2.2-3: Threshold monitoring control NRM fragment**

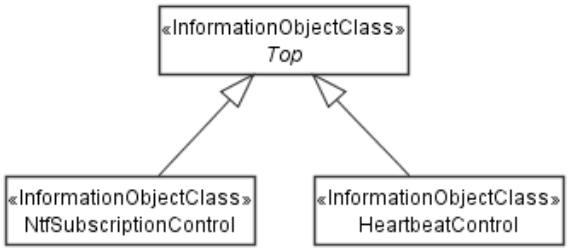



**Figure 4.2.2-4: Notification subscription and heartbeat notification control NRM fragment**

**Figure 4.2.2-5: Void**

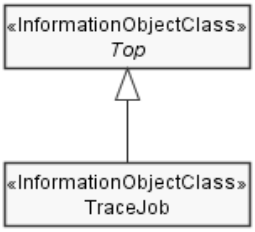



**Figure 4.2.2-6: Trace control NRM fragment**

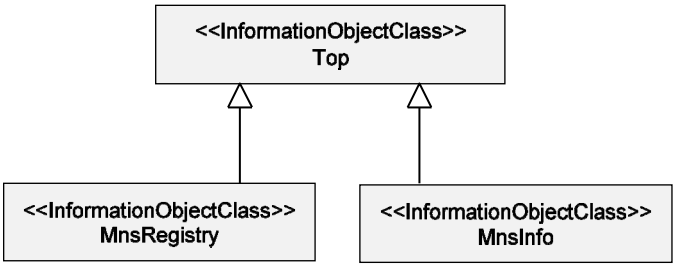



**Figure 4.2.2-7: MnS Registry NRM fragment**

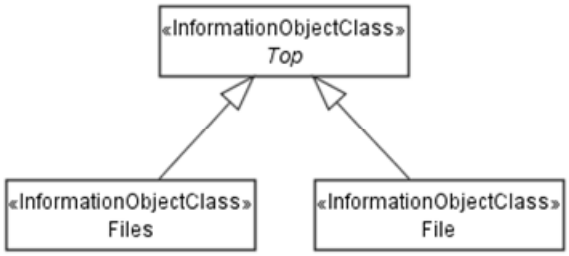



**Figure 4.2.2-8: File retrieval NRM fragment**




**Figure 4.2.2-9: File download NRM fragment**

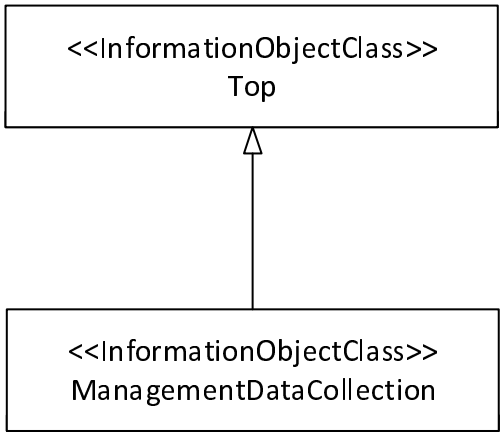



**Figure 4.2.2-10: Management data collection NRM fragment**

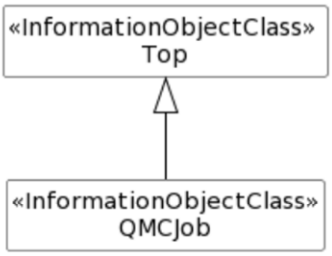



**Figure 4.2.2-11: QoE Measurement Collection NRM fragment**

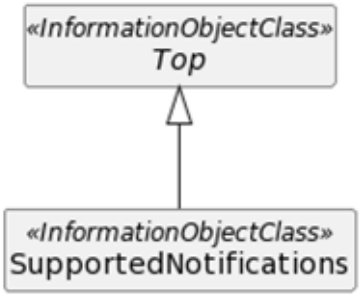



**Figure 4.2.2-12: SupportedNotifications NRM fragment**

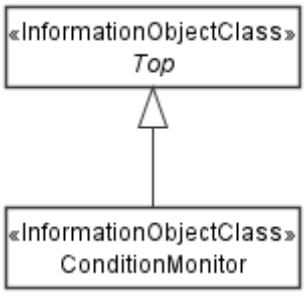



**Figure 4.2.2-13: ConditionMonitor control NRM fragment**

# 4.3 Class definitions

## 4.3.1 Any

### 4.3.1.1 Definition

This class represents the classes (e.g. `IOC`) that are not defined in this specification but are or will be defined in other IRP specification(s).

### 4.3.1.2 Attributes

None

### 4.3.1.3 Attribute constraints

None

### 4.3.1.4 Notifications

This class does not support any notification.

## 4.3.2 Void

## 4.3.2a MnsAgent

### 4.3.2a.1 Definition

The `MnsAgent` represents the MnS producers, incl. the supporting hardware and software, available for a certain management scope that is related to the object name-containing the MnS Agent.

The `MnSAgent` can be name-contained under an IOC as follows:

1) `ManagementNode`;

2) `SubNetwork`, if the `SubNetwork` does not contain a `ManagementNode`;

3) `ManagedElement`, if it is the root element.

In case the `MnsAgent` is name-contained under a `ManagementNode`, the management scope is the complete management scope of the `ManagementNode` or a subset thereof.

In case the MnsAgent is name-contained under a SubNetwork, the management scope is the complete SubNetwork or a subset thereof.

In case the MnsAgent is name-contained under a ManagedElement, the management scope is the complete ManagedElement or a subset thereof.

#### 4.3.2a.2 Attributes

The MnSAgent IOC includes the attributes inherited from Top_ IOC (defined in TS 28.620 [9]), attributes inherited from Top IOC (defined in clause 4.3.8) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| systemDN | M | T | F | F | T |

#### 4.3.2a.3 Attribute constraints

None.

#### 4.3.2a.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions or additions.

### 4.3.3 ManagedElement

#### 4.3.3.1 Definition

This IOC represents telecommunications equipment or TMN entities within the telecommunications network providing support and/or service to the subscriber.
A ManagedElement IOC is used to represent a Network Element defined in TS 32.101[1] including virtualization or non-virtualization scenario. ManagementElement instance is used for communicating with a manager (directly or indirectly) over one or more management interfaces for the purpose of being monitored and/or controlled. ManagedElement may or may not additionally perform element management functionality. A ManagedElement contains equipment that may or may not be geographically distributed.

A telecommunication equipment has software and hardware components. The ManagedElement IOC described above represents the following two cases:

- In the case when the software component is designed to run on dedicated hardware component, the ManagedElement IOC description includes both software and hardware component.

- In the case when the software is designed to run on ETSI NFV defined NFVI in ETSI GS NFV 003 [15], the ManagedElement IOC description would exclude the NFVI component supporting the above mentioned subject software.

A ManagedElement may be contained in either a SubNetwork or in a MeContext instance. A ManagedElement may also exist stand-alone with no parent at all.

The relation of ManagedElement IOC and ManagedFunction IOC can be described as following:

- A ManagedElement instance may have 1..1 containment relationship to a ManagedFunction instance. In this case, the ManagedElement IOC may be used to represent a NE with single ManagedFunction functionality. For example, a ManagedElement is used to represent the 3GPP defined RNC node.

- A ManagedElement instances may have 1..N containment relationship to multiple ManagedFunction IOC instances. In this case, the ManagedElement IOC may be used to represent a NE with combined ManagedFunction functionality (as indicated by the managedElementType attribute and the contained instances of different ManagedFunction IOCs). For example, a ManagedElement is used to represent the combined functionality of 3GPP defined non-split gNB.

NOTE: For some specific functional IOCs a 1..N containment relationship is permitted. The specific functional entities are identified in the NRMs that define subclasses of `ManagedFunction`.

#### 4.3.3.2 Attributes

The `ManagedElement` IOC includes the attributes inherited from `ManagedElement_` IOC (defined in TS 28.620 [9]), attributes inherited from `TopX` IOC (defined in clause 4.3.8) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| vendorName | M | T | F | F | T |
| userDefinedState | M | T | F | F | T |
| swVersion | O | T | T | F | T |
| priorityLabel | O | T | F | F | T |
| supportedPerfMetricGroups | O | T | F | F | T |
| supportedTraceMetrics | O | T | F | F | T |

#### 4.3.3.3 Attribute constraints

Attribute constrains for `dnPrefix`: The attribute `dnPrefix` shall be supported if an instance of `ManagedElement` is the local root instance of the MIB. Otherwise the attribute shall be absent or carry no information.

#### 4.3.3.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC. In addition, the following set of notifications is also valid.

| Name | S | Notes |
|---|---|---|
| notifyFileReady | M | -- |
| notifyFilePreparationError | M | -- |

### 4.3.4 *ManagedFunction*

#### 4.3.4.1 Definition

This IOC is provided for sub-classing only. It provides attribute(s) that are common to functional IOCs. Note that a `ManagedElement` may contain several managed functions, a managed function may contain other managed functions as specified for the specific subclass.. The `ManagedFunction` may be extended in the future if more common characteristics to functional objects are identified.

This IOC can represent a telecommunication function either realized by software running on dedicated hardware or realized by software running on NFVI. Each `ManagedFunction` instance communicates with a manager (directly or indirectly) over one or more management interfaces exposed via its containing ME instance.

#### 4.3.4.2 Attributes

The `ManagedFunction` IOC includes the attributes inherited from `Function_` IOC (defined in TS 28.620 [9]), attributes inherited from `TopX` IOC (defined in clause 4.3.8) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| vnfParametersList | CM | T | T | F | T |
| peeParametersList | CM | T | T | F | T |
| priorityLabel | O | T | T | F | T |
| supportedPerfMetricGroups | O | T | F | F | T |
| supportedTraceMetrics | O | T | F | F | T |

#### 4.3.4.3 Attribute constraints

| Name | Definition |
|---|---|
| vnfParametersList Support Qualifier | Condition: The ManagedFunction instance is realized by one or more VNF instance(s). Otherwise this attribute shall be absent. |
| peeParametersListSupport Qualifier | Condition: The control and monitoring of PEE parameters is supported by the ManagedFunction or sub-class instance. |

#### 4.3.4.4 Notifications

There is no notification defined.

### 4.3.5 ManagementNode

#### 4.3.5.1 Definition

This IOC represents a telecommunications management system (EM) within the TMN that contains functionality for managing a number of ManagedElements (MEs). The management system communicates with the MEs directly or indirectly over one or more interfaces for the purpose of monitoring and/or controlling these MEs.

This class has similar characteristics as the ManagedElement. The main difference between these two classes is that the ManagementNode has a special association to the managed elements that it is responsible for managing.

#### 4.3.5.2 Attributes

The ManagementNode IOC includes the attributes inherited from ManagementSystem_ IOC (defined in TS 28.620 [9]), attributes inherited from TopX IOC (defined in clause 4.3.8) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| vendorName | M | T | F | F | T |
| userDefinedState | M | T | T | F | T |
| locationName | M | T | F | F | T |
| swVersion | M | T | F | F | T |

#### 4.3.5.3 Attribute constraints

None

#### 4.3.5.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC. In addition, the following set of notifications is also valid.

| Name | S | Notes |
|---|---|---|
| notifyFileReady | M | -- |
| notifyFilePreparationError | M | -- |

### 4.3.6 MeContext

#### 4.3.6.1 Definition

This IOC is introduced for naming purposes. It may support creation of unique DNs in scenarios when some MEs have the same RDNs due to the fact that they have been manufacturer pre-configured.
If some MEs have the same RDNs (for the above mentioned reason) and they are contained in the same SubNetwork

instance, some measure shall be taken in order to assure the global uniqueness of DNs for all IOC instances under those MEs. One way could be to set different dnPrefix for those NEs, but that would require either that:

a) all LDNs or DNs are locally modified using the new dnPrefix for the upper portion of the DNs, or

b) a mapping (translation) of the old LDNs or DNs to the new DNs every time they are used externally, e.g. in alarm notifications.

As both the two alternatives above may involve unacceptable drawbacks (as the old RDNs for the MEs then would have to be changed or mapped to new values), using MeContext offers a new alternative to resolve the DN creation. Using MeContext as part of the naming tree (and thus the DN) means that the dnPrefix, including a unique MeContext for each ME, may be directly concatenated with the LDNs, without any need to change or map the existing ME RDNs to new values.

MeContext have 0..N instances. It may exist even if no SubNetwork exists. Every instance of MeContext contains exactly one ManagedElement during steady-state operations.

#### 4.3.6.2 Attributes

The MeContext IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| dnPrefix | CM | T | T | T | T |

#### 4.3.6.3 Attribute constraints

| Name | Definition |
|---|---|
| dnPrefixSupport Qualifier | Condition: The instance of MeContext is the local root instance of the MIB. Otherwise the attribute shall be absent or carry no information. |

#### 4.3.6.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions or additions.

### 4.3.7 SubNetwork

#### 4.3.7.1 Definition

This IOC represents a set of managed entities. There may be zero or more instances of a SubNetwork. It shall be present if either a ManagementNode or multiple ManagedElements are present (i.e. ManagementNode and multiple ManagedElement instances shall have SubNetwork as parent).

The SubNetwork instance not contained in any other instance of SubNetwork is referred to as the "root" SubNetwork instance.

#### 4.3.7.2 Attributes

The SubNetwork IOC includes the attributes inherited from Domain_ IOC (defined in TS 28.620 [9]), attributes inherited from TopX IOC (defined in clause 4.3.8) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| setOfMcc | CM | T | F | F | T |
| priorityLabel | O | T | T | F | T |
| supportedPerfMetricGroups | O | T | F | F | T |
| supportedTraceMetrics | O | T | F | F | T |

#### 4.3.7.3 Attribute constraints

| Name | Definition |
|---|---|
| dnPrefix (inherited from *Domain_*)<br>Support Qualifier | Condition: The instance of SubNetwork is the local root instance of the MIB. Otherwise the attribute shall be absent or carry no information. |
| setOfMccSupport Qualifier | Condition: There is more than one value in setOfMcc of the SubNetwork ; otherwise the support is optional. |

#### 4.3.7.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions or additions

### 4.3.8 TopX

#### 4.3.8.1 Definition

This IOC is provided for sub-classing only. All information object classes defined in all TS that claim to be conformant to TS 32.102 [2] shall inherit from TopX.

#### 4.3.8.2 Attributes

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| objectClass | M | T | T | T | T |
| objectInstance | M | T | T | T | T |

#### 4.3.8.3 Attribute constraints

None

#### 4.3.8.4 Notifications

There is no notification defined.

### 4.3.9 VsDataContainer

#### 4.3.9.1 Definition

The VsDataContainer is a container for vendor specific data. The VsDataContainer is contained by Top and hence optionally name-contained by each IOC.

#### 4.3.9.2 Attributes

The VsDataContainer IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| vsDataType | M | T | F | F | O |
| vsData | M | T | O | F | O |
| vsDataFormatVersion | M | T | F | F | O |

#### 4.3.9.3 Attribute constraints

None

#### 4.3.9.4 Notifications

Support for notification on the change of attribute value is vendor-specific.

### 4.3.10 *Link*

#### 4.3.10.1 Definition

This IOC is provided for sub-classing only. This IOC represents a communication link or reference point between two network entities. The `Link` IOC does not indicate whether the represented communication link or reference point is a physical or logical entity.

For the subclasses of `Link`, the following rules apply:

1) The subclass names shall have the form "Link_<X>_<Y>", where <X> is a string that represents the IOC at one end of the association related to the particular `Link` subclass, and <Y> is a string that represents the IOC at the other end of the association. For the order of the two strings, <X> shall come alphabetically before <Y>.

2) In case <X> and <Y> are YyyFunction IOCs (inheriting from `ManagedFunction` and on first level below ManagedElement), the <X> and <Y> strings shall have the same form as the legal values of the `managedElementTypeList` attribute (see clause 6.1 in TS 28.620 [9]), e.g. "Auc". Otherwise <X> and <Y> shall be the full IOC names.

Thus, two valid examples of `Link` subclass names would be: `Link_As_Cscf` and `Link_Mrfc_Mrfp.`

#### 4.3.10.2 Attributes

The `Link` IOC includes the attributes inherited from `TopologicalLink_` (defined in TS 28.620 [9]), attributes inherited from `TopX` IOC (defined in clause 4.3.8) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| `userLabel` | M | T | T | F | T |
| `linkType` | O | T | F | F | T |
| `protocolVersion` | O | T | F | F | T |

#### 4.3.10.3 Attribute constraints

| Name | Definition |
|---|---|
| aEnd and zEnd (inherited from *`TopologicalLink_`*)<br>Support Qualifier | Condition: The property multiplicity is 1. |

#### 4.3.10.4 Notifications

The common notifications defined in subclause 4.5 are valid for this IOC, without exceptions or additions

### 4.3.11 *EP_RP*

#### 4.3.11.1 Definition

This IOC is provided for sub-classing only. This IOC represents an end point of a communication link or a reference point between two network entities.

For naming the subclasses of `EP_RP`, the following rules shall apply:

- The name of the subclassed IOC shall have the form "EP_<rp>", where <rp> is a string that represents the name of the reference point.

Thus, two valid examples of `EP_RP` subclassed IOC names would be: `EP_S1U` and `EP_X2C`.

#### 4.3.11.2 Attributes

The EP_RP IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| farEndEntity | O | T | F | F | T |
| userLabel | O | T | T | F | T |
| supportedPerfMetricGroups | O | T | F | F | T |

#### 4.3.11.3 Attribute constraints

None

#### 4.3.11.4 Notifications

This class does not support any notification.

### 4.3.12 Void

### 4.3.13 Void

### 4.3.14 Void

### 4.3.15 Void

### 4.3.16 ThresholdMonitor

#### 4.3.16.1 Definition

This IOC represents a threshold monitor for performance metrics. It can be name-contained by SubNetwork, ManagedElement, or ManagedFunction. A threshold monitor checks for threshold crossings of performance metric values related to specified managed objects and generates a notification when that happens.

The ThresholdMonitor is used only when NRM based threshold monitoring is supported.

To activate threshold monitoring, a MnS consumer needs to create a ThresholdMonitor instance on the MnS producer. For ultimate deactivation of threshold monitoring, the MnS consumer should delete the monitor to free up resources on the MnS producer.

For temporary suspension of threshold monitoring, the MnS consumer can manipulate the value of the administrative state attribute. The MnS producer may disable threshold monitoring as well, for example in overload situations. This situation is indicated by the MnS producer with setting the operational state attribute to disabled. When monitoring is resumed the operational state is set again to enabled.

All object instances below and including the instance name-containing the ThresholdMonitor (base object instance) are scoped for performance metricthreshold monitoring. Performance metrics are monitored only on those object instances whose object class matches the object class associated to the performance metrics to be monitored.

The optional attributes objectInstances and rootObjectInstances allow to restrict the scope. When the attribute objectInstances is present, only the object instances identified by this attribute are scoped. When the attribute rootObjectInstances is present, then the subtrees whose root objects are identified by this attribute are scoped. Both attributes may be present at the same time meaning the total scope is equal to the sum of both scopes. Object instances may be scoped by both the objectInstances and rootObjectInstances attributes. This shall not be considered as an error by the MnS producer.

Multiple thresholds can be defined for multiple performance metric sets in a single monitor using `thresholdInfoList`. The attribute `monitorGranularityPeriod` defines the granularity period to be applied. The value is a multiple of a supported granularity period for the measurements being monitored.

A threshold is defined using the attributes `thresholdValue`, `thresholdDirection` and `hysteresis`.

When `hysteresis` is absent or carries no information, a threshold is triggered when the `thresholdValue` is reached or crossed. When `hysteresis` is present, two threshold values are specified for the threshold as follows: A high threshold value equal to the threshold value plus the hysteresis value, and a low threshold value equal to the threshold value minus the hysteresis value. When the monitored performance metric increases, the threshold is triggered when the high threshold value is reached or crossed. When the monitored performance metric decreases, the threshold is triggered when the low threshold value is reached or crossed. The hysteresis ensures that the performance metric value can oscillate around a comparison value without triggering each time the threshold when the threshold value is crossed.Using the `thresholdDirection` attribute a threshold can be configured in such a manner that it is triggered only when the monitored performance metric is going up or down upon reaching or crossing the threshold.

A `ThresholdMonitor` creation request shall be rejected, if the performance metrics requested to be monitored, the requested granularity period, or the requested combination thereof is not supported by the MnS producer. A creation request may fail, when the performance metrics requested to be monitored are not produced by a `PerfMetricJob`.

Creation and deletion of `ThresholdMonitor` instances by MnS consumers is optional; when not supported, `ThresholdMonitor` instances may be created and deleted by the system or be pre-installed.

A threshold crossing event detected by a `ThresholdMonitor` shall trigger a `notifyThresholdCrossing` notification. To subscribe to `notifyThresholdCrossing` notifications the MnS consumer shall specify one or more `ThresholdMonitor` instances in the subscription. All threshold crossings detected by the specified `ThresholdMonitor` instances are sent as `notifyThresholdCrossing` to subscribed MnS consumers (unless filtered out by the `notificationFilter` attribute of `NtfSubscriptionControl`).

### 4.3.16.2 Attributes

The `ThresholdMonitor` IOC includes attributes inherited from `Top` IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| `administrativeState` | M | T | T | F | T |
| `operationalState` | M | T | F | F | T |
| `thresholdInfoList` | M | T | T | F | T |
| `monitorGranularityPeriod` | M | T | T | F | T |
| `objectInstances` | O | T | T | F | F |
| `rootObjectInstances` | O | T | T | F | F |

### 4.3.16.3 Attribute constraints

None.

### 4.3.16.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC.

## 4.3.17 ManagedNFService

### 4.3.17.1 Definition

A ManagedNFService represents a Network Function (NF) service as defined in clause 7 of 3GPP TS 23.501[22].

#### 4.3.17.2 Attributes

The ManagedNFService IOC includes attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| administrativeState | M | T | T | F | T |
| operationalState | M | T | F | F | T |
| userLabel | O | T | T | F | T |
| nFServiceType | M | T | F | T | F |
| sAP | M | T | T | F | T |
| operations | M | T | T | F | T |
| usageState | M | T | F | F | T |
| registrationState | CM | T | F | F | T |

#### 4.3.17.3 Attribute constraints

Attribute constraint for `registrationState`: The attribute `registrationState` should be supported by instance of a ManagedNFService if the service is designed for being published and discovered by other NFs, and need to be registered to a repository function. E.g. Authentication service provided by AUSF should include this attribute. NF management services provided by NRF don't include this attribute.

#### 4.3.17.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions or additions

### 4.3.18 Operation <<dataType>>

#### 4.3.18.1 Definition

This <<dataType>> represents an `Operation`. An `Operation` is comprised of a name, an allowedNFType and an operationSemantics (See TS 23.502 [23]).

#### 4.3.18.2 Attributes

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| name | M | T | F | T | F |
| allowedNFTypes | M | T | T | F | T |
| operationSemantics | M | T | F | T | T |

#### 4.3.18.3 Attribute constraints

None

#### 4.3.18.4 Notifications

The subclause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.19 SAP <<dataType>>

#### 4.3.19.1 Definition

This <<dataType>> represents the access point of a managed NF service which is comprised of a host and a port.

#### 4.3.19.2 Attributes

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| host | M | T | T | F | T |
| port | M | T | T | F | T |

#### 4.3.19.3 Attribute constraints

None

#### 4.3.19.4 Notifications

The subclause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.20 ManagedEntity <<ProxyClass>>

#### 4.3.20.1 Definition

This <<ProxyClass>> represents one or multiple IOCs. The IOCs the <<ProxyClass>> represents are defined where the <<ProxyClass>> is used.

#### 4.3.20.2 Attributes

See respective IOCs.

#### 4.3.20.3 Attribute constraints

See respective IOCs.

#### 4.3.20.4 Notifications

See respective IOCs.

### 4.3.21 HeartbeatControl

#### 4.3.21.1 Definition

MnS consumers (i.e. notification recipients) use heartbeat notifications to monitor the communication channels between themselves and MnS producers configured to emit notifications such as notifyNewAlarm and notifyFileReady.

A HeartbeatControl instance allows controlling the emission of heartbeat notifications by MnS producers. The recipients of heartbeat notifications are specified by the notificationRecipientAddress attribute of the NtfSubscriptionControl instance name containing the HeartbeatControl instance.

Note that the MnS consumer managing the HeartbeatControl instance and the MnS consumer receiving the heartbeat notifications may not be the same.

As a pre-condition for the emission of heartbeat notifications, a HeartbeatControl instance needs to be created. Creation of an instance with an initial non-zero value of the heartbeatNtfPeriod attribute triggers an immediate heartbeat notification emission. Creation of an instance with an initial zero value of the heartbeatNtfPeriod attribute does not trigger an emission of a heartbeat notification. Deletion of an instance does not trigger an emission of a heartbeat notification.

Once the instance is created, heartbeat notifications are emitted with a periodicity defined by the value of the heartbeatNtfPeriod attribute. No heartbeat notifications are emitted if the value is equal to zero. Setting a zero value to a non zero value, or a non zero value to a different non zero value, triggers an immediate heartbeat notification,

that is the base for the new heartbeat period. Setting a non zero value to a zero value stops emitting heartbeats immediately; no final heartbeat notification is sent.

The attribute `triggerHeartbeatNtf` allows MnS consumers to trigger the emission of an immediate additional heartbeat notification. The emission of heartbeat notifications according to the heartbeat period is not impacted by this additional notification.

Creation and deletion of `HeartbeatControl` instances by MnS Consumers is optional; when not supported, the `HeartbeatControl` instances may be created and deleted by the system or be pre-installed.

The emission of heartbeat notifications is fully controlled by `HeartbeatControl` instances. Subscription for heartbeat notifications is not supported by `NtfSubscriptionControl`.

#### 4.3.21.2 Attributes

The `HeartbeatControl` IOC includes attributes inherited from `Top` IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| `heartbeatNtfPeriod` | M | T | T | F | T |
| `triggerHeartbeatNtf` | M | F | T | F | F |

#### 4.3.21.3 Attribute constraints

None.

#### 4.3.21.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC. In addition, the following set of notifications is also valid.

| Name | S | Notes |
|---|---|---|
| `notifyHeartbeat` | M | -- |

### 4.3.22 `NtfSubscriptionControl`

#### 4.3.22.1 Definition

`NtfSubscriptionControl` represents a notification subscription of a notification recipient. It can be name-contained by `SubNetwork` or `ManagedElement`.

The `scope` attribute is used to select the data nodes included in the subscription. The base object instance of the scope (see clause 4.3.23) is the object instance name-containing the `NtfSubscriptionControl` instance. When the `scope` attribute is absent, all objects below and including the base object are scoped.

For most notification types the scope needs to identify a set of managed object instances, such as for alarm notifications. For some notification types one or more attributes, attribute fields or attribute elements may be specified as well, though, such as for attribute value change notifications. Details on this matter are specified together with the definition of the notification type.

Note that a scope may also include objects that are not created yet, for example, when a complete subtree is scoped, or when all objects with a specific object class are scoped. Object instances added after creating the subscription are included in the subscription as well if not explicitly excluded.

The notifications related to the selected data nodes are candidates to be sent to the address specified by the `notificationRecipientAddress` attribute.

The `notificationTypes` attribute and `notificationFilter` attribute allow MnS consumers to control which candidate notifications are sent to the `notificationRecipientAddress`.

If the notificationTypes attribute is present, its value identifies the notification types that are candidates to be sent to the notificationRecipientAddress. If the notificationTypes attribute is absent, notifications of all types are candidates to be sent to notificationRecipientAddress. Notification types supported in the NtfSubscriptionControl.notificationTypes attribute are the ones listed in the attribute SupportedNotifications.notificationTypes.

If supported, the notificationFilter attribute defines a filter that is applied to the set of candidate notifications. The filter is applicable to all parameters of a notification. Only candidate notifications that pass the filter criteria are sent to the notificationRecipientAddress. If the notificationFilter attribute is absent, all candidate notificatios are sent to the notificationRecipientAddress.

To receive notifications, a MnS consumer has to create a NtfSubscriptionControl instance on the MnS producer. A MnS consumer can create a subscription for another MnS consumer since it is not required the notificationRecipientAddress be his own address.

When a MnS consumer does not wish to receive notifications any more the MnS consumer shall delete the corresponding NtfSubscriptionControl instance.

When a subscription is created and the notification scope inludes the created subscription object and the subscribed notification types include notifications reporting object creation (notifyMOICreation or notifyMOIChanges ), the first notification sent related to the new subscription shall report the creation of the NtfSubscriptionControl instance. Likewise, when a subscription is deleted and the notification scope inludes the deleted subscription object and the subscribed notification types include notifications reporting object deletion (notifyMOIDeletion or notifyMOIChanges), the last notification sent related to the subscription shall report the deletion of the NtfSubscriptionControl instance.

Creation and deletion of NtfSubscriptionControl instances by MnS consumers is optional; when not supported, the NtfSubscriptionControl instances may be created and deleted by the system or be pre-installed.

#### 4.3.22.2 Attributes

The NtfSubscriptionControl IOC includes attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| notificationRecipientAddress | M | T | T | F | T |
| notificationTypes | O | T | T | F | T |
| scope | O | T | T | F | T |
| notificationFilter | O | T | T | F | T |

#### 4.3.22.3 Attribute constraints

None.

#### 4.3.22.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions or additions.

### 4.3.23 Scope <<dataType>>

#### 4.3.23.1 Definition

This <<dataType>> allows to select data nodes in an object tree whose root is identified by the so called base managed object instance. The identification of the base object instance is not part of this <<dataType>> and needs to be specified by other means. The base managed object instance is typically a managed object instance in an object tree.

The scopeType and the scopeLevel attributes allow to select managed object instances. Attributes, attribute fields and attribute elements cannot be selected.

The dataNodeSelector attribute allows to select managed object instances, attributes, attribute fields, attribute elements, or attribute field elements. Its value contains a solution set specific expression for specifying the data nodes to be selected.

#### 4.3.23.2 Attributes

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 scopeType | M | T | T | F | T |
| CHOICE_1.2 scopeLevel | O | T | T | F | T |
| CHOICE_2.1 dataNodeSelector | O | T | T | F | T |

#### 4.3.23.3 Attribute constraints

None.

#### 4.3.23.4 Notifications

The subclause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.24 Void

### 4.3.25 Void

### 4.3.27 Void

### 4.3.28 Void

### 4.3.29 *Top*

#### 4.3.29.1 Definition

This IOC is provided for sub-classing only. All information object classes defined in all TS that claim to be conformant to 32.102 [2] and support the Federated Network Information Model (FNIM) concept shall inherit from Top.

#### 4.3.29.2 Attributes

This IOC includes attributes inherited from TopX IOC (defined in clause 4.3.8) and the attributes inherited from Top_ IOC (defined in TS 28.620 [9]).

#### 4.3.29.3 Attribute constraints

None

#### 4.3.29.4 Notifications

There is no notification defined.

### 4.3.30 TraceJob

#### 4.3.30.1 Definition

A TraceJob instance represents the Trace Control and Configuration parameters of a particular Trace Job (see TS 32.421 [29] and TS 32.422 [30] for details). It can be name-contained by SubNetwork, ManagedElement,

ManagedFunction. In case of signalling based trace activation, it shall be name-contained by the UDMFunction, see TS 28.541 [48].

To activate Trace Jobs, a MnS consumer has to create TraceJob object instances on the MnS producer. A MnS consumer can activate a Trace Job for another MnS consumer since it is not required the value of traceCollectionEntityIPAddress or traceReportingConsumerUri to be his own.

For the details of Trace Job activation see clauses 4.1.1.1.2 and 4.1.2.1.2 of TS 32.422 [30].

When a MnS consumer wishes to deactivate a Trace Job, the MnS consumer shall delete the corresponding TraceJob instance. For details of management Trace Job deactivation see clauses 4.1.3.8 to 4.1.3.11 and 4.1.4.10 to 4.1.4.13 of TS 32.422 [30].

The attribute traceReference specifies a globally unique ID and identifies a Trace session. One Trace Session may be activated to multiple Network Elements. The traceReference is populated by the consumer that makes the request for a Trace Session, TS 32.422 [30].

The jobId attribute presents the job identifier of a TraceJob instance. The jobId can be used to associate multiple TraceJob instances. For example, it is possible to configure the same jobId value for multiple TraceJob instances required to produce the data (e.g. RSRP values of M1 and RLF reports) for a specific network analysis.

The attribute traceReportingFormat defines the method for reporting the produced measurements. The selectable options are file-based or stream-based reporting. In case of file-based reporting the attribute traceCollectionEntityIPAddress is used to specify the IP address to which the trace records shall be transferred, while in case of stream-based reporting the attribute traceReportingConsumerUri specifies the streaming target.

The mandatory attribute traceTarget determines the target object of the TraceJob. Dependent on the network element to which the Trace Session is activated different types of the target object are possible. The attribute pLMNTarget defines the PLMN for which sessions shall be selected in the Trace Session in case of management based activation when several PLMNs are supported in the RAN.

The attribute listOfTraceMetrics allows configuration of which metrics shall be recorded.

The attribute jobType specifies the kind of data to collect. In case of TRACE_ONLY, the configuration parameters of attribute traceConfig shall be applied. In case of IMMEDIATE_MDT_ONLY, LOGGED_MDT_ONLY, RLF_REPORT_ONLY, RCEF_REPORT_ONLY and LOGGED_MBSFN_MDT the configuration parameters of attribute mdtConfig or a subset of these shall be applied. In case of 5GC_UE_LEVEL_MEASUREMENTS_ONLY, the configuration parameters of attribute ueCoreMeasConfig shall be applied. In case of any combination of Trace, Immediate MDT, and 5GC UE level measurements, the configuration parameters of the corresponding attributes, traceConfig, mdtConfig and ueCoreMeasConfig are applicable.

Creation and deletion of TraceJob instances by MnS consumers is optional; when not supported, the TraceJob instances may be created and deleted by the system or be pre-installed.

### 4.3.30.2 Attributes

The TraceJob IOC includes attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| jobType | M | T | T | F | T |
| pLMNTarget | CM | T | T | F | T |
| traceReportingConsumerUri | CM | T | T | F | T |
| traceCollectionEntityIPAddress | M | T | T | F | T |
| traceReference | M | T | T | F | T |
| jobId | O | T | T | T | T |
| traceReportingFormat | M | T | T | F | T |
| traceTarget | M | T | T | F | T |
| listOfTraceMetrics | CM | T | T | F | T |
| traceConfig | CM | T | T | F | T |
| mdtConfig | CM | T | T | F | T |
| ueCoreMeasConfig | CM | T | T | F | T |
| nPNTarget | CM | T | T | F | T |

#### 4.3.30.3 Attribute constraints

| Name | Definition |
|---|---|
| pLMNTarget (support qualifier) | This attribute shall be present for management based activation when several PLMNs are supported in the RAN. |
| traceReportingConsumerUri (support qualifier) | This attribute shall be present if streaming trace data reporting is supported. |
| traceConfig (support qualifier) | This attribute shall be present if Trace is supported. |
| mdtConfig (support qualifier) | This attribute shall be present if MDT is supported. |
| ueCoreMeasConfig (support qualifier) | This attribute shall be present if 5GC UE level measurements collection is supported. |
| nPNTarget | This attribute shall be present for management-based activation when several NPNs are supported in the RAN (see TS 38.331 [38]). It is only applicable for NR. |
| listOfTraceMetrics (support qualifier) | This attribute shall be present when configuration of which trace metrics to report is supported. |

#### 4.3.30.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions. In addition, the following set of notifications is also valid.

| Name | S | Notes |
|---|---|---|
| notifyFileReady | M | -- |
| notifyFilePreparationError | M | -- |

### 4.3.31 PerfMetricJob

#### 4.3.31.1 Definition

This IOC represents a performance metric production job. It can be name-contained by SubNetwork, ManagedElement, or ManagedFunction.

To activate the production of the specified performance metrics, a MnS consumer needs to create a PerfMetricJob instance on the MnS producer. For ultimate deactivation of metric production, the MnS consumer should delete the job to free up resources on the MnS producer.

For temporary suspension of metric production, the MnS consumer can manipulate the value of the administrative state attribute. The MnS producer may disable metric production as well, for example in overload situations. This situation is indicated by the MnS producer with setting the operational state attribute to disabled. When production is resumed the operational state is set back to enabled.

The jobId attribute can be used to associate metrics from multiple PerfMetricJob instances. The jobId can be included when reporting performance metrics to allow a MnS consumer to associate received metrics for the same

purpose. For example, it is possible to configure the same `jobId` value for multiple `PerfMetricJob` instances required to produce the measurements for a specific KPI.

The attribute `performanceMetrics` defines the performance metrics to be produced and the attribute `granularityPeriod` defines the granularity period to be applied.

All object instances below and including the instance name-containing the `PerfMetricJob` (base object instance) are scoped for performance metric production. Performance metrics are produced only on those object instances whose object class matches the object class associated to the performance metrics to be produced.

The optional attributes `objectInstances` and `rootObjectInstances` allow to restrict the scope. When the attribute `objectInstances` is present, only the object instances identified by this attribute are scoped. When the attribute `rootObjectInstances` is present, then the subtrees whose root objects are identified by this attribute are scoped. Both attributes may be present at the same time meaning the total scope is equal to the sum of both scopes. Object instances may be scoped by both the `objectInstances` and `rootObjectInstances` attributes. This shall not be considered as an error by the MnS producer.

When the performance metric requires performance metric production on multiple managed objects, which is for example the case for KPIs, the MnS consumer needs to ensure all required objects are scoped. Otherwise, a `PerfMetricJob` creation request shall fail.

The production of the configured performance metrics can be constrained by conditions such that metric production shall be active only if the corresponding conditions are satisfied. These conditions are not configured directly in the `PerfMetricJob`. Instead, the attributes `schedulerRef` and `conditionMonitorRef` are provided allowing a `PerfMetricJob` to refer to a `Scheduler` or `ConditionMonitor` object. Only when the conditions in the referenced object are satisfied shall metric production be enabled.

Multiple `PerfMetricJob` objects can be linked to the same `Scheduler` or `ConditionMonitor` object. This may be necessary, when `PerfMetricJob` objects need to be switched on and off synchronously.

The attribute `reportingCtrl` specifies the method and associated control parameters for reporting the produced measurements to MnS consumers. Three methods are available: file-based reporting with selection of the file location by the MnS producer, file-based reporting with selection of the file location by the MnS consumer, and stream-based reporting.

For file-based reporting, all performance metrics that are produced related to a `PerfMetricJob` instance for a reporting period shall be stored in a single reporting file.

When the administrative state is set to "UNLOCKED" after the creation of a `PerfMetricJob` the first granularity period shall start. When the administrative state is set to "LOCKED" or the operational state to "DISABLED", the ongoing reporting period shall be aborted, for streaming the ongoing granularity period. When the administrative state is set back to "UNLOCKED" or the operational state to "ENABLED" a new reporting period shall start, in case of streaming a new granularity period.

Changes of all other configurable attributes shall take effect only at the beginning of the next reporting period, for streaming at the beginning of the next granularity period.

When the `PerfMetricJob` is deleted, the ongoing reporting period shall be aborted, for streaming the ongoing granularity period.

A `PerfMetricJob` creation request shall be rejected, if the requested performance metrics, the requested granularity period, the requested reporting method, or the requested combination thereof is not supported by the MnS producer.

Creation and deletion of `PerfMetricJob` instances by MnS consumers is optional; when not supported, `PerfMetricJob` instances may be created and deleted by the system or be pre-installed.

When the file retrieval NRM fragment is supported by the MnS producer, the `_linkToFiles` attribute shall be supported, for details on the usage of this attribute see the definition of the file retrieval NRM fragment.

### 4.3.31.2 Attributes

The `PerfMetricJob` IOC includes attributes inherited from `Top` IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| administrativeState | M | T | T | F | T |
| operationalState | M | T | F | F | T |
| jobId | M | T | T | T | T |
| performanceMetrics | M | T | T | F | T |
| granularityPeriod | M | T | T | F | T |
| objectInstances | O | T | T | F | T |
| rootObjectInstances | O | T | T | F | T |
| reportingCtrl | M | T | T | F | T |
| _linkToFiles | CO | T | F | T | F |
| **Attributes related to role** | | | | | |
| CHOICE_1.1 schedulerRef | O | T | T | F | T |
| CHOICE_1.2 conditionMonitorRef | O | T | T | F | T |

### 4.3.31.3 Attribute constraints

| Name | Definition |
|---|---|
| _linkToFiles | This attribute should be supported, when the MnS producer supports the file retrieval NRM fragment. |

### 4.3.31.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC. In addition, the following set of notifications is also valid.

| Name | S | Notes |
|---|---|---|
| notifyFileReady | M | -- |
| notifyFilePreparationError | M | -- |

## 4.3.32 SupportedPerfMetricGroup <<dataType>>

### 4.3.32.1 Definition

This <<dataType>> captures a group of supported performance metrics, and associated (production and monitoring) granularity periods and reporting methods that are supported for the specified performance metric group.

### 4.3.32.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| performanceMetrics | M | T | F | F | T |
| granularityPeriods | M | T | F | F | T |
| reportingMethods | M | T | F | F | T |
| reportingPeriods | M | T | F | F | T |

### 4.3.32.3 Attribute constraints

None

### 4.3.32.4 Notifications

Not applicable.

## 4.3.33 ReportingCtrl <<choice>>

### 4.3.33.1 Definition

This <<choice>> defines the method for reporting collected performance metrics to MnS consumers as well as the parameters for configuring the reporting function. It is a choice between the control parameter required for the reporting methods, whose presence selects the reporting method as follows:

When only the `fileReportingPeriod` attribute is present (CHOICE_1), the MnS producer shall store files on the MnS producer at a location selected by the MnS producer and, on condition that an appropriate subscription is in place, inform the MnS consumer about the availability of new files and the file location using the `notifyFileReady` notification. In case the preparation of a file fails, `notifyFilePreparationError` shall be sent instead.

When the `fileReportingPeriod` and `notificationRecipientAddress` attributes are present (CHOICE_2), then the MnS producer shall behave like described for the case that only the `fileReportingPeriod` is present. In addition, the MnS producer shall create on behalf of the MnS consumer a subscription, using `NtfSubscriptionControl`, for the notification types `notifyMOICreation` and `notifyMOIDeletion` related to the `File` instances that will be produced later. In case an existing subscription does already include the `File` instances to be produced, no new subscription shall be created. The `notificationRecipientAddress` attribute in the created `NtfSubscriptionControl` instance shall be set to the value of the `notificationRecipientAddress` in the related `PerfMetricJob`. This feature is called implicit notification subscription, as opposed to the case where the MnS consumer creates the subscription (explicit notification subscription). When the related `PerfMetricJob` is deleted, the `NtfSubscriptionControl` instance created due to the request for implicit subscription shall be deleted as well.

When only the “fileReportingPeriod” and “fileLocation” attributes are present (CHOICE_3), the MnS producer shall store the files at the location specified by “fileLocation”. The file location may identify any entity such as a file server or a MnS consumer. The identified MnS consumer may or may not be identical to the MnS consumer creating the “PerfMetricJob”. The MnS producer may emit “notifyFileReady” and "notifyFilePreparationError" notifications to inform subscribers that a file has been made available at the location specified by “fileLocation”.

When only the `streamTarget` attribute is present (CHOICE_4), the MnS producer shall stream the data to the location specified by `streamTarget`.

For the file-based reporting methods the `fileReportingPeriod` attribute specifies the time window during which collected measurements are stored into the same file before the file is closed and a new file is opened.

### 4.3.33.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 `fileReportingPeriod` | CM | T | T | F | T |
| CHOICE_2.1 `fileReportingPeriod` | CM | T | T | F | T |
| CHOICE_2.2 `notificationRecipientAddress` | CM | T | T | F | T |
| CHOICE_3.1 `fileReportingPeriod` | CM | T | T | F | T |
| CHOICE_3.2 `fileLocation` | CM | T | T | F | T |
| CHOICE_4.1 `streamTarget` | CM | T | T | F | T |

#### 4.3.33.3 Attribute constraints

| Name | Definition |
|---|---|
| CHOICE_1.1 fileReportingPeriod | This attribute shall be supported, when the MnS producer supports file based reporting and storing files on the MnS producer. |
| CHOICE_2.1 fileReportingPeriod<br>CHOICE_2.2 notificationRecipientAddress | These attributes shall be supported, when the MnS producer supports file based reporting, storing files on the MnS producer and implicit notification subscription. |
| CHOICE_3.1 fileReportingPeriod<br>CHOICE_3.2 fileLocation | These attributes shall be supported, when MnS producer supports file based reporting and storing files on a MnS consumer. |
| CHOICE_4.1 streamTarget | This attribute shall be supported, when the MnS producer supports stream-based reporting. |

#### 4.3.33.4 Notifications

The subclause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.34 ThresholdInfo <<dataType>>

#### 4.3.34.1 Definition

This <<dataType>> defines a single threshold level.

#### 4.3.34.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| performanceMetrics | M | T | T | F | T |
| thresholdDirection | M | T | T | F | T |
| thresholdValue | M | T | T | F | T |
| hysteresis | O | T | T | F | T |

#### 4.3.34.3 Attribute constraints

None

#### 4.3.34.4 Notifications

The subclause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.35 TraceReference <<dataType>>

#### 4.3.35.1 Definition

This <<dataType>> defines a globally unique identifier, which uniquely identifies the Trace Session that is created by the TraceJob. It is composed of the MCC, MNC (resulting in PLMN identifier) and the trace identifier.

#### 4.3.35.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| mcc | M | T | T | T | N/A |
| mnc | M | T | T | T | N/A |
| traceId | M | T | T | T | N/A |

#### 4.3.35.3 Attribute constraints

None.

#### 4.3.35.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.36 AreaConfig <<dataType>>

#### 4.3.36.1 Definition

This <<dataType>> defines the area for which measurement logging should be performed. It is described by a list of cells and a list of frequencies.

#### 4.3.36.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| freqInfo | M | T | T | F | T |
| pciList | M | T | T | F | T |

#### 4.3.36.3 Attribute constraints

None.

#### 4.3.36.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.37 FreqInfo <<dataType>>

#### 4.3.37.1 Definition

This <<dataType>> defines the RF reference frequency and the frequency operating bands used in a cell for a given direction (UL or DL) in FDD or for both UL and DL directions in TDD.

#### 4.3.37.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| arfcn | M | T | T | F | T |
| freqBands | M | T | T | F | T |

#### 4.3.37.3 Attribute constraints

None.

#### 4.3.37.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

## 4.3.38 AreaScope <<dataType>>

### 4.3.38.1 Definition

This <<dataType>> defines an area scope.

The Area Scope parameter in LTE and NR contains one of the following:

- list of Cells, identified by E-UTRAN-CGI or NG-RAN CGI. Maximum 32 CGI can be defined.
- list of Tracking Area, identified by TAC. Maximum of 8 TAC can be defined.
- list of Tracking Area Identity, identified by TAC with associated plmn-Identity perTAC-List containing the PLMN identity for each TAC. Maximum of 8 TAI can be defined.
    - list of NPN IDs in NR. It is either a list of PNI-NPNs identified by CAG ID with associated plmn-Identity (Maximum 256 PNI-NPNs can be defined) or a list of SNPN by Network ID with associated plmn-Identity (Maximum 16 SNPNs can be defined).

### 4.3.38.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 eutraCellIdList | CM | T | T | F | T |
| CHOICE_2.1 nrCellIdList | CM | T | T | F | T |
| CHOICE_2.2 nPNIdentityList | O | T | T | F | T |
| CHOICE_3.1 tacList | M | T | T | F | T |
| CHOICE_3.2 cAGIdList | O | T | T | F | T |
| CHOICE_4.1 taiList | M | T | T | F | T |
| CHOICE_4.2 nPNIdentityList | O | T | T | F | T |
| CHOICE_5.1 nPNIdentityList | O | T | T | F | T |

### 4.3.38.3 Attribute constraints

| Name | Definition |
|---|---|
| eutraCellIdList | This attribute shall be supported, when the system allows supporting by E-UTRAN cells. |
| nrCellIdList | This attribute shall be supported, when the system allows supporting by NR cells. |

### 4.3.38.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

## 4.3.39 Tai <<dataType>>

### 4.3.39.1 Definition

This <<dataType>> defines a Tracking Area Identity (TAI) as specified in clause 28.6 of TS 23.003 [5], clause 8.2 of TS 38.300 [33] and clause 9.3.3.11 of TS 38.413 [34]. It is composed of the PLMN identifier (PLMN-Id, which is composed of the MCC and MNC) and the Tracking Area Code (TAC).

### 4.3.39.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| mcc | M | T | T | T | N/A |
| mnc | M | T | T | T | N/A |
| tac | M | T | T | T | N/A |

#### 4.3.39.3 Attribute constraints

None.

#### 4.3.39.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.40 MbsfnArea <<dataType>>

#### 4.3.40.1 Definition

This <<dataType>> defines a MBSFN area. It is composed of the MBSFN Area identifier and the carrier frequency (EARFCN).

#### 4.3.40.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| mbsfnAreaId | M | T | T | F | T |
| earfcn | M | T | T | F | T |

#### 4.3.40.3 Attribute constraints

None.

#### 4.3.40.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.41 MnsRegistry

#### 4.3.41.1 Definition

This IOC is a container for MnsInfo IOC-s. It can be contained only by SubNetwork IOC. A SubNetwork IOC can contain only one instance of MnsRegistry.

The IOC is instantiated by the system.

#### 4.3.41.2 Attributes

The MnsRegistry IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29).

#### 4.3.41.3 Attribute constraints

None.

#### 4.3.41.4 Notifications

None.

## 4.3.42 MnsInfo

### 4.3.42.1 Definition

This IOC represents an available Management Service (MnS) and provides the data required to support its discovery. It is name-contained by MnsRegistry.

The present document does not specify how MnsInfo objects are created and maintained.

This information is used by the consumer to discover the producers of specific Management Services and to derive the addresses of the Management Service.

Attributes mnsLabel, mnsType, and mnsVersion are used to describe the Management Service.

Attribute mnsAddress is used to provide addressing information for the Management Service operations.

Attribute mnsScope is used to provide information about the management scope of the Management Service. The management scope is defined as the set of managed object instances that can be accessed using the Management Service..

### 4.3.42.2 Attributes

The MnsInfo IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| mnsLabel | M | T | F | F | T |
| mnsType | M | T | F | F | T |
| mnsVersion | M | T | F | F | T |
| mnsAddress | M | T | F | F | T |
| mnsScope | M | T | F | F | T |

### 4.3.42.3 Attribute constraints

None.

### 4.3.42.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions or additions.

## 4.3.43 ProcessMonitor <<dataType>>

### 4.3.43.1 Definition

This <<dataType>> provides attributes to monitor the progress of processes with specific purpose and limited lifetime running on MnS producers. It may be used as data type for dedicated progress monitor attributes when specifying the management representation of these processes. The attributes in this clause are defined in a generic way. For some attributes specialisations may be provided when specifying a concrete process representation.

If a management operation on some IOCs triggers an associated asynchronous process (whose progress shall be monitored), this should also result in creating an attribute named processMonitor (of type ProcessMonitor) in these IOC(s). The processMonitor attribute may be accompanied by use-case specific additional data items.

The progress of the process is described by the status and progressPercentage attributes. Additional textual qualifications for the Status attribute may be provided by the progessStateInfo and resultStateInfo attributes.

When the process is instantiated, the `status` is set to "NOT_RUNNING" and the `progressPercentage` to "0". The MnS producer decides when to start executing the process and to transition into the "RUNNING" state. This time is captured in the `startTime` attribute. Alternatively, the process may start to execute directly upon its instantiation. One alternative must be selected when using this <<dataType>>.

During the "RUNNING" state the `progressPercentage` attribute may be repeatedly updated. The exact semantic of this attribute is subject to further specialisation. The `progessStateInfo` attribute may be used to provide additional textual information in the "NOT_RUNNING", "CANCELLING" and "RUNNING" states. Further specialisation of `progessStateInfo` may be provided where this <<dataType>> is used.

Upon successful completion of the process, the `status`attribute is set to "FINISHED", the `progressPercentage` to 100%. The time is captured in the `endTime` attribute. Additional textual information may be provided in the `resultStateInfo` attribute. The type of `resultStateInfo` in this <<dataType>> definition is `String`. Further specialisation of `resultStateInfo` may be provided where this <<dataType>> is used.

In case the process fails to complete successfully, the `status` attribute is set to "FAILED" or "PARTIALLY_FAILED", the current value of `progressPercentage` is frozen, and the time captured in `endTime`. The `resultStateInfo` specifies the reason for the failure. Specific failure reasons may be specified where the data type defined in this clause is used. The exact semantic of failure may be subject for further specialisation as well.

In case the process is cancelled, the `status` attribute is first set to "CANCELLING" and when the process is really cancelled then to "CANCELLED". The transition to "CANCELLED" is captured in the `endTime` attribute. The value of `progressPercentage` is frozen. Additional textual information may be provided in the `resultStateInfo` attribute.

The `resultStateInfo` attribute is provided only for additional textual qualification of the states "FINISHED", "FAILED", "PARTIALLY_FAILED" or "CANCELLED". It shall not be used for making the outcome, that the process may produce in case of success, available.

The process may have to be completed within a certain time after its creation, for example because required data may not be available any more after a certain time, or the process outcome is needed until a certain time and when not provided by this time is not needed any more. The time until the MnS producer automatically cancels the process is indicated by the `timer` attribute.

### 4.3.43.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| id | M | T | F | F | T |
| status | M | T | F | F | T |
| progressPercentage | O | T | F | F | T |
| progressStateInfo | O | T | F | F | T |
| resultStateInfo | O | T | F | F | T |
| startTime | O | T | F | F | T |
| endTime | O | T | F | F | T |
| timer | O | T | T | F | F |

### 4.3.43.3 Attribute constraints

None.

### 4.3.43.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

## 4.3.44 Files

### 4.3.44.1 Definition

This IOC represents a collection of files. It can be name-contained by SubNetwork, ManagedElement, PerfMetricJob or TraceJob. The Files object name-contains File objects, that represent the files of the collection. File collections allow to structure related files under a common root.

Instances of Files are created by MnS producers. They shall be created at latest when the first file of the collection becomes available for retrieval by MnS consumers.

The attributes of Files represent properties of the file collection and not properties of individual files.

When the file retrieval NRM fragment is used together with a data collection job (PerfMetricJob or TraceJob) the following provisions shall apply:

- The Files object shall be created at the same time as the object representing the data collection job.
- The attributes jobRef and jobId shall be supported and present in a Files instance. They shall identify the job that the files in the file collection relate to.
- A Files instance shall contain files related to one and only one job.
- The files produced by one job shall be contained in one and only one Files instance.
- The job object shall support an attribute with a link to the created Files instance (_linkToFiles).
- The attribute _linkToFiles shall be returned in the job creation response, if the stage 3 protocol supports returning attributes in an object creation response.
- The MnS producer decides where to name-contain the Files instance related to a job.

The attribute _linkToFiles allows a MnS consumer to create simple and targeted subscriptions for notifyFileReady and notifyFilePreparationError, or notifyMOICreation, notifyMOIChanges, and notifyMOIDeletion related to File instances created or deleted under the Files instance of a specific job. The subscription needs to scope simply objects one level below the Files object.

In addition, the attribute _linkToFiles allows for simple deployments not relying on notifications for reporting the availability of new files, where the MnS consumer polls regularly for new files under Files.

### 4.3.44.2 Attributes

The Files IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| numberOfFiles | M | T | F | F | F |
| **Attributes related to roles** | | | | | |
| jobRef | CM | T | F | T | F |
| jobId | CM | T | F | T | F |

### 4.3.44.3 Attribute constraints

| Name | Definition |
|---|---|
| jobRef Support Qualifier | Condition: This attribute shall be supported when PerfMetricJob or TraceJob are supported. |
| jobId Support Qualifier | Condition: This attribute shall be supported when PerfMetricJob or TraceJob are supported. |

#### 4.3.44.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions or additions.

### 4.3.45 File

#### 4.3.45.1 Definition

This IOC represents a file. It is name-contained by `Files`.

When a file becomes available on a MnS producer for retrieval by a MnS consumer, the MnS producer shall create a `File` instance representing that file.

The time of creation shall be captured by the MnS producer in the `fileReadyTime` attribute. The MnS producer shall keep the file at least until the time specified by `fileExpirationTime`. After that time the MnS producer may delete the `File` instance. The `fileExpirationTime` is determined by the MnS producer based on considerations such as available storage space or file retention policies.

The attributes `fileSize`, `fileCompression`, `fileDataType` and `fileFormat` describe the file properties.

The MnS producer shall make the file available by one of the following means:

- For retrieval via a file transfer protocol. The `fileLocation` attribute indicates the address from where the file can be retrieved. The address includes the file transfer protocol (schema). Allowed file transfer protocols are "sftp", "ftpes" and "https".

- For retrieval of the file contents via a CM read operation. The attribute `fileContent` is provided for retrieving the actual file content. In this case, the name of the file is equal to the identity of the `File` instance.

When the file retrieval NRM fragment is used together with a data collection job (`PerfMetricJob` or `TraceJob`) the following provisions shall apply:

- The attributes `jobRef` and `jobId` shall be supported and present. They shall identify the job that the file is related to.

The attributes `jobRef` and `jobId` allow to set notification filters in the subscription in such a way that only `notifyMOICreation`, `notifyMOIDeletion` and `notifyMOIChanges` notifications are sent to subscribed MnS consumers if the created or deleted `File` instance represents data related to jobs the subscribed MnS consumer created or is interested in.

Upon creation of a `File` instance, a notification of type `notifyMOICreation` or `notifyMOIChanges` shall be emitted to subscribed MnS consumers as usual. For the case that the file contains performance metric data (`fileDataType` is "PERFORMANCE") the MnS producer shall emit either a notification of type `notifyMOICreation` or `notifyMOIChanges` or of type `notifyFileReady`. The MnS consumer selects the notification type he wishes to receive with the subscription created on the MnS producer.

The `objectClass` and `objectInstance` parameters in the notification header of `notifyFileReady` shall identify the new `File` instance, instead of the related `PerfMetricJob`, `TraceJob`, `ManagedElement` or `ManagementNode` as described in TS 28.532 [27], clause 11.6.1.1.1 for the case that `notifyFileReady` is used as part of the file data reporting MnS.

The notification `notifyFilePreparationError` shall be supported as well by the `File` object. It shall be sent when an error occurs during the preparation of the file. No `notifyFileReady` or `notifyMOICreation` or `notifyMOIChanges` shall be sent in that case. The `objectClass` and `objectInstance` parameters of the notification header shall identify the new `File` instance representing the corrupted file, instead of the related `PerfMetricJob`, `TraceJob`, `ManagedElement` or `ManagementNode` as described in 3GPP TS 28.532 [27], clause 11.6.1.1.1 for the case that `notifyFilePreparationError` is used as part of the file data reporting MnS. When the file is not created at all or deleted, the `objectClass` and `objectInstance` parameters of the notification header are populated as described in 3GPP TS 28.532 [27], clause 11.6.1.1.1. Note that to receive `notifyFilePreparationError` in that case the notification subscription needs to include these objects in its scope.

### 4.3.45.2 Attributes

The File IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 fileLocation | M | T | F | T | F |
| CHOICE_2.1 fileContent | M | T | F | T | F |
| fileCompression | M | T | F | T | F |
| fileSize | O | T | F | T | F |
| fileDataType | M | T | F | T | F |
| fileFormat | O | T | F | T | F |
| fileReadyTime | M | T | F | T | F |
| fileExpirationTime | O | T | F | T | F |
| **Attributes related to roles** | | | | | |
| jobRef | CM | T | F | T | F |
| jobId | CM | T | F | T | F |

### 4.3.45.3 Attribute constraints

| Name | Definition |
|---|---|
| jobRef<br>Support Qualifier | Condition: This attribute shall be supported when PerfMetricJob or TraceJob are supported. |
| jobIdSupport Qualifier | Condition: This attribute shall be supported when PerfMetricJob or TraceJob are supported. |

### 4.3.45.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC. In addition, the following set of notifications is also valid.

| Name | S | Notes |
|---|---|---|
| notifyFileReady | M | |
| notifyFilePreparationError | M | |

## 4.3.46 FileDownloadJob

### 4.3.46.1 Definition

The FileDownloadJob represents a job on a MnS producer that downloads a file to the MnS producer. It can be name-contained by ManagedElement or SubNetwork.

A FileDownloadJob is created by a MnS consumer to request that the MnS producer download a file from a specified location. The creation request contains the information required by the MnS producer to download the file, namely the attribute fileLocation.

The creation request may contain as well a notificationRecipientAddress. If present, this attribute instructs the MnS producer to create, on behalf of the MnS consumer, a subscription for attribute value change notifications of the new FileDownloadJob (implicit notification subscription). In case the MnS producer supports the notification type notifyMOIChanges, the created subscription shall be for this type, otherwise for notifyMOIAttributeValueChanges. The MnS consumer needs to be prepared to receive either of them. The notificationRecipientAddress attribute of the created NtfSubscriptionControl object shall be set to the value of the notificationRecipientAddress in the FileDownloadJob creation request.

The jobMonitor attribute represents the status of a file download job and includes information the MnS consumer can use to monitor the progress and result of the file download job. The data type of this attribute is ProcessMonitor . The following specialisations are provided for this <<dataType>> for the file download job:

- The status attribute values are "NOT_STARTED", "RUNNING", "CANCELLING", "FINISHED, "FAILED" and "CANCELLED". The values "SUSPENDED" and "PARTIALLY_FAILED" are not used.

- The MnS consumer can set the value of the timer attribute to specify the time by which the file download is expected to complete, i.e. to indicate how long the file is available for download. If the timer expires before the MnS producer has finished the job the status is set to "FAILED" and resultStateInfo is set to "TIMER_EXPIRED".

- The "progessPercentage" attribute indicates how much percent of the file is already downloaded as measured by downloaded bytes from total file size in bytes.

- No specialisations are provided for the progressStateInfo attribute. Vendor specific information may be provided though.

- For the case that the status is equal to "FAILED" the resultStateInfo attribute shall indicate one of the following failure reasons: "UNKNOWN", "NO_STORAGE", "LOW_MEMROY", "NO_CONNECTION_TO_REMOTE_SERVER", "FILE_NOT_AVAILABLE", "DNS_CANNOT_BE_RESOLVED", "TIMER_EXPIRED", "OTHER".

- For the case that the status is equal to "FINISHED" or "CANCELLED" no specialisations are provided for the resultStateInfo attribute. Vendor specific information may be provided though.

Once the job is complete with jobstatus equal to "FINISHED", "CANCELLED", or "FAILED" the MnS consumer shall delete the FileDownloadJob. The MnS producer may also delete the FileDownloadJob.

### 4.3.46.2 Attributes

The FileDownloadJob IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| fileLocation | M | T | T | T | F |
| notificationRecipientAddress | O | T | T | T | F |
| cancelJob | M | T | T | F | T |
| jobMonitor | M | T | T | F | T |

### 4.3.46.3 Attribute constraints

None.

### 4.3.46.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions or additions.

## 4.3.47 ManagementDataCollection

### 4.3.47.1 Definition

This IOC represents a management data collection request job. The requested data could be of kind Trace, MDT (Minimization of Drive Test), RLF (Radio Link Failure) report, RCEF (RRC Connection Establishment Failure) report, PM (performance measurements), KPI (end-to-end key performance indicators) or a combination of these.

The attribute managementData defines the management data which shall be reported. This may either include a list of data categories or a list of management data identified with their name. For further details see clause 4.3.50. The targetNodeFilter attribute can be used to target object instance(s) producing the required management data. It is assumed that the consumer may not have detailed knowledge of the network and hence may not identify the exact object instance producing the required management data. In this case consumer can request management data, specified by 3GPP, produced by certain object instance (s) based on a particular location, the domain (CN or RAN) of theobject instances, and the handled traffic (CP or UP) of the object instances.

To activate the production of the requested data, a MnS consumer has to create a `ManagementDataCollection` object instance on the MnS producer.

The MnS producer may derive multiple jobs `PerfMetricJob`, `TraceJob`) from a single `ManagementDataCollection` job for collecting the required management data. If the MnS producer receives the collected data from multiple sources, it consolidate the data into a set of management data for reporting.

The attribute `collectionTimeWindow` specifies the time window for which the management data should be reported.

The attribute `reportingCtrl` specifies the method and associated control parameters for reporting the produced management data to MnS consumers. Three methods are available: file-based reporting with selection of the file location by the MnS producer, file-based reporting with selection of the file location by the MnS consumer and stream-based reporting.

The attribute `dataScope` configures, whether the management data should be reported per S-NSSAI or per 5QI or per PLMN, if applicable.

### 4.3.47.2 Attributes

The `ManagementDataCollection` IOC includes the attributes inherited from `Top` IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| managementData | M | T | T | T | N/A |
| targetNodeFilter | M | T | T | T | N/A |
| collectionTimeWindow | M | T | T | T | N/A |
| reportingCtrl | M | T | T | T | N/A |
| dataScope | O | T | T | T | N/A |

### 4.3.47.3 Attribute constraints

None.

### 4.3.47.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC. In addition, the following set of notifications is also valid.

| Name | S | Notes |
|---|---|---|
| notifyFileReady | M | -- |
| notifyFilePreparationError | M | -- |

## 4.3.48 `TimeWindow <<choice>>`

### 4.3.48.1 Definition

This choice defines a time window.

It is a choice between the control parameters required to define the time window as follows:

When `startTime` and `endTime` is present (CHOICE_1), the time window starts when `startTime` is reached and ends when `endTime` is reached.

When only the `startTime` attribute is present (CHOICE_2), the time window starts when `startTime` is reached and runs until deletion of the managed object instance including this `TimeWindow`.

When only the `endTime` attribute is present (CHOICE_3), the time window starts when the managed object instance including this `TimeWindow` is created and ends when `endTime` is reached.

#### 4.3.48.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 startTime | CM | T | T | T | T |
| CHOICE_1.2 endTime | CM | T | T | T | T |
| CHOICE_2.1 startTime | CM | T | T | T | T |
| CHOICE_3.1 endTime | CM | T | T | T | T |

#### 4.3.48.3 Attribute constraints

| Name | Definition |
|---|---|
| CHOICE_1.1 startTime<br>CHOICE_1.2 endTime | These attributes shall be supported, when the MnS consumer configures the start and end time of the time window.<br>These attributes are supported for ManagementDataCollection IOC. |
| CHOICE_2.1 startTime | This attribute shall be supported, if the MnS consumer indicates only the start time of a time window and the end time is defined by the deletion of the managed object instance.<br>This attribute is not supported for ManagementDataCollection IOC. |
| CHOICE_3.1 endTime | This attribute shall be supported, if the MnS consumer indicates only the end time of a time window and the start time is defined by the creation of the managed object instance.<br>This attribute is not supported for ManagementDataCollection IOC. |

#### 4.3.48.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.49 NodeFilter <<dataType>>

#### 4.3.49.1 Definition

This <<dataType>> defines several selection criteria for the target node(s) i.e., the node(s) producing the requested management data.

The attribute areaOfInterest determines the location for which the management data is collected. The system translates the area into the target managed objects. The location is either configured by a list of TAI, a list of cells (identified either by NG-RAN CGI, E-UTRAN CGI or UTRAN CGI) or by a geographical area. The geographical area will be mapped to the cells providing coverage for this area. The cell coverage status at the time of the request is used for the mapping. Managed objects providing service to these cells are considered as target managed objects. Furthermore, an object which name contains or is associated to a managed object providing service to the considered cell, is considered as target managed object as well.

The attribute networkDomain is used to select a particular domain (e.g. RAN, CN) for which the management data is collected. The system translates this information into the target managed objects. Managed objects from this selected particular domain (e.g RAN, CN) are considered as target managed objects. Furthermore, an object which name contains or is associated to a managed object of that domain, is considered as target managed object as well.

The attribute cpUpType is used to select the traffic type (CP, UP) for which the management data is collected. The system translates this information into the target managed objects. Managed objects catering particular traffic type (CP, UP) are considered as target managed objects. Furthermore, an object which name contains or is associated to a managed object of that traffic type, shall be considered as target managed object as well.

The attribute `sst` is used to select the SST (Slice/Service Type) TS 23.501 [22] for which the management data is collected. The system translates this information into the target managed objects. Managed objects related to particular SST will be considered as target managed objects.

If it is not possible to select the target node(s) (based on a particular selection criteria) deterministically, the selection criteria should not be used.

#### 4.3.49.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| areaOfInterest | O | T | T | T | N/A |
| networkDomain | O | T | T | T | N/A |
| cpUpType | O | T | T | T | N/A |
| sst | O | T | T | T | N/A |

#### 4.3.49.3 Attribute constraints

None.

#### 4.3.49.4 Notifications

The subclause 4.5 of the <<IOC>> using this `<<dataType>>` as one of its attributes, shall be applicable.

### 4.3.50 ManagementData <<choice>>

#### 4.3.50.1 Definition

This `<<choice>>` defines the management data which is requested. It is a choice between

- a list of data categories (attribute `mgtDataCategory`) This may include "COVERAGE", "CAPACITY", "MOBILITY", "ENERGY_EFFICIENCY", "ACCESSIBILITY" etc. The mapping of exact measurement with the requested category will be done at the producer and is implementation specific.
- a list of management data identified with their name (attribute `mgtDataName`). The management data name presents a specific single measurement (e.g. by selecting "RRU.PrbTotDl", see TS 28.552 [20] or "immediateMdt.nr.m1", see TS 32.422 [30]) or a set of measurements (e.g. measurement families such as RRU (radio resource utilization) or MM (mobility management) in case of PM, see TS 28.552 [20], or group of measurements such as "immediateMdt.nr" in case of MDT, see TS 32.422 [30]).

#### 4.3.50.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 mgtDataCategory | M | T | T | T | N/A |
| CHOICE_2.1 mgtDataName | M | T | T | T | N/A |

#### 4.3.50.3 Attribute constraints

None.

#### 4.3.50.4 Notifications

The clause 4.5 of the <<IOC>> using this `<<choice>>` as one of its attributes, shall be applicable.

### 4.3.51 AreaOfInterest <<choice>>

#### 4.3.51.1 Definition

This <<choice>> defines the area which shall be considered for the service.

#### 4.3.51.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 geoAreaToCellMapping | M | T | T | T | N/A |
| CHOICE_2.1 taiList | M | T | T | T | N/A |
| CHOICE_3.1 nrCellIdList | CM | T | T | T | N/A |
| CHOICE_4.1 eutraCellIdList | CM | T | T | T | N/A |
| CHOICE_5.1 utraCellIdList | CM | T | T | T | N/A |

#### 4.3.51.3 Attribute constraints

| Name | Definition |
|---|---|
| nrCellIdList | This attribute shall be supported, when the system supports scoping by NR cells. |
| eutraCellIdList | This attribute shall be supported, when the system supports scoping by E-UTRAN cells. |
| utraCellIdList | This attribute shall be supported, when the system supports scoping by UTRAN cells. |

#### 4.3.51.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.52 GeoAreaToCellMapping <<dataType>>

#### 4.3.52.1 Definition

This <<dataType>> contains a geographical area and an association threshold. The geo-area is defined as a polygon using the attribute geoArea.

The MnS producer shall map the geographical area to cells. There are two evaluation criteria whether a cell belongs to a geographical area or not. If attribute associationThreshold is absent, the location of the base station antenna determines the belonging. If attribute associationThreshold is configured, the coverage area is considered. The attribute associationThreshold determines the lower boundary of the coverage ratio. For example, if the associationThreshold is configured to 60%, a cell shall be considered as included in the geographical area if at least 60% of the coverage area of that cell overlaps with the specified geographical area.

The mapping of the geographical area to cells is performed at instantiation of the IOC.

#### 4.3.52.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| geoArea | M | T | T | F | N/A |
| associationThreshold | O | T | T | T | N/A |

#### 4.3.52.3 Attribute constraints

None.

#### 4.3.52.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

## 4.3.53 GeoCoordinate <<dataType>>

### 4.3.53.1 Definition

This <<dataType>> defines a geographical location on earth with the altitude.

### 4.3.53.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| latitude | M | T | T | F | T |
| longitude | M | T | T | F | T |
| altitude | O | T | T | F | T |

### 4.3.53.3 Attribute constraints

None.

### 4.3.53.4 Notifications

The clause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

## 4.3.54 QMCJob

### 4.3.54.1 Definition

The QoE Measurement Collection provides capability for collecting QoE information from:

- UEs which are in the specified area in case of Management Based Activation or

- an individual UE in case of Signalling Based Activation.

The QoE Measurement Collection enables collection of application layer measurements from the UE for specified end user service type. The supported service types are:

- Streaming services, see TS 26.247 [51].
- MTSI services, see TS 26.114 [52].
- VR services, see TS 26.118 [53].

A QMCJob instance represents the job for collecting QoE measurements according to the job parameters. For details of the QoE measurement collection configuration parameters see clause 5 of TS 28.405 [50]. A QMCJob instance can be name-contained by SubNetwork or ManagedElement.

A QMC Job is activated by creating a QMCJob object instance in the MnS producer. For details of Management Based Activation of QoE Measurement Collection see clause 4.5 and for details of Signalling Based Activation of QoE Measurement Collection see clause 4.6 of TS 28.405 [50]. The attributes areaScope and pLMNTarget are only relevant when Management Based Activation is used and the attribute qoETarget is only relevant when Signalling Based Activation is used. All other attributes are common for both Management Based Activation and Signalling Based Activation.

The areaScope attribute defines the area scope of QoE, which is specified in clause 5.4 of TS 28.405 [50].

When a MnS consumer wishes to deactivate a QMC Job, the MnS consumer shall delete the corresponding QMCJob instance.

NOTE: If the reporting is ongoing, when a request to delete a QMCJob instance is received, the reporting does not end. The QMCJob instance is deleted, when the last reporting for the QMC Job expires.

The jobId attribute presents the job identifier of a QMCJob instance. The jobId can be used to associate multiple QMCJob instances. For example, it is possible to configure the same jobId value for multiple QMCJob instances required to produce the data (e.g. Streaming services and MTSI reports) for a specific network analysis.

#### 4.3.54.2 Attributes

The QMCJob IOC includes attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute Name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| serviceType | M | T | T | F | T |
| areaScope | CM | T | T | F | T |
| qoECollectionEntityAddress | M | T | T | F | T |
| pLMNTarget | CM | T | T | F | T |
| qoETarget | CM | T | T | F | T |
| qoEReference | M | T | T | F | T |
| jobId | O | T | T | T | T |
| sliceScope | CM | T | T | F | T |
| qMCConfigFile | M | T | T | F | T |
| mDTAlignmentInformation | O | T | T | F | T |
| availableRANqoEMetrics | O | T | T | F | T |

#### 4.3.54.3 Attribute constraints

| Name | Definition |
|---|---|
| areaScope | Condition: This attribute shall be supported when the QMC is targeting specific area(s). |
| pLMNTarget | Condition: This attribute shall be supported when Management Based Activation is used and if network sharing is deployed. |
| qoETarget | Condition: This attribute shall be supported when Signalling Based Activation is used. |
| sliceScope | Condition: This attribute shall be supported when the QMC is targeting specific slice(s). |

#### 4.3.54.4 Notifications

The common notifications defined in clauses 4.5.1 and 4.5.2 are valid for this IOC, without exceptions.

### 4.3.55 GeoArea <<choice>>

#### 4.3.55.1 Definition

This <<choice>> defines a geographical area.

A set of geo-coordinates representing the corners of a polygon configured in the attribute geoPolygon is one choice.

This <<choice>> can be enhanced by further choices in the future.

#### 4.3.55.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 geoPolygon | M | T | T | F | N/A |

### 4.3.56 ExcessPacketDelayThresholds <<dataType>>

#### 4.3.56.1 Definition

This <<dataType>> defines an excess packet delay threshold information to enable the calculation of the PDCP Excess Packet Delay in the uplink in case of M6 uplink measurements are requested. The excess packet delay threshold

information is specified with the 5QI value and excess packet delay threshold value. The excess packet delay thresholds attribute is specified in clause 5.10.41 of TS 32.422 [30].

### 4.3.56.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| fiveQIValue | M | T | T | F | T |
| excessPacketDelayThresholdValue | M | T | T | F | T |

### 4.3.56.3 Attribute constraints

None

### 4.3.56.4 Notifications

The subclause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

## 4.3.57 TraceConfig <<dataType>>

### 4.3.57.1 Definition

This <<dataType>> defines the configuration parameters of IOC TraceJob which are specific for Trace or any combination of Trace.

The attribute listOfNeTypes specifies the network elements to be traced. The optional attribute listOfInterfaces allows to specify the individual interfaces of the network elements to be recorded.

The attribute traceDepth allows to configure the level of detail of the information which shall be recorded.

For trace the reporting is event based, where the triggering event is configured with attribute triggeringEvent. For each triggering event the first and last message (start/stop triggering event) to record are specified.

### 4.3.57.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| listOfInterfaces | O | T | T | F | T |
| listOfNeTypes | CM | T | T | F | T |
| traceDepth | M | T | T | F | T |
| triggeringEvents | M | T | T | F | T |

### 4.3.57.3 Attribute constraints

| Name | Definition |
|---|---|
| listOfNeTypes (support qualifier) | This attribute shall be present only for Trace with Signalling Based Activation |

### 4.3.57.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions.

## 4.3.58 MdtConfig <<dataType>>

### 4.3.58.1 Definition

This <<dataType>> defines the configuration parameters of IOC TraceJob which are specific for MDT or any combination of MDT.

The attribute anonymizationOfMdtData specifies the level of anonymization of MDT data.

The attribute areaScope defines the area scope of MDT, which is specified in clause 5.10.2 of TS 32.422 [30].

The optional attribute sensorInformation allows to specify the sensor information to include.

Based on the value configured for attribute jobType in IOC TraceJob, the attributes immediateMdtConfig or loggedMdtConfig are available: In case of IMMEDIATE_MDT_ONLY or any combination of Immediate MDT the attribute immediateMdtConfig is applicable. In case of LOGGED_MDT_ONLY or LOGGED_MBSFN_MDT the attribute loggedMdtConfig is applicable.

The optional attribute plmnList allows to specify the PLMNs where measurements are collected, status indication and log reporting is allowed.

### 4.3.58.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| anonymizationOfMdtData | M | T | T | F | T |
| areaScope | M | T | T | F | T |
| sensorInformation | CO | T | T | F | T |
| immediateMdtConfig | CM | T | T | F | T |
| loggedMdtConfig | CM | T | T | F | T |
| mNOnly | CO | T | T | F | T |
| plmnList | CO | T | T | F | T |

### 4.3.58.3 Attribute constraints

| Name | Definition |
|---|---|
| sensorInformation | This attribute may be present only if NR is supported. |
| immediateMdtConfig | This attribute shall be present only if Immediate MDT is supported. |
| loggedMdtConfig | This attribute shall be present only if Logged MDT is supported. |
| mNOnly | This attribute may be present if signalling based MDT for NR is supported and MN only for MDT is supported. |
| plmnList | This attribute may be present only if multiple PLMNs are supported. |

### 4.3.58.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions.

## 4.3.59 ImmediateMdtConfig <<dataType>>

### 4.3.59.1 Definition

This <<dataType>> defines the configuration parameters of IOC TraceJob which are specific for Immediate MDT or any combination of Immediate MDT.

The optional attribute positioningMethod allows to specify the positioning methods to use.

The following attributes are conditional available based on the measurements configured in listOfMeasurements:

- reportInterval (conditional for M1 in LTE or NR and M1/M2 in UMTS),
- reportAmount (conditional for M1/M2 in UMTS),
- reportAmountM1LTE (conditional for M1 in LTE),
- reportAmountM4LTE (conditional for M4 in LTE),
- reportAmountM5LTE (conditional for M5 in LTE),
- reportAmountM6LTE (conditional for M6 in LTE),
- reportAmountM7LTE (conditional for M7 in LTE),
- reportAmountM1NR (conditional for M1 in NR),
- reportAmountM4NR (conditional for M4 in NR),
- reportAmountM5NR (conditional for M5 in NR),
- reportAmountM6NR (conditional for M6 in NR),
- reportAmountM7NR (conditional for M7 in NR),
- reportingTrigger (conditional for M1 in LTE or NR and M1/M2 in UMTS),
- eventThreshold (conditional for A2 event reporting or A2 event triggered periodic reporting),
- collectionPeriodRRMNR (conditional for M4 and M5 in NR),
- collectionPeriodM6NR (conditional for M6 in NR),
- collectionPeriodM7NR (conditional for M7 in NR),
- collectionPeriodRRMLTE (conditional for M3 in LTE),
- measurementPeriodLTE (conditional for M4 and M5 in LTE),
- collectionPeriodM6LTE (conditional for M6 in LTE),
- collectionPeriodM7LTE (conditional for M7 in LTE),
- collectionPeriodRRMUMTS (conditional for M4 and M5 in UMTS),
- measurementPeriodUMTS (conditional for M6 and M7 in UMTS),
- measurementQuantity (conditional for 1F event reporting).
- beamLevelMeasurement (conditional for M1 in NR),
- excessPacketDelayThresholds (conditional for M6 UL measurement in NR).

For immediate MDT, the measurement reporting is dependent on the configured measurements:

- For measurement M1 in LTE or NR, it is possible to select between periodical, event triggered, event triggered periodic reporting or reporting according to all configured RRM event triggers. For M1 and M2 measurement in UMTS, it is possible to select between periodical, event triggered reporting or reporting according to all configured RRM event triggers. Parameter reportingTrigger determines which of the reporting methods is selected and in case of event triggered or event-triggered periodic, which is the decisive event type. For periodical reporting, parameter reportInterval and one of reportAmount, reportAmountM1LTE and reportAmountM1NR, for UMTS, LTE or NR, respectively, determine the interval between two successive reports and the number of reports. This means the periodical reporting terminates after reportAmount, reportAmountM1LTE or reportAmountM1NR reports have been sent as long as the corresponding attribute is configured with a value different from infinity. For event-triggered periodic reporting, these two parameters apply in addition to parameter eventThreshold which determines the threshold of the event. In this case up to reportAmountM1LTE or reportAmountM1NR reports are sent with a periodicity of reportInterval after the entering condition is fulfilled. The reporting is stopped, if the leaving condition is fulfulled and is restarted if the configured event reoccurs. For event based reporting, there is only one report sent after the event occurs. The parameters to configure are reportingTrigger and eventThreshold. In case of UMTS and 1F event reporting, additionally parameter measurementQuantity is necessary in order to determine for which measurement(s) the event threshold is applicable. Parameter beamLevelMeasurement determines whether beam level measurements shall be included in case of NR.

- For measurement M2 in NR or LTE, reporting is according to RRM configuration, see TS 38.321 [36], TS 36.321 [37] and TS 38.331 [38], TS 36.331 [39].

- For measurement M4 in UMTS, reporting is either according to RRM configuration, see TS 25.321 [40] and TS 25.331 [41] or periodic or event triggered periodic using parameter collectionPeriodRRMUMTS and eventThresholdUphUMTS.

- For measurement M3 in UMTS, the reporting is done upon availability, see TS 37.320 [43].

- For measurements M4, M5, M6 and M7 in NR, for measurements M3, M4, M5, M6 and M7 in LTE and for measurements M5, M6 and M7 in UMTS periodical reporting is applied. The configurable parameter is the

interval between two measurements (collectionPeriodRrmNR, collectionPeriodM6NR, collectionPeriodM7NR, collectionPeriodRrmLTE, measurementPeriodLTE, collectionPeriodM6LTE, collectionPeriodM7LTE, collectionPeriodRrmUMTS, measurementPeriodUMTS) and the number of reports (reportAmountM4NR, reportAmountM5NR, reportAmountM6NR, reportAmountM7NR, reportAmountM4LTE, reportAmountM5LTE, reportAmountM6LTE, reportAmountM7LTE). If no collection period is configured for M5 in UMTS, all available measurements are logged according to RRM configuration.

- Measurements M8 and M9 in NR or LTE are reported according to configured M1 and/or M6 related UE measurement reporting.

### 4.3.59.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| listOfMeasurements | M | T | T | F | T |
| reportingTrigger | CM | T | T | F | T |
| reportInterval | CM | T | T | F | T |
| reportAmount | CM | T | T | F | T |
| eventThreshold | CM | T | T | F | T |
| collectionPeriodRRMNR | CM | T | T | F | T |
| collectionPeriodM6NR | CM | T | T | F | T |
| collectionPeriodM7NR | CM | T | T | F | T |
| collectionPeriodRRMLTE | CM | T | T | F | T |
| measurementPeriodLTE | CM | T | T | F | T |
| collectionPeriodM6LTE | CM | T | T | F | T |
| collectionPeriodM7LTE | CM | T | T | F | T |
| eventThresholdUphUMTS | CO | T | T | F | T |
| collectionPeriodRRMUMTS | CM | T | T | F | T |
| measurementPeriodUMTS | CM | T | T | F | T |
| measurementQuantity | CM | T | T | F | T |
| beamLevelMeasurement | CM | T | T | F | T |
| positioningMethod | O | T | T | F | T |
| excessPacketDelayThresholds | CO | T | T | F | T |
| reportAmountM1LTE | CM | T | T | F | T |
| reportAmountM4LTE | CM | T | T | F | T |
| reportAmountM5LTE | CM | T | T | F | T |
| reportAmountM6LTE | CM | T | T | F | T |
| reportAmountM7LTE | CM | T | T | F | T |
| reportAmountM1NR | CM | T | T | F | T |
| reportAmountM4NR | CM | T | T | F | T |
| reportAmountM5NR | CM | T | T | F | T |
| reportAmountM6NR | CM | T | T | F | T |
| reportAmountM7NR | CM | T | T | F | T |

#### 4.3.59.3 Attribute constraints

| Name | Definition |
|---|---|
| reportingTrigger (support qualifier) | This attribute shall be present only if measurement set for M1 (in UMTS, LTE and NR) or M2 (only in UMTS) is supported. |
| reportInterval (support qualifier) | This attribute shall be present when these conditions are met: measurement set for M1 (in UMTS, LTE and NR) or M2 (only in UMTS) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmount (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M1/M2 (in UMTS) is supported; periodic measurements or event triggered periodic measurements is supported. |
| eventThreshold (support qualifier) | This attribute shall be present only if A2 event reporting (in LTE and NR) or 1F/1I event reporting (in UMTS) is supported. |
| collectionPeriodRRMNR (support qualifier) | This attribute shall be present only if measurement set for M4 (in NR) or M5 (in NR) is supported. |
| collectionPeriodM6NR (support qualifier) | This attribute shall be present only if measurement set for M6 (in NR) is supported. |
| collectionPeriodM7NR (support qualifier) | This attribute shall be present only if measurement set for M7 (in NR) is supported. |
| collectionPeriodRRMLTE (support qualifier) | This attribute shall be present only if measurement set for M2 (in LTE) or M3 (in LTE) is supported. |
| measurementPeriodLTE (support qualifier) | This attribute shall be present only if measurement set for M4 (in LTE) or M5 (in LTE) is supported. |
| measurementPeriodM6LT**E** (support qualifier) | This attribute shall be present only if measurement set for M6 (in LTE) is supported. |
| measurementPeriodM7LTE (support qualifier) | This attribute shall be present only if measurement set for M7 (in LTE) is supported. |
| eventThresholdUphUMTS (support qualifier) | This attribute may be present only if measurement set for M4 (in UMTS) is supported. |
| collectionPeriodRRMUMTS (support qualifier) | This attribute shall be present only if measurement set for M3 (in UMTS), M4 (in UMTS) or M5 (in UMTS) is supported. |
| measurementPeriodUMTS (support qualifier) | This attribute shall be present only if measurement set for M6 (in UMTS) or M7 (in UMTS) is supported. |
| measurementQuantity (support qualifier) | This attribute shall be present only if 1F event reporting is supported. |
| beamLevelMeasurement (support qualifier) | This attribute shall be present only if measurement set for M1 (in NR) is supported. |
| excessPacketDelayThresholds | This attribute may be present only if measurement set for M6 (for UL in NR) is supported. |
| reportAmountM1LTE (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M1 (in LTE) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmountM4LTE (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M4 (in LTE) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmountM5LTE (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M5 (in LTE) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmountM6LTE (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M6 (in LTE) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmountM7LTE (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M7 (in LTE) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmountM1NR (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M1 (in NR) is supported; periodic measurements or event triggered periodic measurements is supported. |

| | |
|---|---|
| reportAmountM4NR (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M4 (in NR) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmountM5NR (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M5 (in NR) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmountM6NR (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M6 (in NR) is supported; periodic measurements or event triggered periodic measurements is supported. |
| reportAmountM7NR (support qualifier) | This attribute shall be present only when these conditions are met: measurement set for M7 (in NR) is supported; periodic measurements or event triggered periodic measurements is supported. |

### 4.3.59.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions.

## 4.3.60 LoggedMdtConfig <<dataType>>

### 4.3.60.1 Definition

This <<dataType>> defines the configuration parameters of IOC TraceJob which are specific for Logged MDT or Logged MBSFN MDT.

Based on the value configured for attribute jobType in IOC TraceJob, different attributes are available. In case of LOGGED_MDT_ONLY, the attributes reportType, eventListForEventTriggeredMeasurement, eventThresholdL1, hysteresisL1, timeToTriggerL1, areaConfigurationForNeighCell and npnIdentityList are applicable. In case of LOGGED_MBSFN_MDT, the attribute mbsfnAreaList is applicable.The optional attribute areaConfigurationForNeighCell allows to specify the area for which UE is requested to perform measurements logging for neighbour cells which have list of frequencies

For logged MDT in UMTS and LTE, the reporting is periodical. Parameter loggingInterval determines the interval between the reports and parameter loggingDuration determines how long the configuration is valid meaning after this duration has passed no further reports are sent. In NR, the reporting can be periodical or event based, determined by parameter reportType. For periodical reporting the same parameters as in LTE and UMTS apply. For event based reporting, parameter eventListForEventTriggeredMeasurement configures the event type, namely ‘out of coverage’ or ‘L1 event’. In case ‘L1 event’ is selected as event type, the logging is performed according to parameter loggingInterval at regular intervals only when the conditions indicated by eventThresholdL1, hysteresisL1, timeToTriggerL1 (defining the thresholds, hysteresis and time to trigger) are met and if UE is ‘camped normally’ state (TS 38.331 [38], TS 38.304 [42]). In case ‘out of coverage’ is selected as event type, the logging is performed according to parameter loggingInterval at regular intervals only when the UE is in ‘any cell selection’ state. Furthermore, logging is performed immediately upon transition from the ‘any cell selection’ state to the ‘camped normally’ state ( TS 38.331 [38], TS 38.304 [42]).

### 4.3.60.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| traceCollectionEntityId | M | T | T | F | T |
| loggingDuration | M | T | T | F | T |
| loggingInterval | M | T | T | F | T |
| reportType | CM | T | T | F | T |
| eventListForEventTriggeredMeasurement | CM | T | T | F | T |
| eventThresholdL1 | CM | T | T | F | T |
| hysteresisL1 | CM | T | T | F | T |
| timeToTriggerL1 | CM | T | T | F | T |
| areaConfigurationForNeighCell | CO | T | T | F | T |
| mbsfnAreaList | CM | T | T | F | T |
| nPNIdentityList | CM | T | T | F | T |

### 4.3.60.3 Attribute constraints

| Name | Definition |
|---|---|
| reportType (support qualifier) | This attribute shall be present only if NR is supported. |
| eventListForEventTriggeredMeasurement (support qualifier) | This attribute shall be present only if NR is supported. |
| eventThresholdL1 (support qualifier) | This attribute shall be present only if NR is supported. |
| hysteresisL1 (support qualifier) | This attribute shall be present only if NR is supported. |
| timeToTriggerL1 (support qualifier) | This attribute shall be present only if NR is supported. |
| areaConfigurationForNeighCell (support qualifier) | This attribute may be present only if NR is supported. |
| mbsfnAreaList (support qualifier) | This attribute shall be present only if E-UTRAN is supported. |
| nPNIdentityList (support qualifier) | This attribute shall be present only when these conditions are met: NR is supported; several NPNs are supported in the RAN (see TS 38.331 [38]). |

### 4.3.60.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions.

## 4.3.61 SupportedNotifications

### 4.3.61.1 Definition

SupportedNotifications represents the notification related capabilities of a MnS producer. It can be name-contained by SubNetwork or ManagedElement.

The notificationTypes attribute lists notification Type values supported by the MnSProducer. Specific IOCs can be the source of a specific but not necessary every supported notificationType.

The notificationProtocols attribute identifies the notification transport protocols supported by a MnS producer.

### 4.3.61.2 Attributes

The SupportedNotifications IOC includes attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| notificationTypes | M | T | F | F | T |
| notificationProtocols | M | T | F | F | T |

#### 4.3.61.3 Attribute constraints

None

#### 4.3.61.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC

### 4.3.62 Scheduler

#### 4.3.62.1 Definition

This IOC defines a time scheduler. It can be name-contained by SubNetwork or ManagedElement.

The attribute schedulingTimes allows to configure one or several active time intervals. The active intervals can be configured to occur once or recurring periodically.

The boolean attribute schedulerStatus switches between TRUE and FALSE depending upon whether the configured time constraints are fulfilled or not. This attribute makes the internal Scheduler status observable.

#### 4.3.62.2 Attributes

The Scheduler IOC includes the attributes inherited from Top IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| schedulingTimes | M | T | T | F | T |
| schedulerStatus | M | T | F | F | T |

#### 4.3.62.3 Attribute constraints

None.

#### 4.3.62.4 Notifications

The configuration notifications defined in clause 4.5.2 are valid for this IOC.

### 4.3.63 SchedulingTime <<choice>>

#### 4.3.63.1 Definition

This <<choice>> defines the scheduling time and allows to configure one of four possible scheduling methods:

One time interval: The attribute timeWindow presents the active scheduling time. A duration more than one day may be configured.

Daily periodicity: Several active intervals per day can be configured in attribute timeIntervals. The active scheduling times recur each day.

Weekly periodicity: Several active intervals for one day can be configured in attribute timeIntervals. The active scheduling times recur on the days of the weeks configured by attribute daysOfWeek

Monthly periodicity: Several active intervals for one day can be configured in attribute timeIntervals. The active scheduling times recur on the days of the months configured by attribute daysOfMonth.

#### 4.3.63.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| CHOICE_1.1 timeWindow | CM | T | T | F | T |
| CHOICE_2.1 timeIntervals | CM | T | T | F | T |
| CHOICE_3.1 timeIntervals | CM | T | T | F | T |
| CHOICE_3.2 daysOfWeek | CM | T | T | F | T |
| CHOICE_4.1 timeIntervals | CM | T | T | F | T |
| CHOICE_4.2 daysOfMonth | CM | T | T | F | T |

#### 4.3.63.3 Attribute constraints

| Name | Definition |
|---|---|
| CHOICE_1.1 timeWindow | This attribute shall be supported, when the MnS producer supports a management activity for a configured one-time interval. |
| CHOICE_2.1 timeIntervals | This attribute shall be supported, when the MnS producer supports daily repetitive interval-based functionality. |
| CHOICE_3.1 timeIntervals<br>CHOICE_3.2 daysOfWeek | This attribute shall be supported, when the MnS producer supports weekly repetitive interval-based functionality. |
| CHOICE_4.1 timeIntervals<br>CHOICE_4.2 daysOfMonth | This attribute shall be supported, when the MnS producer supports monthly repetitive interval-based functionality. |

#### 4.3.63.4 Notifications

The subclause 4.5 of the <<IOC>> using this <<choice>> as one of its attributes, shall be applicable.

### 4.3.64 TimeInterval <<dataType>>

#### 4.3.64.1 Definition

This <<dataType>> defines a time interval within one day. If the whole day shall be selected, intervalStart shall be set to 00:00:00 and intervalEnd shall be set to 23:59:59.

#### 4.3.64.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| intervalStart | M | T | T | F | T |
| intervalEnd | M | T | T | F | T |

#### 4.3.64.3 Attribute constraints

None.

#### 4.3.64.4 Notifications

The subclause 4.5 of the <<IOC>> using this <<dataType>> as one of its attributes, shall be applicable.

### 4.3.65 ConditionMonitor

#### 4.3.65.1 Definition

This IOC defines one or several conditions and monitors whether these conditions are satisfied. It can be name-contained by SubNetwork or ManagedElement.

The attribute `condition` allows to configure one or several conditions. Possible conditions include but are not limited to scheduling requirements or parameter settings e.g. evaluation if a configuration parameter is above a certain threshold or has a certain values.

The boolean attribute `conditionStatus` switches between TRUE and FALSE depending upon whether the configured conditions are fulfilled or not. This attribute makes the internal `ConditionMonitor` status observable.

### 4.3.65.2 Attributes

The `ConditionMonitor` IOC includes the attributes inherited from `Top` IOC (defined in clause 4.3.29) and the following attributes:

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| `condition` | M | T | T | F | T |
| `conditionStatus` | M | T | F | F | T |

### 4.3.65.3 Attribute constraints

None.

### 4.3.65.4 Notifications

The configuration notifications defined in clause 4.5.2 are valid for this IOC.

## 4.3.66 `NpnId <<choice>>`

### 4.3.66.1 Definition

This `<<choice>>` represents the NPN supported by the <<IOC>> using this `<<choice>>` as one of its attributes in a Non-Public Network use case.

### 4.3.66.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| `plmnId` | M | T | T | F | T |
| CHOICE_1 `cAGIdList` | CM | T | T | F | T |
| CHOICE_2 `nIDList` | CM | T | T | F | T |

### 4.3.66.3 Attribute constraints

| Name | Definition |
|---|---|
| CHOICE_1 `cAGIdList` | This attribute shall be supported when the NPN use case corresponding to PNI-NPN is supported in the RAN (see TS 38.331 [38]) . |
| CHOICE_2 `nIDList` | This attribute shall be supported when the NPN use case corresponding to SNPN is supported in the RAN (see TS 38.331 [38]). |

### 4.3.66.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions.

## 4.3.67 UECoreMeasConfig <<dataType>>

### 4.3.67.1 Definition

This <<dataType>> defines the configuration parameters of IOC TraceJob which are specific for 5GC UE level measurements collection.

The attribute ueCoreMeasurements defines the measurements to be produced, and the attribute ueCoreMeasGranularityPeriod defines the granularity period to be applied.

All object instances below and including the instance name-containing the TraceJob (base object instance) are scoped for measurements collection and production. The 5GC UE level measurements are produced only on those object instances whose object class matches the object class associated to the measurements to be produced.

The optional attributes objectInstances and rootObjectInstances allow to restrict the scope. When the attribute objectInstances is present, only the object instances identified by this attribute are scoped. When the attribute rootObjectInstances is present, then the subordinated objects whose root objects are identified by this attribute are scoped. Both attributes may be present at the same time meaning the total scope is equal to the sum of both scopes. Object instances may be scoped by both the objectInstances and rootObjectInstances attributes. When they are present, this shall not be considered as an error by the MnS producer.

Changes of all other configurable attributes shall take effect only at the beginning of the next granularity period.

### 4.3.67.2 Attributes

| Attribute name | S | isReadable | isWritable | isInvariant | isNotifyable |
|---|---|---|---|---|---|
| ueCoreMeasurements | M | T | T | F | T |
| ueCoreMeasGranularityPeriod | O | T | T | F | T |
| nfTypeToMeasure | CM | T | T | F | T |
| objectInstances | O | T | T | F | T |
| rootObjectInstances | O | T | T | F | T |

### 4.3.67.3 Attribute constraints

| Name | Definition |
|---|---|
| nfTypeToMeasure (support qualifier) | This attribute shall be supported if the signalling activation of 5GC UE level measurements collected are supported. |

### 4.3.67.4 Notifications

The common notifications defined in clause 4.5 are valid for this IOC, without exceptions.

## 4.4 Attribute definitions

### 4.4.1 Attribute properties

The following table defines the properties of attributes specified in the present document.

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| numberOfFiles | Number of files in a file collection.<br><br>allowedValues: NA | Type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileLocation | Location of the file incl. the file transfer protocol, and the file name.<br><br>The allowed file transfer protocols are:<br>- sftp<br>- ftpes<br>- https<br><br>Examples:<br>"sftp://companyA.com/datastore/fileName.xml",<br>"https://companyA.com/ManagedElement=1/Files=1/File=1" | Type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileCompression | Name of the algorithm used for compressing the file. An absent fileCompression parameter indicates the file is not compressed. The MnS producer selects the compression algorithm. It is encouraged to use popular algorithms such as GZIP. | Type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileSize | Size of the file.<br><br>Unit is byte.<br><br>allowedValues: non-negative integers | Type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileDataType | Type of the management data stored in the file.<br><br>AllowedValues:<br>- "PERFORMANCE"<br>- "TRACE"<br>- "ANALYTICS"<br>- "PROPRIETARY"<br><br>The value "PERFORMANCE" refers to measurements and KPIs. | Type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileFormat | Identifier of the schema (incl. its version) used to produce the file content. If there is no schema for the file or it is not available, the fileFormat parameter is absent. | Type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileReadyTime | Date and time, when the file was closed (the last time) and made available on the MnS producer. The file content will not be changed anymore. | Type: DateTime<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileExpirationTime | Date and time after which the file may be deleted. | Type: DateTime<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileContent | File content as a Base64 encoded string according to RFC 4648 [60] section 4. | Type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| jobMonitor | Provides monitoring for the file download job. The data type of this attribute is the "ProcessMonitor" as defined in clause 4.3.43 with the specialisations defined in clause 4.3.46.1.<br><br>allowedValues: N/A | Type: ProcessMonitor<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| cancelJob | Setting this attribute to "TRUE" cancels the file download job. As specified in the definition of "ProcessMonitor", cancellation is possible in the "NOT_STARTED" and "RUNNING" state. Setting the attribute to "FALSE" has no observable result.<br><br>allowedValues: TRUE, FALSE | Type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: FALSE<br>isNullable: False |
| FileDownloadJob.jobMonitor.resultStateInfo | Provides the following specialisation for the "resultStateInfo" attribute of the "ProcessMonitor" data type for the "FileDownloadJob".<br><br>In the event the file download fails, and the "status" is equal to "FAILED", it provides the reason for the failure.<br><br>allowedValues for "status" = "FAILED":<br>- NULL<br>- UNKNOWN<br>- NO_STORAGE<br>- LOW_MEMORY<br>- NO_CONNECTION_TO_REMOTE_SERVER<br>- FILE_NOT_AVAILABLE<br>- DNS_CANNOT_BE_RESOLVED<br>- TIMER_EXPIRED<br>- OTHER<br><br>The allowed values for "FINISHED" or "CANCELLED" are vendor specific. | Type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| heartbeatNtfPeriod | Periodicity of the heartbeat notification emission. The value of zero has the special meaning of stopping the heartbeat notification emission.<br><br>Unit is in seconds.<br><br>AllowedValues: non-negative integers | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: 0<br>isNullable: False |
| triggerHeartbeatNtf | Setting this attribute to TRUE triggers an immediate additional heartbeat notification emission. Setting the value to FALSE has no observable result.<br><br>The periodicity of notifyHeartbeat emission is not changed.<br><br>allowedValues: TRUE, FALSE | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: FALSE<br>isNullable: False |
| notificationRecipientAddress | Address of the notification recipient.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| notificationTypes | List of notification types.<br><br>Below is a list of notificationType values that are defined in 3GPP specifications. Other notification types defined by SDOs or enterprises may also be supported.<br><br>allowedValues:<br>- notifyMOICreation<br>- notifyMOIDeletion<br>- notifyMOIAttributeValueChanges<br>- notifyMOIChanges<br>- notifyEvent<br>- notifyNewAlarm<br>- notifyChangedAlarm<br>- notifyAckStateChanged<br>- notifyComments<br>- notifyCorrelatedNotificationChanged<br>- notifyChangedAlarmGeneral<br>- notifyClearedAlarm<br>- notifyAlarmListRebuilt<br>- notifyPotentialFaultyAlarmList<br>- notifyFileReady<br>- notifyFilePreparationError<br>- notifyThresholdCrossing | type: ENUM<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| notificationFilter | Filter to be applied to candidate notifications identified by the notificationTypes attribute. Only notifications that pass the filter criteria are forwarded to the notification recipient. All other notifications are discarded.<br>The filter can be applied to any field of a notification.<br><br>allowedValues: N/A | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| notificationProtocols | List of protocols supported for notifications.<br>TS 28.532 [27] defines options<br>Restful HTTP and Restful HTTP aligned with VES<br>Other values defined by SDOs or enterprises may also be supported.<br><br>allowedValues:<br>- HTTP<br>- HTTP_VES_ENCAPS | type: ENUM<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| scope | Scopes (selects) data nodes in an object tree.<br><br>allowedValues: N/A | type: Scope<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| scopeType | If the optional scopeLevel attribute is not supported or absent, allowed values of scopeType are BASE_ONLY and BASE_ALL.<br><br>The value BASE_ONLY indicates only the base object is selected.<br><br>The value BASE_ALL indicates the base object and all of its subordinate objects (incl. the leaf objects) are selected.<br><br>If the scopeLevel attribute is supported and present, allowed values of scopeType are BASE_NTH_LEVEL and BASE_SUBTREE.<br><br>The value BASE_NTH_LEVEL indicates all objects on the level, which is specified by the scopeLevel attribute, below the base object are selected. The base object is at scopeLevel zero.<br><br>The value BASE_SUBTREE indicates the base object and all subordinate objects down to and including the objects on the level, which is specified by the scopeLevel attribute, are selected. The base object is at scopeLevel zero.<br><br>allowedValues: N/A | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| scopeLevel | See definition of scopeType attribute.<br><br>allowedValues: N/A | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| dataNodeSelector | The dataNodeSelector attribute allows to select one or more managed object instances, attributes, attribute fields or attribute elements. Its value contains a solution set specific expression for selecting the nodes.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| farEndEntity | The value of this attribute shall be the Distinguished Name of the far end network entity to which the reference point is related.<br><br>allowedValues: N/A | type: DN<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| linkType | This attribute defines the type of the Link.<br><br>allowedValues: Signalling, Bearer, OAM&P, Other or multiple combinations of this type. | type: String<br>multiplicity: 0..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| locationName | The physical location of this entity (e.g. an address).<br><br>allowedValues: N/A | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| monitorGranularityPeriod | Granularity period used to monitor performance metrics for threshold crossings. The period is defined in seconds.<br><br>See Note 5<br><br>allowedValues: a multiple of a supported GP of the associated performance metrics | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| reportingPeriods | Reporting periods supported for the associated performance metrics. The period is defined in seconds.<br><br>allowedValues: Integer with a minimum value of 1 | type: Integer<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| thresholdInfoList | List of threshold infos. | type: ThresholdInfo<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| thresholdValue | Value against which the monitored performance metric is compared at a threshold level in case the hysteresis is zero.<br><br>allowedValues: float or integer | type: Float or Integer<br>multiplicity: 1<br>isOrdered: NA<br>isUnique: NA<br>defaultValue: None<br>isNullable: False |
| hysteresis | Hysteresis of a threshold. If this attribute is present the monitored performance metric is not compared against the threshold value as specified by the thresholdValue attribute but against a high and low threshold value given by<br><br>highThresholdValue- = thresholdValue + hysteresis<br>lowThresholdValue = thresholdValue - hysteresis<br><br>When going up, the threshold is triggered when the performance metric reaches or crosses the high threshold value. When going down, the threshold is triggered when the performance metric reaches or crosses the low threshold value.<br><br>A hysteresis may be present only when the monitored performance metric is not of type counter that can go up only. If present for a performance metric of type counter, it shall be ignored.<br><br>allowedValues: non-negative float or integer | type: Float or Integer<br>multiplicity: 0..1<br>isOrdered: NA<br>isUnique: NA<br>defaultValue: None<br>isNullable: False |
| thresholdDirection | Direction of a threshold indicating the direction for which a threshold crossing triggers a threshold.<br><br>When the threshold direction is configured to "UP", the associated treshold is triggered only when the performance metric value is going up upon reaching or crossing the threshold value. The treshold is not triggered, when the performance metric is going down upon reaching or crossing the threshold value.<br><br>Vice versa, when the threshold direction is configured to "DOWN", the associated treshold is triggered only when the performance metric is going down upon reaching or crossing the threshold value. The treshold is not triggered, when the performance metric is going up upon reaching or crossing the threshold value.<br><br>When the threshold direction is set to "UP_AND_DOWN" the treshold is active in both direcions.<br><br>In case a threshold with hysteresis is configured, the threshold direction attribute shall be set to "UP_AND_DOWN".<br><br>allowedValues:<br>- UP<br>- DOWN<br>- UP_AND_DOWN | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| objectClass | Class of a managed object instance.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| objectInstance | Managed object instance identified by its DN.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| objectInstances | List of managed object instances. Each object instance is identified by its DN.<br><br>allowedValues: N/A | type: DN<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |

| | | |
|---|---|---|
| peeParametersList | This attribute contains the parameter list for the control and monitoring of power, energy and environmental parameters of ManagedFunction instance(s). This list contains the following parameters:<br><br>- siteIdentification<br>- siteLatitude (optional)<br>- siteLongitude (optional)<br>- siteAltitude (optional)<br>- siteDescription<br>- equipmentType<br>- environmentType<br>- powerInterface<br><br>siteIdentification: The identification of the site where the ManagedFunction resides.<br><br>allowedValues: N/A<br><br>siteLatitude: The latitude of the site where the ManagedFunction instance resides, based on World Geodetic System (1984 version) global reference frame (WGS 84). Positive values correspond to the northern hemisphere. This attribute is optional for BTSFunction, RNCFunction , GNBDUFunction and NRSectorCarrier instance(s).<br><br>allowedValues: -90.0000 to +90.0000<br><br>siteLongitude: The longitude of the site where the managedFunction instance resides, based on World Geodetic System (1984 version) global reference frame (WGS 84). Positive values correspond to degrees east of 0 degrees longitude. This attribute is optional for BTSFunction, RNCFunction, GNBDUFunction and NRSectorCarrier instance(s).<br><br>allowedValues: -180.0000 to +180.0000<br><br>siteAltitude: The altitude of the site where the ManagedFunction instance resides, in unit of meter. This attribute is optional for BTSFunction, RNCFunction, GNBDUFunction and NRSectorCarrier instance(s).<br><br>siteDescription: An operator defined description of the site where the ManagedFunction instance resides.<br><br>allowedValues: N/A<br><br>equipmentType: The type of equipment where the ManagedFunction instance resides.<br><br>allowedValues: see clause 4.4.1 of ETSI ES 202 336-12 [18].<br><br>environmentType: The type of environment where the ManagedFunction instance resides.<br><br>allowedValues: see clause 4.4.1 of ETSI ES 202 336-12 [18].<br><br>powerInterface: The type of power. | type: String<br>multiplicity: 0..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| | allowedValues: see clause 4.4.1 of ETSI ES 202 336-12 [18]. | |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| priorityLabel | This is a label that consumer would assign a value on a concrete instance of the managed object. The management system takes the value of this attribute into account. The effect of this attribute value to the subject managed entity is not standardized | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| protocolVersion | Versions(s) and additional descriptive information for the protocol(s) used for the associated communication link. Syntax and semantic is not specified.<br><br>allowedValues: N/A | type: String<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| setOfMcc | Set of Mobile Country Code (MCC). The MCC uniquely identifies the country of domicile of the mobile subscriber. MCC is part of the IMSI (TS 23.003 [5])<br><br>This list contains all the MCC values in subordinate object instances to this SubNetwork instance.<br><br>allowedValues: See clause 2.3 of TS 23.003 [5] for MCC allocation principles. | type: Integer<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| swVersion | The software version of the ManagementNode or ManagedElement (this is used for determining which version of the vendor specific information is valid for the ManagementNode or ManagedElement).<br><br>allowedValues: N/A | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| systemDN | Distinguished Name (DN) of a MnSAgent.<br><br>allowedValues: N/A | type: DN<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| userDefinedState | An operator defined state for operator specific usage.<br><br>allowedValues: N/A | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| userLabel | A user-friendly (and user assignable) name of this object.<br><br>allowedValues: N/A | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| vendorName | The name of the vendor.<br><br>allowedValues: N/A | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| vnfParametersList | This attribute contains the parameter set of the VNF instance(s) corresponding to an NE. Each entry in the list contains:<br>- vnfInstanceId<br>- vnfdId (optional)<br>- flavourId (optional)<br>- autoScalable (optional)<br><br>vnfInstanceId: VNF instance identifier (vnfInstanceId, see section 9.4.2 of ETSI GS NFV-IFA 008 [16]).<br><br>See Note 1.<br><br>vnfdId: Identifier of the VNFD on which the VNF instance is based, see section 9.4.2 of [A]. This attribute is optional.<br>Note: the value of this attribute is identical to that of the same attribute in clause 9.4.2 of ETSI GS NFV-IFA 008 [A].<br><br>flavourId: Identifier of the VNF Deployment Flavour applied to this VNF instance, see section 9.4.3 of ETSI GS NFV-IFA 008 [16]. This attribute is optional.<br>Note: the value of this attribute is identical to that of the same attribute in clause 9.4.3 of ETSI GS NFV-IFA 008 [16].<br><br>autoScalable: Indicator of whether the auto-scaling of this VNF instance is enabled or disabled. The type is Boolean.<br>This attribute is optional.<br><br>See Note2.<br><br>The presence of this attribute indicates that the ManagedFunction represented by the MOI is a virtualized function.<br><br>See Note 3.<br><br>allowedValues: N/A<br><br>A string length of zero for vnfInstanceId means the VNF instance(s) corresponding to the MOI does not exist (e.g. has not been instantiated yet, has already been terminated). | type: String<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| vsData | Vendor specific attributes of the type vsDataType. The attribute definitions including constraints (value ranges, data types, etc.) are specified in a vendor specific data format file.<br><br>allowedValues: -- | type: --<br>multiplicity: --<br>isOrdered: --<br>isUnique: --<br>defaultValue: --<br>isNullable: False |
| vsDataFormatVersion | Name of the data format file, including version.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| vsDataType | Type of vendor specific data contained by this instance, e.g. relation specific algorithm parameters, cell specific parameters for power control or re-selection or a timer. The type itself is also vendor specific.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| supportedPerfMetricGroups | A set of performance metric groups. When this attribute is contained in a managed object it may define performance metrics for this object and all descendant objects.<br><br>allowedValues: N/A | type: SupportedPerfMetricGroup<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| performanceMetrics | List of performance metrics identified by name<br><br>allowedValues:.<br><br>Performance metrics include measurements defined in TS 28.552 [20] and KPIs defined in TS 28.554 [28].<br><br>For measurements defined in TS 28.552 [20] the name is constructed as the bullet e) of the measurement definition with allowed measurement type.<br><br>For KPIs defined in TS 28.554 [28] the name is defined in the KPI definitions template, see chapter 5 in TS 28.554 [28], as the component designated with a).<br><br>For non-3GPP specified measurements the name is defined elsewhere. | type: String<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| supportedTraceMetrics | List of trace metrics. When this attribute is contained in a managed object it defines the trace metrics supported for this object and all descendant objects.<br><br>Trace metrics include trace messages, MDT measurements (Immediate MDT, Logged MDT, Logged MBSFN MDT), RLF and RCEF reports, see TS 32.422 [30]. Trace metrics are identified with their metric identifier. The metric identifier is constructed as defined in clause 10 of TS 32.422 [30].<br><br>For non-3GPP specified trace metrics the name is defined elsewhere.<br><br>allowedValues: N/A | type: String<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| listOfTraceMetrics | List of trace metrics identified by name.<br><br>Includes trace messages, MDT measurements (Immediate MDT, Logged MDT, Logged MBSFN MDT), RLF and RCEF reports, see TS 32.422 [30]. Trace messages are identified with their message identifier. Trace metric identifier is constructed as defined in clause 10 of TS 32.422 [30].<br><br>For non-3GPP specified trace metrics the name is defined elsewhere.<br><br>allowedValues: N/A | type: String<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| rootObjectInstances | List of object instances. Each object instance is identified by its DN and designates the root of a subtree that contains the root object and all descendant objects. | type: DN<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| reportingMethods | List of reporting methods for performance metrics<br><br>allowedValues:<br>- "FILE_BASED_LOC_SET_BY_PRODUCER",<br>- "FILE_BASED_LOC_SET_BY_CONSUMER",<br>- "STREAM_BASED" | type: ENUM<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| nFServiceType | The parameter defines the type of the managed NF service instance<br><br>allowedValues: See clause 7.2 of TS 23.501[22] | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| operations | This parameter defines set of operations supported by the managed NF service instance.<br><br>allowedValues: See TS 23.502[23] for supporting operations | type: Operation<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| Operation.name | This parameter defines the name of the operation of the managed NF service instance.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| allowedNFTypes | This parameter identifies the type of network functions allowed to access the operation of the managed NF service instance.<br><br>allowedValues: See TS 23.501[22] for NF types | type: ENUM<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| operationSemantics | This paramerter identifies the semantics type of the operation. See TS 23.502[23]<br><br>allowedValues: “Request/Response”, “Subscribe/Notify”. | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| sAP | This parameter specifies the service access point of the managed NF service instance.<br><br>allowedValues: N/A | type: SAP<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| host | This parameter specifies the host address of the managed NF service instance. It can be FQDN (See TS 23.003 [5]) or an IPv4 address (See RFC 791 [24]) or an IPv6 address (See RFC 2373 [25]).<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| port | This parameter specifies the transport port of the managed NF service instance.<br><br>allowedValues: 1 - 65535 | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| usageState | Usage state of a managed object instance. It describes whether the resource is actively in use at a specific instant, and if so, whether or not it has spare capacity for additional users at that instant.<br><br>allowedValues: "IDLE", "ACTIVE", "BUSY".<br>The meaning of these values is as defined in 3GPP TS 28.625 [21] and ITU-T X.731 [19]. | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| registrationState | This parameter defines the registration status of the managed NF service instance.<br><br>allowedValues: "Registered", "Deregistered". | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: Deregistered<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| jobRef | Object instance of the PerfMetricJob or TraceJob that produced the file.<br><br>allowedValues: NA | Type: Dn<br>multiplicity: 0..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| jobId | Identifier of a PerfMetricJob, a TraceJob or a QMCJob. | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| granularityPeriod | Granularity period used to produce performance metrics. The period is defined in seconds.<br><br>See Note 4.<br><br>allowedValues: Integer with a minimum value of 1 | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| granularityPeriods | Granularity periods supported for the production of associated performance metrics. The period is defined in seconds.<br><br>allowedValues: Integer with a minimum value of 1 | type: Integer<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| reportingCtrl | Selecting the reporting method and defining associated control parameters. | type: ReportingCtrl<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| fileReportingPeriod | For the file-based reporting method this is the time window during which collected measurements are stored into the same file before the file is closed and a new file is opened. The period is defined in minutes.<br><br>allowedValues: Multiples of granularityPeriod | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| _linkToFiles | Link to a Files object.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| streamTarget | The stream target for the stream-based reporting method.<br><br>allowedValues: N/A | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| administrativeState | Administrative state of a managed object instance. The administrative state describes the permission to use or prohibition against using the object instance. The adminstrative state is set by the MnS consumer.<br><br>allowedValues: LOCKED, UNLOCKED. | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: LOCKED<br>isNullable: False |
| operationalState | Operational state of manged object instance. The operational state describes if an object instance is operable ("ENABLED") or inoperable ("DISABLED"). This state is set by the object instance or the MnS producer and is hence READ-ONLY.<br><br>allowedValues: ENABLED, DISABLED. | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue:<br>DISABLED<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| jobType | It specifies whether the TraceJob represents only MDT, Trace, RLF, RCEF or 5GC UE level measurements job, or a combined job. It also defines the MDT mode.<br>See the clause 5.9a of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: TRACE_ONLY<br>isNullable: False |
| traceConfig | The set of parameters specific for trace configuration. | type: TraceConfig<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mdtConfig | The set of parameters specific for MDT configuration. | type: MdtConfig<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| immediateMdtConfig | The set of parameters specific for Immediate MDT configuration. | type: ImmediateMdtConfig<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| loggedMdtConfig | The set of parameters specific for Logged MDT and Logged MBSFN MDT configuration. | type: LoggedMdtConfig<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| listOfInterfaces | It specifies the interfaces that need to be traced. The attribute is applicable only for Trace.<br>See the clause 5.5 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| listOfNeTypes | It specifies the network element types where the trace should be activated. The attribute is applicable only for Trace with Signalling Based Trace activation.<br>See the clause 5.4 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| pLMNTarget | It specifies which PLMN that the subscriber of the session to be recorded uses as selected PLMN. | type: PlmnId<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| traceReportingConsumerUri | It specifies the Uniform Resource Identifier (URI) of the Streaming Trace data reporting MnS consumer (a.k.a. streaming target).<br>See the clause 5.9 c of TS 32.422 [30] for additional details on the allowed values. | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| traceCollectionEntityIPAddress | It specifies the address of the Trace Collection Entity when the attribute traceReportingFormat is configured for the file-based reporting. The attribute is applicable for both Trace and MDT.<br>See the clause 5.9 of TS 32.422 [30] for additional details on the allowed values. | type: IpAddress<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| traceDepth | It specifies the trace depth. The attribute is applicable only for Trace.<br>See the clause 5.3 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: MAXIMUM<br>isNullable: False |
| traceReference | A globally unique identifier, which uniquely identifies the Trace Session that is created by the TraceJob.<br>In case of shared network, it is the MCC and<br>MNC of the Participating Operator that request the trace session that shall be provided.<br>The attribute is applicable for both Trace and MDT.<br>See the clause 5.6 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: TraceReference<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| traceReportingFormat | It specifies the trace reporting format - streaming trace reporting or file-based trace reporting.<br><br>allowedValues: FILE-BASED, STREAMING | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: FILE-BASED<br>isNullable: False |

| | | |
|---|---|---|
| traceTarget | It specifies the target object of the Trace, MDT and 5GC UE level measurements collection. The attribute is applicable for Trace, MDT, and 5GC UE level measurements collection. This attribute includes the ID type of the target as an enumeration and the ID value(s).<br><br>The traceTarget shall be "PUBLIC_ID" in case of a Management Based Activation is done to an SCSCFFunction (Serving Call Session Control Function) or PCSCFFunction (Proxy Call Session Control Function) (TS 28.705[44]). The traceTarget shall be "UTRAN_CELL" only in case of the UTRAN cell traffic trace function.<br>The traceTarget shall be "E-UTRAN_CELL" only in case of E-UTRAN cell traffic trace function.<br>The traceTarget shall be "NG-RAN_CELL" only in case of NR cell traffic trace function.<br>The traceTarget shall be either "IMSI", "IMEI" or "IMEISV" if the Trace Session is activated to any of the following ManagedEntity(ies):<br>- HSSFunction (Home Subscriber Server) (TS 28.705 [44])<br>- MscServerFunction (Mobile Switching Centre Server) (TS 28.702 [45])<br>- SgsnFunction (Serving GPRS Support Node) (TS 28.702[45])<br>- GgsnFunction (Gateway GPRS Support Node) (TS 28.702[45])<br>- BmscFunction (Broadcast Multicast Service Centre) (TS 28.702[45])<br>- RncFunction (Radio Network Controller) (TS 28.652[46])<br>- MmeFunction (Mobility Management Entity) (TS 28.708[47])<br>- ServingGWFunction (Serving Gateway) (TS 28.708[47])<br><br>- PGWFunction (PDN Gateway) (TS 28.708[47]).<br>The traceTarget shall be either “SUPI” or “IMEISV” if the Trace Session is activated to any of the following ManagedEntity(ies) (TS 28.541[48]):<br>- AFFunction<br>- AMFFunction<br>- AUSFunction<br>- NEFFunction<br>- NRFFunction<br>- NSSFFunction<br>- PCFFunction<br>- SMFFunction<br>- UPFFunction<br>- UDMFunction<br><br>In case of signalling based MDT, the traceTarget attribute shall be able to carry "PUBLIC_ID", "IMSI", "IMEI", "IMEISV)" or "SUPI".<br>In case of management based Immediate MDT, the traceTarget attribute shall be null value.<br>In case of management based Logged MDT, the traceTarget attribute shall carry an "eNB" or a "gNB" or an "RNC". The Logged MDT should be initiated on the specified eNB/gNB/RNC in traceTarget.<br>In case of RLF reporting, or RCEF reporting, the traceTarget attribute shall be null value.<br><br>In case of signalling based 5GC UE level measurements collection, the traceTarget attribute shall be able to carry "IMEISV" or "SUPI".<br>In case of management based 5GC UE level measurements collection, the traceTarget attribute shall be able to carry the corresponding Measured UE Identifier as defined by the bullet g) of the 5GC UE level measurements (see TS 28.558 [57]) when the TraceJob is created at the subject ManagedEntity. | type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: No<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| triggeringEvents | It specifies the triggering event parameter of the trace session. The attribute is applicable only for Trace.<br>See the clause 5.1 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| anonymizationOfMdtData | It specifies the level of anonymization of MDT data. This attribute is only applicable for management based activation.<br>See the clause 5.10.12 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue:<br>NO_IDENTITY<br>isNullable: False |
| areaConfigurationForNeighCell | It specifies the area for which UE is requested to perform measurement logging for neighbour cells which have list of frequencies. If it is not configured, the UE shall perform measurement logging for all the neighbour cells.<br>Applicable only to NR Logged MDT.<br>See the clause 5.10.26 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: AreaConfig<br>multiplicity:*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| areaScope | It specifies the area where data shall be collected. | type: AreaScope<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| collectionPeriodRRMLTE | It specifies the collection period for collecting RRM configured measurement samples for M3 in LTE. The attribute is applicable only for Immediate MDT. See the clause 5.10.20 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| collectionPeriodRRMUMTS | It specifies the collection period for collecting RRM configured measurement samples for M3, M4, M5 in UMTS. The attribute is applicable only for Immediate MDT. See the clause 5.10.21 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| eventListForEventTriggeredMeasurement | It specifies event types for event triggered measurement in the case of logged NR MDT. Each trace session may configure at most one event. The UE shall perform logging of measurements only upon certain condition being fulfilled:<br>- Out of coverage.<br>- A2 event.<br>See the clause 5.10.28 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| eventThreshold | It specifies the threshold which should trigger<br>the reporting in case A2 event reporting in LTE and NR or 1F/1l event in UMTS. The attribute is applicable only for Immediate MDT and when reportingTrigger is configured for A2 event in LTE and NR or 1F event or 1l event in UMTS.<br>See the clauses 5.10.7 and 5.10.7a of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| listOfMeasurements | It specifies the UE measurements that shall be collected in an Immediate MDT job. The attribute is applicable only for Immediate MDT.<br>See the clause 5.10.3 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| loggingDuration | It specifies how long the MDT configuration is valid at the UE in case of Logged MDT. The attribute is applicable only for Logged MDT and Logged MBSFN MDT.<br>See the clause 5.10.9 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| loggingInterval | It specifies the periodicity for Logged MDT. The attribute is applicable only for Logged MDT and Logged MBSFN MDT. See the clause 5.10.8 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| eventThresholdL1 | It specifies the threshold which should trigger the reporting in case of event based reporting of logged NR MDT. The attribute is applicable only for Logged MDT and when reportType is configured for event triggered reporting and when eventListEventForTriggeredMeasurement is configured for L1 event. See the clause 5.10.36 of TS 32.422 [30] for additional details on the allowed values. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| hysteresisL1 | It specifies the hysteresis used within the entry and leave condition of the L1 event based reporting of logged NR MDT. The attribute is applicable only for Logged MDT, when reportType is configured for event triggered reporting and when eventListForEventTriggeredMeasurement is configured for L1 event. See the clause 5.10.37 of TS 32.422 [30] for additional details on the allowed values. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| timeToTriggerL1 | It specifies the threshold which should trigger the reporting in case of event based reporting of logged NR MDT. The attribute is applicable only for Logged MDT, when reportType is configured for event triggered reporting and when eventListForEventTriggeredMeasurement is configured for L1 event. See the clauses 5.10.38 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mbsfnAreaList | The MBSFN Area consists of a MBSFN Area ID and Carrier Frequency (EARFCN). The target MBSFN area List can have up to 8 entries. This parameter is applicable only if the job type is Logged MBSFN MDT. See the clause 5.10.25 of TS 32.422 [30] for additional details on the allowed values. | type: MbsfnArea<br>multiplicity: 0..8<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| measurementPeriodLTE | It specifies the collection period for the Data Volume (M4) and Scheduled IP throughput measurements (M5) for LTE MDT taken by the eNB. The attribute is applicable only for Immediate MDT. See the clause 5.10.23 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| collectionPeriodM6LTE | It specifies the collection period for the Packet Delay measurement (M6) for MDT taken by the eNB. The attribute is applicable only for Immediate MDT. See the clause 5.10.32 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| collectionPeriodM7LTE | It specifies the collection period for the Packet Loss Rate measurement (M7) for LTE MDT taken by the eNB. The attribute is applicable only for Immediate MDT. See the clause 5.10.33 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| measurementPeriodUMTS | It specifies the collection period for the Data Volume (M6) and Throughput measurements (M7) for UMTS MDT taken by RNC. The attribute is applicable only for Immediate MDT. See the clause 5.10.22 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| collectionPeriodRRMNR | It specifies the collection period for collecting RRM configured measurement samples for M4, M5 in NR. The attribute is applicable only for Immediate MDT. See the clause 5.10.30 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| collectionPeriodM6NR | It specifies the collection period for the Packet Delay measurement (M6) for NR MDT taken by the gNB. The attribute is applicable only for Immediate MDT.<br>See the clause 5.10.34 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| collectionPeriodM7NR | It specifies the collection period for the Packet Loss Rate measurement (M7) for NR MDT taken by the gNB. The attribute is applicable only for Immediate MDT.<br>See the clause 5.10.35 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: True |
| beamLevelMeasurement | This indicates whether the NR M1 beam level measurements shall be included or not.<br>See the clause 5.10.40 of TS 32.422 [30] for additional details.<br><br>The default value is "FALSE".<br><br>allowedValues: TRUE, FALSE | type: Boolean<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: FALSE<br>isNullable: False |
| eventThresholdUphUMTS | It specifies the threshold which should trigger the reporting in case of event-triggered periodic reporting for M4 (UE power headroom measurement) in UMTS.<br>See the clause 5.10.39 of TS 32.422 [30] for additional details on the allowed values. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| measurementQuantity | It specifies the measurements that are collected in an MDT job for a UMTS MDT configured for event triggered reporting.<br>See the clause 5.10.15 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| plmnList | It indicates the PLMNs where measurement collection, status indication and log reporting are allowed.<br>See the clause 5.10.24 of TS 32.422 [30] for additional details on the allowed values. | type: PlmnId<br>multiplicity: 0..16<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| positioningMethod | It specifies what positioning method should be used in the MDT job.<br>See the clause 5.10.19 of TS 32.422 [30] for additional details on the allowed values. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmount | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and when reportingTrigger is configured for periodical measurements.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM1LTE | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for LTE.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM4LTE | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for LTE.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| reportAmountM5LTE | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for LTE.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM6LTE | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for LTE.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM7LTE | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for LTE.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM1NR | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for NR. In case this attribute is not used, it carries a null semantic.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM4NR | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for NR.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM5NR | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for NR.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM6NR | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for NR.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportAmountM7NR | It specifies the number of measurement reports that shall be taken for periodic reporting while the UE is in connected mode. The attribute is applicable only for Immediate MDT and combined Trace and Immediate MDT and when reportingTrigger is configured for periodical measurements and applicable only for NR.<br>See the clause 5.10.6 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| reportingTrigger | It specifies whether periodic or event based measurements should be collected. The attribute is applicable only for Immediate MDT and when the listOfMeasurements is configured for M1 (for UMTS, LTE and NR) or M2 (only for UMTS).<br>See the clause 5.10.4 of TS 32.422 [30] for additional details on the allowed values. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportInterval | It specifies the interval between the periodical measurements that shall be taken when the UE is in connected mode. The attribute is applicable only for Immediate MDT and when reportingTrigger is configured for periodical measurements.<br>See the clause 5.10.5 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| reportType | It specifies report type for logged NR MDT as:<br>- periodical.<br>- event triggered.<br>See the clause 5.10.27 of TS 32.422 [30] for additional details on the allowed values. | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| sensorInformation | It specifies which sensor information shall be included in logged NR MDT and immediate NR MDT measurement if they are available. The following sensor measurement can be included or excluded for the UE:<br>- BAROMETRIC_PRESSURE.<br>- UE_SPEED.<br>- UE_ORIENTATION.<br>See the clause 5.10.29 of 3GPP TS 32.422 [30] for additional details. | type: ENUM<br>multiplicity:*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| traceCollectionEntityId | It specifies the TCE Id which is sent to the UE in Logged MDT.<br>See the clause 5.10.11 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mcc | Mobile Country Code<br><br>see clause 2.3 of TS 23.003 [5] for MCC, String with pattern: '^[0-9]{3}$' | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mnc | Mobile Network<br><br>Mobile Network Code, see clause 2.3 of TS 23.003 [5] for MNC, String with pattern: '^[0-9]{2,3}$' | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| traceId | An identifier, which identifies the Trace (together with MCC and MNC). This is a 3 byte Octet String.<br><br>See the clause 5.6 of 3GPP TS 32.422 [30] for additional details on the allowed values. | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| freqInfo | It specifies the carrier frequency and bands used in a cell. | type: FreqInfo<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| arfcn | RF Reference Frequency as defined in TS 38.104 [35], clause 5.4.2.1. The frequency provided identifies the absolute frequency position of the reference resource block (Common RB 0) of the carrier. Its lowest subcarrier is also known as Point A.<br><br>allowedValues: 0, 1, …,3279165 | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| freqBands | List of NR frequency operating bands. Primary NR Operating Band as defined in TS 38.104 [35], clause 5.4.2.3.<br>The value 1 corresponds to n1, value 2 corresponds to NR operating band n2, etc.<br><br>allowedValues: 1, 2, …,1024 | type: Integer<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| pciList | List of neighbour cells subject for MDT scope.<br><br>allowedValues: 0, 1, …,1007 | type: Integer<br>multiplicity: 1..32<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| tac | Tracking Area Code<br><br>2 or 3-octet string identifying a tracking area code as specified in clause 9.3.3.10 of 3GPP TS 38.413 [34], in hexadecimal representation. Each character in the string shall take a value of "0" to "9", "a" to "f" or "A" to "F" and shall represent 4 bits. The most significant character representing the 4 most significant bits of the TAC shall appear first in the string, and the character representing the 4 least significant bit of the TAC shall appear last in the string.<br><br>pattern: '(^[A-Fa-f0-9]{4}$)\|(^[A-Fa-f0-9]{6}$)'<br><br>Examples:<br>A legacy TAC 0x4305 shall be encoded as "4305".<br>An extended TAC 0x63F84B shall be encoded as "63F84B" | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| utraCellIdList | List of UTRAN cells identified by UTRAN CGI. It identifies a list of UTRAN cells containing up to 32 UTRAN cells. | type: String<br>multiplicity: 1..32<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| eutraCellIdList | List of E-UTRAN cells identified by E-UTRAN-CGI. It identifies a list of UTRAN cells containing up to 32 E-UTRAN cells. | type: String<br>multiplicity: 1..32<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| nrCellIdList | List of NR cells identified by NG-RAN CGI<br><br>NR cell:<br>36-bit string identifying an NR Cell Id as specified in clause 9.3.1.7 of 3GPP TS 38.413 [34], in hexadecimal representation. Each character in the string shall take a value of "0" to "9", "a" to "f" or "A" to "F" and shall represent 4 bits. The most significant character representing the 4 most significant bits of the Cell Id shall appear first in the string, and the character representing the 4 least significant bit of the Cell Id shall appear last in the string.<br><br>Pattern: '^[A-Fa-f0-9]{9}$'<br><br>Example:<br>An NR Cell Id 0x225BD6007 shall be encoded as "225BD6007". | type: String<br>multiplicity: 1..32<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| tacList | Tracking Area Code list<br><br>It identifies a list of TACs containing up to 8 TACs | type: String<br>multiplicity: 1..8<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| taiList | Tracking Area Identity list<br><br>allowedValues: As defined by the data type | type: Tai<br>multiplicity: 1..8<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| mbsfnAreaId | MBSFN Area Identifier<br><br>AllowedValues: 1, 2, … | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| earfcn | Carrier Frequency<br><br>AllowedValues: 1, 2, … | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mnsLabel | Human-readable name of management service. | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mnsType | Type of management service.<br><br>allowedValues: ProvMnS, FaultSupervisionMnS, StreamingDataReportingMnS, FileDataReportingMnS | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mnsVersion | Version of management service. | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mnsAddress | Addressing information for Management Service operations. | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| ProcessMonitor.id | Id of the process. It is unique within a single multivalue attribute of type ProcessMonitor. | Type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| ProcessMonitor.status | This attribute represents the status of the associated process, whether it fails, succeeds etc. It does not represent the returned values of a successfully finished process.<br><br>allowedValues:<br>- NOT_STARTED<br>- RUNNING<br>- CANCELLING<br>- FINISHED<br>- FAILED<br>- PARTIALLY_FAILED<br>- CANCELLED | Type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| ProcessMonitor.progressPercentage | Progress of the process as percentage.<br><br>Allowed values: integer between 0 and 100 | Type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| ProcessMonitor.progressStateInfo | Additional textual qualification of the states "NOT_STARTED", "CANCELLING" and "RUNNING".<br><br>For specific processes, specific well-defined strings (e.g. string patterns or enums) may be defined as a specialisation.<br><br>allowedValues: N/A | Type: String<br>multiplicity: 0..*<br>isOrdered: True<br>isUnique: False<br>defaultValue: None<br>isNullable: False |
| ProcessMonitor.resultStateInfo | Additional textual qualification of the states "FINISHED", "FAILED", "PARTIALLY_FAILED and "CANCELLED". For example, in the "FAILED" or "PARTIALLY_FAILED" state this attribute may be used to provide error reasons.<br><br>This attribute shall not be used to make the outcome of the process available for retrieval, if any. For this purpose, dedicated attributes shall be specified when specifying the representation of a specific process.<br><br>For specific processes, specific well-defined strings (e.g. string patterns or enums) may be defined as a specialisation.<br><br>allowedValues: N/A | Type: String<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| ProcessMonitor.startTime | Start time of the associated process, i.e. the time when the status changed from "NOT_STARTED" to "RUNNING".<br><br>allowedValues: N/A | Type: DateTime<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| ProcessMonitor.endTime | Date and time when status changed to SUCCESS, CANCELLED, FAILED or PARTIALLY_FAILED. If the time is in the future, it is the estimated time the process will end.<br><br>allowedValues: N/A | Type: DateTime<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| ProcessMonitor.timer | Time until the associated process is automatically cancelled.<br>If set, the system decreases the timer with time. When it reaches zero the cancellation of the associated process is initiated by the MnS_Producer.<br>If not set, there is no time limit for the process.<br>Once the timer is set, the consumer can not change it anymore.<br>If the consumer has not set the timer the MnS Producer may set it.<br>Unit is minutes.<br><br>allowedValues: Positive integers | Type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mnsScope | This attribute list contains the DNs of the managed object instances that can be accessed using the Management Service. If a complete SubNetwork can be accessed using the Management Service, this attribute may contain the DN of the SubNetwork instead of the DNs of the individual managed entities within the SubNetwork.<br><br>If a complete ManagedElement can be accessed using the Management Service, this attribute may contain the DN of the ManagedElement instead of the DNs of the individual managed entities within the ManagedElement. | type: DN<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| managementData | This attribute defines the list of management data that are requested. | Type: ManagementData<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| mgtDataCategory | This attributes defines the type of management data that are requested.<br><br>Allowed values for data category are COVERAGE, CAPACITY, ENERGY_EFFICIENCY, MOBILITY, ACCESSIBILITY. The data categories will map to certain measurement families defined in TS 28.552 [2], see below. In addition to the below mappings, MnS producer may map the provided categories to any additional proprietary management data, as appropriate.<br><br>The COVERAGE category will map to measurement families of MR (measurements related to Measurement Report) and L1M (measurements related to Layer 1 Measurement).<br>The CAPACITY category will map to measurement family RRU (measurements related to Radio Resource Utilization).<br>The ENERGY_EFFICIENCY category will map to measurement family PEE (measurements related to Power, Energy and Environment).<br>The MOBILITY category will map to measurement family MM (measurements related to Mobility Management).<br>The ACCESSIBILITY category will map to measurement family CE (measurements related to Connection Establishment).<br><br>Allowed values: COVERAGE, CAPACITY, SERVICE EXPERIENCE, TRACE, ENERGY EFFICIENCY, MOBILITY, ACCESSIBILITY<br><br>See NOTE 7. | type: ENUM<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| mgtDataName | A list of management data identified by name.<br><br>allowedValues:<br>The list may include metrics or set of metrics defined in TS 28.552 [20], TS 28.554 [28] and TS 32.422 [30].<br><br>For performance measurements defined in TS 28.552 [20] the name is constructed as the bullet e) of the measurement definition with allowed measurement type.<br><br>For trace metrics (including trace messages, MDT measurements (Immediate MDT, Logged MDT, Logged MBSFN MDT), RLF and RCEF reports) defined in TS 32.422 [30], the name (metric identifier) is defined in clause 10 of TS 32.422 [30].<br><br>For non-3GPP specified managment data the name is defined elsewhere. | type: string<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| targetNodeFilter | Set of information to target the Object Instance to collect the management data from. | type: NodeFilter<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: No<br>isNullable: False |
| areaOfInterest | It specifies a location(s) from where the management data shall be collected. | type: AreaOfInterest<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: No<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| geoAreaToCellMapping | It specifies the geographical area from where the management data shall be collected and the mapping to cells.<br><br>allowedValues: N/A | type: GeoAreaToCellMapping<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| geoPolygon | It specifies the geographical area with a polygon. The polygon is specified by its corners.<br><br>allowedValues: N/A | type: GeoCoordinate<br>multiplicity: 1..*<br>isOrdered: True<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| geoArea | It specifies the geographical area using the cordinates of the corners of a convex polygon.<br><br>allowedValues: N/A | type: GeoArea<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| latitude | Latitude based on World Geodetic System (1984 version) global reference frame (WGS 84). Positive values correspond to the northern hemisphere.<br><br>AllowedValues: -90.0000, …+90.0000 | type: float<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| longitude | Longitude based on World Geodetic System (1984 version) global reference frame (WGS 84). Positive values correspond to degrees east of 0 degrees longitude.<br><br>AllowedValues: -180.0000, … +180.0000 | type: float<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| altitude | It is the vertical distance between the point of interest from the mean sea level measured in metres. | type: Float<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| associationThreshold | It specifies the threshold of coverage area in percentage whether a cell belongs to the geographical area or not.<br>If this attribute is absent, the location of the base station antenna determines whether a cell belongs to the geographical area or not.<br><br>Allowed values: 1,…,100 | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| networkDomain | It specifies the network domain of the target node. This will also result in collecting appropriate management data from the nodes belonging to the specified domain.<br><br>Allowed Values: CN, RAN | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: N/A<br>isNullable: False |
| cpUpType | It specifies the traffic type of the target node. This will also result in collecting appropriate management data from the nodes handling the specified traffic (e.g AMF for CP and UPF for UP).<br><br>Allowed Values: CP, UP | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: N/A<br>isNullable: False |
| sst | It specifies the slice service type (SST) of which the slice subnet should be targeted. Please refer to TS 23.501 [22]. | type: Integer<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: N/A<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| collectionTimeWindow | Collection time window for which the management data should be reported. | type: TimeWindow<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: N/A<br>isNullable: False |
| startTime | It indicates the time (in "date-time" format) when the management activity shall be started.<br><br>AllowedValues: N/A. | type: DateTime<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| endTime | It indicates the time (in "date-time" format) when the management activityshall be stopped.<br><br>AllowedValues: N/A. | type: DateTime<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| timeWindow | Time window for which the configured management activity shall be active. | type: TimeWindow<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| timeIntervals | List of intervals within one day for which the service shall be active. | type: TimeInterval<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| intervalStart | It indicates the time (in "full-time" format) when the service shall be started.<br><br>Data type "FullTime" defines the time as specified by "full-time" in RFC3339 [54].<br><br>AllowedValues: N/A. | type: FullTime<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| intervalEnd | It indicates the time (in "full-time" format) when the service shall be stopped.<br><br>"FullTime" defines the time as specified by "full-time" in RFC3339 [54].<br><br>AllowedValues: N/A. | type: FullTime<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| daysOfWeek | It indicates the days on which the service shall be scheduled in case of weekly repetition. The intervals per day are configured by attribute timeIntervals.<br><br>allowedValues:<br>- MONDAY<br>- TUESDAY<br>- WEDNESDAY<br>- THURSDAY<br>- FRIDAY<br>- SATURDAY<br>- SUNDAY | type: ENUM<br>multiplicity: 1..7<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| daysOfMonth | It indicates the days in a month on which the service shall be scheduled in case of monthly repetition. Value 0 presents the last day of the month. The intervals per day are configured by attribute timeIntervals.<br><br>allowedValues: 0, 1, …31 | type: Integer<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| schedulingTimes | It defines the active scheduling times. | type: SchedulingTime<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| schedulerStatus | Switches between TRUE and FALSE depending upon whether the configured time constraints are fulfilled or not. | type: Boolean<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| conditionStatus | Switches between TRUE and FALSE depending upon whether the configured constraints are fulfilled or not. | type: Boolean<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| schedulerRef | Pointer to a Scheduler object. | type: DN<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| conditionMonitorRef | Pointer to a ConditionMonitor object. | type: Dn<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| condition | Logical expression of one or several condition(s).<br><br>The actual syntax and capabilities of condition is SS specific. However, each SS should support condition consisting of one or several assertions that may be grouped using the logical operators AND, OR and NOT. Only if the whole expression of condition evaluates TRUE, the attribute conditionStatus will be TRUE.<br><br>Each assertion is a pointer to a Boolean parameter or a logical expression of attribute existence or attribute value comparison ("equal to X, less than Y" etc.).<br><br>An empty string is not allowed.<br><br>allowedValues: N/A | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| dataScope | It specifies whether the required data is reported per S-NSSAI or per 5QI or per PLMN.<br><br>allowedValues: SNSSAI, 5QI, PLMN | type: ENUM<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| serviceType | Specifies an end user service type for QoE measurements.<br><br>allowedValues: DASH, MTSI, VR | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| qoECollectionEntityAddress | Specifies the address to which the QMC records shall be transferred. Ipv4 or Ipv6 address(es) may be used. | type: IpAddress<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| qoETarget | Specifies the target object of the QMC in case of signalling based QMC. The qoETarget attribute shall be able to carry "IMSI" or "SUPI". | type: String<br>multiplicity: 0..1<br>isOrdered:N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| qoEReference | Identifies the QoE measurement collection job in the Managed Elements and in the measurement collection entity.<br><br>The QoE reference shall be globally unique therefore it is composed as follows:<br><br>MCC+MNC+QMC ID, where the MCC and MNC are coming with the QMC activation request from the management system to identify one PLMN containing the management system, and QMC ID is a 3 byte Octet String.<br><br>The QMC ID is generated by the management system or the operator. | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| sliceScope | Contains a list of S-NSSAIs (Single Network Slice Selection Assistance Information). A Network Slice is identified by S-NSSAI. | type: S-NSSAI<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| qMCConfigFile | Provides a reference to a file including the parameters for configuration of application layer measurements, known as Container for Application Layer Measurement Configuration | Type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| excessPacketDelayThresholds | Excess packet delay thresholds info for M6 UL measurement. | type: ExcessPacketDelayThresholds<br>multiplicity: 0..255<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| fiveQIValue | It indicates 5QI value.<br><br>allowedValues: 0 - 255 | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| excessPacketDelayThresholdValue | Value of excess packet delay threshold for M6 UL measurement.<br><br>allowedValues: 0.25ms, 0.5ms, 1ms, 2ms, 4ms, 5ms, 10ms, 20ms, 30ms, 40ms, 50ms, 60ms, 70ms, 80ms, 90ms, 100ms, 150ms, 300ms, 500ms, … | type: ENUM<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mDTAlignmentInformation | This parameter indicates the MDT measurements with which alignment of QoE measurement is required. This parameter is optional and is valid for NR only. | Type: TraceReference<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| availableRANqoEMetrics | This parameter indicates available RAN visible QoE metrics to the gNB. This parameter is optional and is valid for NR only.<br><br>allowedValues: appLayerBufferLevelList, playoutDelayForMediaStartup | Type: ENUM<br>multiplicity: 0..2<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| dnPrefix | It carries the DN Prefix information or no information. See Annex C of TS 32.300 [13] for one usage of this attribute.<br><br>allowedValues: N/A | type: DN<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| nPNIdentityList | It defines which NPNs that can be served by the NR cell, and which CAG IDs or NIDs can be supported by the NR cell for corresponding PNI-NPN or SNPN in case of the cell is NPN-only cell.<br>(NPN-Identity referring to TS 38.331 [38])<br><br>allowedValues: N/A | type: NpnId<br>multiplicity: *<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| cAGIdList | It identifies a CAG list containing up to 256 CAG-identifiers per UE or up to 12 CAG-identifiers per cell, see TS 38.331 [38].<br>CAG ID is used to combine with PLMN ID to identify a PNI-NPN.<br>CAG ID is a hexadecimal range with size 32 bit.<br><br>allowedValues: N/A | type: String<br>multiplicity: 0..256<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| nIDList | It identifies a list of NIDs containing up to 16 NIDs, see TS 38.331 [38].<br>NID is used to combine with PLMN ID to identify an SNPN.<br>NID is a hexadecimal range with size 44 bit. | type: String<br>multiplicity: 0..16<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| nPNTarget | It defines which NPN that the subscriber of the session to be recorded uses as selected NPN.<br>There is maximum one CAG ID present in cAGIdList in case of PNI-NPN or maximum one NID present in nIDList in case of SNPN | type: NpnId<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| ueCoreMeasConfig | The set of parameters specific for 5GC UE level measurements configuration. | type: UECoreMeasConfig<br>multiplicity: 0..1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| ueCoreMeasurements | List of 5GC UE level measurements identified by name.<br><br>allowedValues:<br>The list may include 5GC UE level measurements defined in TS 28.558 [57], or vendor specific measurements.<br><br>For 5GC UE level measurements defined in TS 28.558 [57], the name is constructed as the bullet e) of the measurement definition with allowed measurement type.<br><br>For non-3GPP specified 5GC UE level measurements the name is defined elsewhere. | type: String<br>multiplicity: 1..*<br>isOrdered: False<br>isUnique: True<br>defaultValue: None<br>isNullable: False |
| ueCoreMeasGranularityPeriod | Granularity period used to produce 5GC UE level measurements. The period is defined in milliseconds (ms).<br><br>See Note 8.<br><br>allowedValues: Integer with a minimum value of 10 | type: Integer<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |

| Attribute Name | Documentation and Allowed Values | Properties |
|---|---|---|
| nfTypeToMeasure | It indicates the type of NE to produce the 5GC UE level measurements.<br><br>allowedValues: The NF types represented by the measured object classes as defined by f) of the 5GC UE level measurements specified in TS 28.558 [57]. | type: String<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: None<br>isNullable: False |
| mNOnly | This indicates whether the MDT configuration is for MN only or not.<br><br>The value "FALSE" means the MDT configuration is for both MN and SN.<br><br>The value “TRUE” means the MDT configuration is for MN only. | type: Boolean<br>multiplicity: 1<br>isOrdered: N/A<br>isUnique: N/A<br>defaultValue: FALSE<br>isNullable: False |

NOTE 1: The value of this attribute is identical to that of the same attribute in clause 9.4.2 of ETSI GS NFV-IFA 008 [16].

NOTE 2: The value of this attribute is identical to that of the attribute isAutoscaleEnabled included in vnfConfigurableProperty in clause 9.4.2 of ETSI GS NFV-IFA 008 [16].

NOTE 3: The presence of the attribute vnfParametersList, whose vnfInstanceId with a string length of zero, in createMO operation can trigger the instantiation of the related VNF/VNFC instances.

NOTE 4: The GP defines the measurement data production rate. The supported rates are dependent on the capacity of the producer involved (e.g. the processing power of the producer, the complexity of the measurement type involved etc) and therefore, it cannot be standardized for all producers involved. The supported GPs reflects the agreement between producer and the consumer involved.

NOTE 5: The monitoring granularity period defines the measurements monitoring period. The supported monitoring periods are dependent on the capacity of the producer involved (e.g. the processing power of the producer, the complexity of the measurement type involved etc) and therefore, it cannot be standardized for all producers involved. The supported monitoring GPs reflect the agreement between producer and the consumer involved.

NOTE 6: The supported threshold levels are dependent on the capacity of the producer involved (e.g. the processing power of the producer, number of measurements being measured by the producer at the time, the complexity of the measurement type involved etc) and therefore, it cannot be standardized for all producers involved. The supported levels can only reflect the negotiated agreement between producer and the consumer involved.

NOTE 7: The above values can be further extended by the implementations, as appropriate.

NOTE 8: The ueCoreMeasGranularityPeriod defines the measurement data production rate. The supported rates are dependent on the capacity of the producer involved (e.g. the processing power of the producer, the complexity of the measurement type involved etc) and therefore, it cannot be standardized for all producers involved. The supported Granularity periods reflects the agreement between producer and the consumer involved.

### 4.4.2 Constraints

None

## 4.5 Common notifications

### 4.5.1 Alarm notifications

This clause presents a list of notifications, defined in TS 28.111 [58], that a MnS consumer can receive. The notification header attribute objectClass/objectInstance, defined in TS 32.302 [3], captures the DN of an instance of an IOC defined in the present document.

| Name | S | Notes |
|---|---|---|
| notifyNewAlarm | M | |
| notifyClearedAlarm | M | |
| notifyChangedAlarm | O | |
| notifyChangedAlarmGeneral | O | |
| notifyCorrelatedNotificationChanged | O | |
| notifyAckStateChanged | O | |
| notifyComments | O | |
| notifyPotentialFaultyAlarmList | O | |
| notifyAlarmListRebuilt | M | |

## 4.5.2 Configuration notifications

This clause presents a list of notifications, defined in TS 28.532 [27], that a MnS consumer can receive. The notification header attribute objectClass/objectInstance, defined in TS 32.302 [3], captures the DN of an instance of an IOC defined in the present document.

| Name | S | Notes |
|---|---|---|
| notifyMOICreation | O | |
| notifyMOIDeletion | O | |
| notifyMOIAttributeValueChanges | O | |
| notifyMOIChanges | O | |
| notifyEvent | O | |

## 4.5.3 Threshold Crossing notifications

This clause presents a list of notifications, defined in TS 28.532 [27], that a MnS consumer can receive. The notification header attribute objectClass/objectInstance, defined in TS 32.302 [3], captures the DN of an instance of an IOC defined in the present document.

| Name | S | Notes |
|---|---|---|
| notifyThresholdCrossing | CM | Mandatory if NRM based threshold monitoring is supported. |

# Annex A (informative): Alternate class diagram

This class diagram combines the Figure 4.2.1-1 of this document with Figure 1 of TS 28.620 [9], the class diagram of UIM.

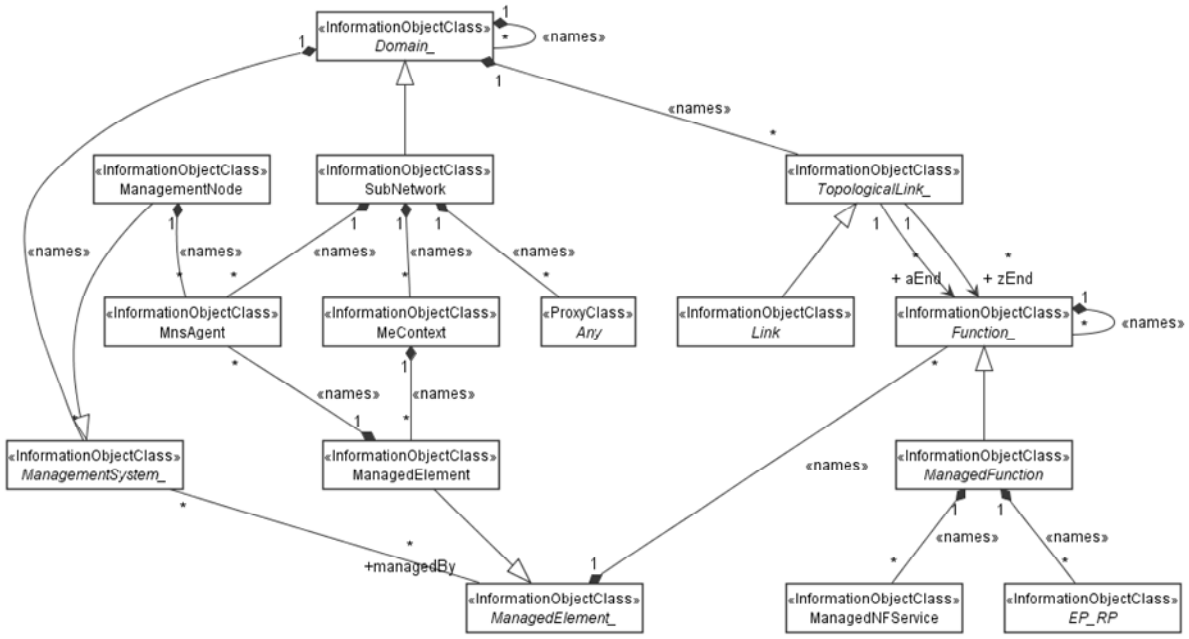



**Figure A-1: Alternate class diagram**

# Annex B (informative): PlanUML for figures

## B.1 Relationships

```
@startuml Figure 4.2.1-1 NRM fragment
hide empty members
hide circle
skinparam class {
BackgroundColor White
ArrowColor Black
BorderColor Black
}
skinparam ClassStereotypeFontStyle normal
top to bottom direction
skinparam nodesep 40

abstract class ManagedFunction <<InformationObjectClass>>
abstract class EP_RP <<InformationObjectClass>>
class ManagedNFService <<InformationObjectClass>>
EP_RP "*" -u-* "1" ManagedFunction: <<names>>
ManagedNFService "*" -u-* "1" ManagedFunction: <<names>>

class SubNetwork <<InformationObjectClass>>
class ManagementNode <<InformationObjectClass>>
class MnsAgent   <<InformationObjectClass>>
class MeContext  <<InformationObjectClass>>
abstract class Any        <<ProxyClass>>
class ManagedElement <<InformationObjectClass>>

ManagementNode "*" -u-* "1" SubNetwork: <<names>>
MnsAgent       "*" -u-* "1" SubNetwork: <<names>>
MeContext      "*" -u-* "1" SubNetwork: <<names>>
Any            "*" -u-* "1" SubNetwork: <<names>>
'SubNetwork     "*" --* "1" SubNetwork: <<names>>
MeContext -[hidden]l- MnsAgent
MnsAgent       "*" -l-* "1" ManagementNode: <<names>>
MnsAgent       "*" -d-* "1" ManagedElement: <<names>>

ManagedElement "*" -u-* "1" MeContext: <<names>>
ManagedFunction -[hidden]u- ManagedElement

@enduml
```

**Source code for Figure 4.2.1-1 NRM fragment**

```
@startuml Figure 4.2.1-12: QoE Measurement Collection NRM fragment
skinparam monochrome true
skinparam ClassStereotypeFontStyle normal
class ManagedEntity <<ProxyClass>>
class QMCJob <<InformationObjectClass>>
hide empty members
hide circle
ManagedEntity “1” *- “*” QMCJob : <<names>>
note top of ManagedEntity
Represents the following IOCs:
SubNetwork, or
ManagedElement.
End note
@enduml
```

**Source code for Figure 4.2.1-12: QoE Measurement Collection NRM fragment**

```
@startuml Figure 4.2.1-13: SupportedNotifications NRM fragment
skinparam monochrome true
skinparam ClassStereotypeFontStyle normal
class ManagedEntity <<ProxyClass>>
class SupportedNotifications <<InformationObjectClass>>
hide empty members
hide circle
ManagedEntity “1” *- “1” SupportedNotifications : <<names>>
```

```
note top of ManagedEntity
Represents the following IOCs:
SubNetwork, or
ManagedElement.
End note
@enduml
```

**Source code for Figure 4.2.1-13: SupportedNotifications NRM fragment**

```
@startuml Figure 4.2.1-14: Scheduler NRM fragment
skinparam monochrome true
skinparam ClassStereotypeFontStyle normal
class ManagedEntity <<ProxyClass>>
class Scheduler <<InformationObjectClass>>
hide empty members
hide circle
ManagedEntity “1” *- “*” Scheduler : <<names>>
note top of ManagedEntity
Represents the following IOCs:
SubNetwork, or
ManagedElement.
End note
@enduml
```

**Source code for Figure 4.2.1-14: Scheduler NRM fragment**

```
@startuml Figure 4.2.1-15: Condition monitor NRM fragment
skinparam monochrome true
skinparam ClassStereotypeFontStyle normal
class ManagedEntity <<ProxyClass>>
class ConditionMonitor <<InformationObjectClass>>
hide empty members
hide circle
ManagedEntity “1” *- “*” ConditionMonitor: <<names>>
note top of ManagedEntity
Represents the following IOCs:
SubNetwork, or
ManagedElement.
End note
@enduml
```

**Source code for Figure 4.2.1-15: Condition monitor NRM fragment**

# B.2 Inheritance

```
@startuml Figure 4.2.2-12: SupportedNotifications NRM fragment
skinparam monochrome true
abstract class Top <<InformationObjectClass>> {
}
class SupportedNotifications <<InformationObjectClass>> {
}
hide empty members
hide circle

Top  <|-- SupportedNotifications
@endum
```

**Source code for Figure 4.2.2-12: SupportedNotifications NRM fragment**

# Annex C (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **TSG #** | **TSG Doc.** | **CR** | **Rev** | **Subject/Comment** | **Old** | **New** |
| 2012-12 | | | | | New version after approval | 2.0.0 | 11.0.0 |
| 2012-02 | | | | | MCC update of TOC | 11.0.0 | 11.0.1 |
| 2014-06 | SA#64 | SP-140332 | 001 | - | Correction of reference | 11.0.1 | 11.1.0 |
| | | SP-140358 | 002 | - | Remove the feature support statements | | |
| 2014-09 | SA#65 | | | | Upgrade to Rel-12 | 11.1.0 | 12.0.0 |
| 2015-12 | SA#70 | SP-150691 | 005 | 1 | Add missing id attribute for 28.622 | 12.0.0 | 12.1.0 |
| 2016-01 | | | | | Upgrade to Rel-13 (MCC) | 12.1.0 | 13.0.0 |

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **Meeting** | **TDoc** | **CR** | **Rev** | **Cat** | **Subject/Comment** | **New version** |
| 2016-12 | SA#74 | SP-160853 | 0010 | - | A | Clarification on the need to show VsDataContainer self-containing itself several times | 13.1.0 |
| 2017-03 | SA#75 | SP-170139 | 0012 | 2 | A | Clarify notification triggered by VsDataContainer change | 13.2.0 |
| 2017-03 | SA#75 | SP-170143 | 0015 | 1 | B | Modify definitions of ME and MF to support virtualized network element | 14.0.0 |
| 2017-03 | SA#75 | SP-170142 | 0016 | 3 | B | Adding an attribute for ManagedFunction to support management of virtualized NE | 14.0.0 |
| 2017-06 | SA#76 | SP-170510 | 0019 | 2 | B | Add VNFInfo related attributes in IOC ManagedFunction | 14.1.0 |
| 2018-01 | SA#78 | SP-170969 | 0021 | - | F | Missing note in table of Attribute Properties | 14.2.0 |
| 2018-03 | SA#79 | SP-180060 | 0022 | - | B | Add new attribute peeParametersList to IOC ManagedFunction | 15.0.0 |
| 2018-06 | SA#80 | SP-180421 | 0024 | 1 | B | Remove references to Itf-N | 15.1.0 |
| 2018-12 | SA#82 | SP-181156 | 0027 | - | F | Add the missing NRM fragment supporting network performance management | 15.2.0 |
| 2018-12 | SA#82 | SP-181042 | 0028 | 1 | F | Replace MF with ManagedFunction | 15.2.0 |
| 2018-12 | SA#82 | SP-181042 | 0029 | 1 | F | Update NRM root IOCs to support slice priority | 15.2.0 |
| 2019-06 | SA#84 | SP-190371 | 0031 | 2 | B | Add IOCs for threshold monitoring control | 16.0.0 |
| 2019-06 | SA#84 | SP-190373 | 0033 | 2 | B | Update generic NRM Information Service to support Managed NF Service Object | 16.0.0 |
| 2019-09 | SA#85 | SP-190744 | 0038 | 2 | A | Update class definition with inheritance information | 16.1.0 |
| 2019-09 | SA#85 | SP-190744 | 0043 | 1 | A | Correct PMControl (Add report period attribute and disambiguate the delivery method attributes) | 16.1.0 |
| 2019-09 | SA#85 | SP-190751 | 0044 | - | A | Correct NR definition to avoid misalignment with RAN2 and add NRM definition | 16.1.0 |
| 2019-09 | SA#85 | SP-190744 | 0046 | 1 | A | Correct definitions of granularity period. | 16.1.0 |
| 2019-09 | SA#85 | | | | | Correction in implementation of CR0043 | 16.1.1 |
| 2019-12 | SA#86 | SP-191158 | 0057 | 2 | A | Correct definition of network resource | 16.2.0 |
| 2019-12 | SA#86 | SP-191173 | 0059 | - | A | Add measurementsList attribute into related IOCs | 16.2.0 |
| 2019-12 | SA#86 | SP-191166 | 0062 | 2 | B | Add heartbeat control NRM fragment | 16.2.0 |
| 2019-12 | SA#86 | SP-191166 | 0063 | 2 | B | Add notification subscription control fragment | 16.2.0 |
| 2020-03 | SA#87E | SP-200169 | 0066 | - | B | Add configurable FM. | 16.3.0 |
| 2020-03 | SA#87E | SP-200163 | 0069 | 1 | B | Add configurable KPI control NRM | 16.3.0 |
| 2020-03 | SA#87E | SP-200169 | 0071 | 1 | F | Correct definition of HeartbeatControl and attribute NotificationType | 16.3.0 |
| 2020-07 | SA#88-e | SP-200489 | 0074 | 1 | F | Add TOP_ as parent IOC | 16.4.0 |
| 2020-07 | SA#88-e | SP-200489 | 0075 | 1 | F | Update concept of ME and MF | 16.4.0 |
| 2020-07 | SA#88-e | SP-200489 | 0076 | - | F | Update the attribute priorityLabel for several IOCs | 16.4.0 |
| 2020-07 | SA#88-e | SP-200489 | 0077 | - | F | Updated MF description with nested clarification | 16.4.0 |
| 2020-07 | SA#88-e | SP-200483 | 0078 | 1 | B | Add trace control NRM fragment stage 2 | 16.4.0 |
| 2020-07 | SA#88-e | SP-200484 | 0080 | 1 | D | Fix inconsistent formatting | 16.4.0 |
| 2020-07 | SA#88-e | SP-200490 | 0083 | 1 | F | Combine class diagrams of subscription and heartbeat NRM control fragments (stage 2) | 16.4.0 |
| 2020-07 | SA#88-e | SP-200490 | 0084 | 1 | F | Update PM control fragment (stage 2) | 16.4.0 |
| 2020-07 | SA#88-e | SP-200490 | 0085 | - | F | Clarify usage of the VsDataContainer (stage 2) | 16.4.0 |
| 2020-07 | SA#88-e | SP-200490 | 0086 | 1 | F | Update FM control fragment (stage 2) | 16.4.0 |
| 2020-09 | SA#89e | SP-200729 | 0087 | 1 | F | Correct ThresholdMonitor definition (stage 2) | 16.5.0 |
| 2020-09 | SA#89e | SP-200729 | 0088 | - | F | Correct HeartbeatControl definition and some other smaller issues (stage 2) | 16.5.0 |
| 2020-09 | SA#90e | SP-201063 | 0089 | 1 | F | Add new MDT specific parameter collection period for NR aligning with 32.422 | 16.6.0 |
| 2020-09 | SA#90e | SP-201057 | 0090 | 1 | F | Remove thresholdLevel attribute from ThresholdMonitor (stage 2) | 16.6.0 |
| 2020-09 | SA#90e | SP-201057 | 0091 | 1 | F | Update the perfMetricJobGroupId attribute | 16.6.0 |
| 2020-09 | SA#90e | SP-201057 | 0092 | - | F | Remove value handling from the granularityPeriod description. | 16.6.0 |
| 2020-09 | SA#90e | SP-201088 | 0093 | - | F | Correct the attributes description of the IOCs inherited from Top and Top_ | 16.6.0 |
| 2020-09 | SA#90e | SP-201063 | 0094 | | F | Correct 5G trace parameter for trace control | 16.6.0 |
| 2020-09 | SA#90e | SP-201089 | 0095 | - | F | Update notifyThresholdCrossing to be a common notification. | 16.6.0 |
| 2021-03 | SA#91e | SP-210150 | 0097 | - | F | Correct notification support table for ManagedElement and ManagementNode | 16.7.0 |
| 2021-03 | SA#91e | SP-210153 | 0099 | 1 | F | Correction of attribute properties and IOC inheritance description | 16.7.0 |
| 2021-04 | SA#91e | | | | | Editorial cleanup with the help of the Rapporteur | 16.7.1 |
| 2021-06 | SA#92e | SP-210406 | 0096 | 3 | F | Replace legacy IRPAgent with MnsAgent (stage 2) | 16.8.0 |
| 2021-06 | SA#92e | SP-210397 | 0100 | 1 | F | Addition, adaptation and cleanup of Trace/MDT related parameters (stage2) | 16.8.0 |
| 2021-06 | SA#92e | SP-210416 | 0102 | - | F | Align different (abbreviated) names for support qualifier to S | 16.8.0 |
| 2021-06 | SA#92e | SP-210406 | 0103 | 1 | F | Clarify a subscription is required for notifyFileReady | 16.8.0 |
| 2021-06 | SA#92e | SP-210406 | 0104 | 1 | F | Clarify definition of PerfMetricJob | 16.8.0 |
| 2021-06 | SA#92e | SP-210406 | 0105 | - | F | Clarify the notification filter applies to all parameters of a notification | 16.8.0 |
| 2021-06 | SA#92e | SP-210406 | 0106 | - | F | Correct common notifications table | 16.8.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2021-06 | SA#92e | | | | | Editorial fix on tables and fonts | 16.8.1 |
| 2021-09 | SA#93e | SP-210879 | 0110 | 1 | A | Correction for vnfParametersList | 16.9.0 |
| 2021-09 | SA#93e | SP-210885 | 0111 | 1 | F | Add missing MnsAgent to class and inheritance diagrams | 16.9.0 |
| 2021-09 | SA#93e | SP-210871 | 0112 | - | F | Add missing notification type “notifyClearedAlarm” to the attribute “notificationTypes” | 16.9.0 |
| 2021-09 | SA#93e | SP-210871 | 0113 | 1 | F | Fix the issue caused by the updated NetworkSliceSubnet inheritence relationship | 16.9.0 |
| 2021-09 | SA#93e | SP-210865 | 0115 | - | F | Correction and clarification of reporting in TraceJob (stage2) | 16.9.0 |
| 2021-09 | SA#93e | SP-210865 | 0116 | - | F | Adaptation and cleanup of Trace/MDT related parameters (stage2) | 16.9.0 |
| 2021-12 | SA#94e | SP-211458 | 0121 | - | F | Introduce missing references | 16.10.0 |
| 2021-12 | SA#94e | SP-211478 | 0124 | - | A | Update Scope to be applicable for SBMA | 16.10.0 |
| 2021-12 | SA#94e | SP-211475 | 0125 | 1 | F | Clarify behavior of NtfSubscriptionControl | 16.10.0 |
| 2021-12 | SA#94e | SP-211467 | 0122 | - | B | Add support for MnS Discovery | 17.0.0 |
| 2022-03 | SA#95e | SP-220168 | 0126 | 1 | C | Asynchronous operation NRM additions | 17.1.0 |
| 2022-03 | SA#95e | SP-220179 | 0127 | 1 | A | Alarm Record changes | 17.1.0 |
| 2022-03 | SA#95e | SP-220179 | 0128 | 1 | A | Notification Subscription changes | 17.1.0 |
| 2022-03 | SA#95e | SP-220177 | 0131 | 1 | B | Enhance NRM with geographical information supporting MDA | 17.1.0 |
| 2022-03 | SA#95e | SP-220163 | 0133 | 1 | B | Add support for discovery of managed entities | 17.1.0 |
| 2022-03 | SA#95e | SP-220183 | 0134 | 1 | B | Add attribute to configure an identifier of a TraceJob | 17.1.0 |
| 2022-03 | SA#95e | SP-220171 | 0141 | - | B | Add parameter to configure beam level measurements in NR MDT | 17.1.0 |
| 2022-03 | SA#95e | SP-220183 | 0147 | - | B | Add stage2 definition for file management | 17.1.0 |
| 2022-03 | SA#95e | | | | | Adding a missing parenthesis in clause 4.4.1 (misimplemented CR) | 17.1.1 |
| 2022-06 | SA#96 | SP-220510 | 0151 | 1 | A | Correct isOrdered-isUnique for multivalue attributes | 17.2.0 |
| 2022-06 | SA#96 | SP-220516 | 0154 | - | A | Alignment of attribute names of TraceJob IOC to TS 32.422 (stage 2) | 17.2.0 |
| 2022-06 | SA#96 | SP-220510 | 0156 | - | A | Clean up of attribute properties | 17.2.0 |
| 2022-06 | SA#96 | SP-220510 | 0158 | 1 | A | Alarm Handling Clarifications | 17.2.0 |
| 2022-06 | SA#96 | SP-220505 | 0166 | - | B | Add stage 2 for management data collection and discovery | 17.2.0 |
| 2022-09 | SA#97e | SP-220863 | 0170 | 1 | F | Include already approved changes or enhancements of attribute properties for IOC ManagementDataCollectio | 17.3.0 |
| 2022-09 | SA#97e | SP-220864 | 0172 | - | A | Correction of attribute names of IOC TraceJob in the attribute property table | 17.3.0 |
| 2022-09 | SA#97e | SP-220865 | 0176 | - | F | Correcting Support Qualifier for jobId attribute | 17.3.0 |
| 2022-09 | SA#97e | SP-220855 | 0174 | - | B | Adding QMC job | 18.0.0 |
| 2023-01 | SA#98e | SP-221186 | 0178 | - | A | Correcting attribute definitions | 18.1.0 |
| 2023-01 | SA#98e | SP-221186 | 0181 | 1 | A | Correct description for ManagementDataCollection IOC | 18.1.0 |
| 2023-01 | SA#98e | SP-221187 | 0183 | 2 | A | Adding a new data type to represent GeoArea via convex polygon - Stage 2 | 18.1.0 |
| 2023-01 | SA#98e | SP-221167 | 0186 | 2 | A | Add missing notifyMOIChanges in Files and File IOC | 18.1.0 |
| 2023-01 | SA#98e | SP-221167 | 0189 | 1 | A | Correct inheritance diagram of the file download NRM fragment | 18.1.0 |
| 2023-01 | SA#98e | SP-221200 | 0192 | - | A | Removing reference to non-existing clause in 32.422 | 18.1.0 |
| 2023-01 | SA#98e | SP-221170 | 0195 | 1 | A | Update MnsAgent Definition | 18.1.0 |
| 2023-01 | SA#98e | SP-221186 | 0197 | 3 | A | Correct ManagementDataCollection definition | 18.1.0 |
| 2023-01 | SA#98e | SP-221176 | 0205 | 1 | C | JobID for QMCJob | 18.1.0 |
| 2023-01 | SA#98e | SP-221176 | 0206 | - | B | Definition of parameters MDT Alignment Information and Available RAN Visible QoE Metrics | 18.1.0 |
| 2023-01 | SA#98e | SP-221197 | 0207 | | A | Correct M6 Delay Threshold to align with TS 38.314 and TS 38.413 | 18.1.0 |
| 2023-03 | SA#99 | SP-230210 | 0212 | 1 | A | Correcting traceRecordingSessionReference property. Aligning with 32.422. | 18.2.0 |
| 2023-03 | SA#99 | SP-230207 | 0217 | - | A | Adding altitude to GeoArea datatype | 18.2.0 |
| 2023-03 | SA#99 | SP-230210 | 0218 | - | A | Correcting attribute constraints for Trace Job | 18.2.0 |
| 2023-03 | SA#99 | SP-230207 | 0220 | 1 | A | Correct issues for generic NRM Fragment | 18.2.0 |
| 2023-03 | SA#99 | SP-230204 | 0223 | 1 | B | TS 28.622 add missing UML diagram for QoE Measurement Collection | 18.2.0 |
| 2023-03 | SA#99 | SP-230211 | 0227 | 1 | A | Remove unused create link subscription attribute definition. | 18.2.0 |
| 2023-03 | SA#99 | SP-230208 | 0230 | 1 | A | Clarify reporting and monitoring period usage in SupportedPerfMetricGroup datatype. | 18.2.0 |
| 2023-03 | SA#99 | SP-230199 | 0233 | - | A | Correction of reference list | 18.2.0 |
| 2023-03 | SA#99 | SP-230202 | 0241 | 1 | A | Correction of attribute dnPrefix | 18.2.0 |
| 2023-03 | SA#99 | SP-230204 | 0242 | - | F | Correction of attribute availableRANqoEMetrics properties | 18.2.0 |
| 2023-06 | SA#100 | SP-230651 | 0249 | - | F | Adding the missing definition of the attribute excessPacketDelayThreshold | 18.3.0 |
| 2023-06 | SA#100 | SP-230649 | 0250 | - | A | Correcting attributes properties for Excess Packet Delay Threshold | 18.3.0 |
| 2023-06 | SA#100 | SP-230648 | 0254 | - | A | Clean up of incorrect use of multiplicity isOrdered isUnique and isNullable in attribute properties table | 18.3.0 |
| 2023-06 | SA#100 | SP-230648 | 0257 | - | A | Correction to missing Notification and Attribute constraints clauses | 18.3.0 |
| 2023-06 | SA#100 | SP-230647 | 0261 | - | A | Add clarification on TS version applicable for the IRP framework (partially implemented, MCC) | 18.3.0 |
| 2023-06 | SA#100 | SP-230681 | 0264 | 1 | A | Clarify how to subscribe to notifyThresholdCrossing | 18.3.0 |
| 2023-06 | SA#100 | SP-230649 | 0268 | - | A | Correction of attribute syntax | 18.3.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2023-09 | SA#101 | SP-230944 | 0244 | 4 | A | Clarify MnsRegistry handling | 18.4.0 |
| 2023-09 | SA#101 | SP-230938 | 0270 | - | B | Enhance the applicable notifications for Trace Job | 18.4.0 |
| 2023-09 | SA#101 | SP-230938 | 0271 | - | C | Name containment of Trace job in case of Signalling based activation | 18.4.0 |
| 2023-09 | SA#101 | SP-230938 | 0272 | - | B | Report Amount for M4, M5, M6 and M7 measurements in LTE | 18.4.0 |
| 2023-09 | SA#101 | SP-230938 | 0273 | - | B | Re-structuring Trace job | 18.4.0 |
| 2023-09 | SA#101 | SP-230939 | 0274 | - | F | Editorial Corrections | 18.4.0 |
| 2023-09 | SA#101 | SP-230960 | 0276 | - | C | Introduce MnS Producer Notification Capabilility | 18.4.0 |
| 2023-09 | SA#101 | SP-230942 | 0279 | 1 | A | Clarify HeartbeatControl IOC definition | 18.4.0 |
| 2023-09 | SA#101 | SP-230943 | 0283 | - | A | Remove the IOCs which are not applicable for SBMA | 18.4.0 |
| 2023-12 | SA#102 | SP-231453 | 0287 | - | F | Rel-18 CR TS 28.622 Aligning ReportAmount LTE parameters with ImmediateMDT datatype | 18.5.0 |
| 2023-12 | SA#102 | SP-231453 | 0288 | - | B | Rel-18 CR TS 28.622 Report Amount parameter in NR | 18.5.0 |
| 2023-12 | SA#102 | SP-231452 | 0293 | - | A | Rel-18 CR 28.622 Clarify MnS scope value for Managed Elements | 18.5.0 |
| 2023-12 | SA#102 | SP-231458 | 0297 | 1 | B | Rel-18 CR 28.622 Add NRM fragment for scheduler and condition monitor (stage 2) | 18.5.0 |
| 2023-12 | SA#102 | SP-231488 | 0300 | - | A | Correction of IOC ManagedNFService attribute values | 18.5.0 |
| 2023-12 | SA#102 | SP-231477 | 0308 | 1 | B | Rel-18 CR TS 28.622 Enhance the ManagementDataCollection to support request management data per PLMN | 18.5.0 |
| 2023-12 | SA#102 | SP-231453 | 0309 | 1 | B | Rel-18 CR TS28.622 Adding NPN Area Scope of MDT | 18.5.0 |
| 2023-12 | SA#102 | SP-231471 | 0318 | 1 | A | Rel-18 CR 28.622 Add measurement bin support to NRM | 18.5.0 |
| 2024-03 | SA#103 | SP-240186 | 0320 | 1 | F | TS28.622 corrections to headings for Figure title | 18.6.0 |
| 2024-03 | SA#103 | SP-240168 | 0325 | - | B | Rel-18 CR 28.622 Add new method for specifying the scope of subscriptions | 18.6.0 |
| 2024-03 | SA#103 | SP-240180 | 0326 | 1 | B | Enhance TraceJob for UE level measurements collection | 18.6.0 |
| 2024-03 | SA#103 | SP-240168 | 0324 | 1 | B | Rel-18 CR 28.622 Remove FM related parts | 18.6.0 |
| 2024-06 | SA#104 | SP-240821 | 0349 | - | F | Rel-18 CR TS 28.622 Fix references to non-existing attributes | 18.7.0 |
| 2024-06 | SA#104 | SP-240824 | 0352 | 1 | F | Rel-18 CR TS 28.622 Fix description of MdtConfig dataType attributes | 18.7.0 |
| 2024-06 | SA#104 | SP-240824 | 0353 | 1 | F | Rel-18 CR TS 28.622 Fix description of attributes in npnId dataType | 18.7.0 |
| 2024-06 | SA#104 | SP-240824 | 0354 | 1 | F | Rel-18 CR TS 28.622 Update contraints descriptions for attributes of npnId dataType | 18.7.0 |
| 2024-06 | SA#104 | SP-240813 | 0356 | 1 | A | TS28.622 Rel18 correction to using ENUM and Union as dataType | 18.7.0 |
| 2024-06 | SA#104 | SP-240806 | 0361 | 1 | A | Rel-18 CR 28.622 Add missing trace message support to trace job (stage 2) | 18.7.0 |
| 2024-06 | SA#104 | SP-240813 | 0365 | 1 | A | R18 CR 28.622 Trace Report Format Correction | 18.7.0 |
| 2024-06 | SA#104 | SP-240809 | 0370 | 1 | F | Rel-18 CR TS 28.622 Change NpnId from dataType to choice to align with TS 38.331 | 18.7.0 |
| 2024-06 | SA#104 | SP-240813 | 0373 | - | A | Rel-18 CR TS 28.622 Correct definitions for granularityPeriods and monitorGranularityPeriod | 18.7.0 |
| 2024-06 | SA#104 | SP-240813 | 0376 | - | A | Rel-18 CR TS 28.622 remove notifications which are not defined in SBMA | 18.7.0 |
| 2024-06 | SA#104 | SP-240822 | 0381 | - | A | Rel-18 CR 28.622 Correct CR implementation error regarding applicable TS versions | 18.7.0 |
| 2024-06 | SA#104 | SP-240824 | 0382 | 1 | F | Rel-18 CR 28.622 Correction of attribute name according to specified name style | 18.7.0 |
| 2024-06 | SA#104 | SP-240824 | 0383 | 1 | F | Rel-18 CR 28.622 Clarification on usage of reportAmount attributes in ImmediateMdtConfig | 18.7.0 |
| 2024-06 | SA#104 | SP-240837 | 0384 | - | A | Rel-18 CR 28.622 Correct reference to specification of name of PMs and KPIs for attribute performanceMetrics | 18.7.0 |
| 2024-06 | SA#104 | SP-240806 | 0388 | - | A | Rel-18 CR TS 28.622 Remove notifyFileDeletion as notification type (stage 2) | 18.7.0 |
| 2024-06 | SA#104 | SP-240824 | 0392 | 1 | F | Rel-18 CR TS 28.622 Update Trace attributes | 18.7.0 |
| 2024-06 | SA#104 | SP-240821 | 0394 | - | F | Rel-18 CR TS 28.622 Add missing notification clause for AreaOfInterest | 18.7.0 |
| 2024-06 | SA#104 | SP-240821 | 0396 | - | F | Rel-18 CR 28.622 Correction of attribute name appearance | 18.7.0 |
| 2024-06 | SA#104 | SP-240806 | 0399 | - | A | Rel-18 CR 28.622 Remove invalid clauses | 18.7.0 |
| 2024-06 | SA#104 | SP-240806 | 0401 | - | A | Rel-18 CR 28.622 Fix trace attribute definition (stage 2) | 18.7.0 |
| 2024-06 | SA#104 | SP-240818 | 0402 | - | F | Rel-18 CR 28.622 Clarification of attribute name for 5GC UE measurements | 18.7.0 |
| 2024-09 | SA#105 | SP-241172 | 0406 | 1 | A | Rel-18 CR 28.622 Correct the definition for Link and EP_RP | 18.8.0 |
| 2024-09 | SA#105 | SP-241179 | 0408 | 1 | F | Rel-18 CR TS 28.622 Update the wrong reference | 18.8.0 |
| 2024-09 | SA#105 | SP-241175 | 0416 | 1 | F | Rel-18 CR 28.622 Correction on MDT configuration in MR-DC | 18.8.0 |
| 2024-09 | SA#105 | SP-241175 | 0418 | 1 | F | Rel-18 CR 28.622 correction on reportAmount | 18.8.0 |
| 2024-09 | SA#105 | SP-241194 | 0422 | 1 | F | Rel-18 CR TS 28.622 add area scope in QMC | 18.8.0 |
| 2024-09 | SA#105 | SP-241166 | 0425 | 2 | A | Rel-18 CR 28.622 Fix list of trace metrics attribute description | 18.8.0 |
| 2024-09 | SA#105 | SP-241168 | 0429 | 1 | A | Rel-18 CR 28.622 Cleanup of TraceJob | 18.8.0 |
| 2024-09 | SA#105 | SP-241168 | 0433 | 1 | A | Rel-18 CR 28.622 Correction of TraceJob attributes MBSFN Area List and Area Configuration For Neighboring Cells (stage 2) | 18.8.0 |
| 2024-09 | SA#105 | SP-241175 | 0435 | 1 | F | Rel-18 CR TS 28.622 Rectifying the incorrect attribute name areaConfigurationForNeighCell | 18.8.0 |

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| 2024-09 | SA#105 | SP-241168 | 0441 | 1 | A | Rel18 CR TS 28.622 Removing unnecessary statement on MnsAgent usage in SBMA | 18.8.0 |
| 2024-09 | SA#105 | SP-241168 | 0444 | - | A | Rel-18 CR TS 28.622 Correction to using data types | 18.8.0 |
| 2024-12 | SA#106 | SP-241643 | 0465 | - | F | Correction to class diagram | 18.9.0 |
| 2024-12 | SA#106 | SP-241636 | 0469 | - | A | Rel-18 CR 28.622 Correction of attribute "jobType" of TraceJob IOC | 18.9.0 |
| 2024-12 | SA#106 | SP-241656 | 0471 | 1 | F | Rel-18 CR 28.622 Corrections for 5GC UE level measurements in TraceJob IOC | 18.9.0 |
| 2024-12 | SA#106 | SP-241634 | 0479 | 1 | A | Rel-18 CR TS 28.622 Correction of undefined data types | 18.9.0 |
| 2024-12 | SA#106 | SP-241650 | 0480 | 1 | F | Rel-18 CR TS 28.622 Rapporteur cleanup | 18.9.0 |
| 2024-12 | SA#106 | SP-241659 | 0485 | - | F | Rel-18 CR 28.622 Correction on MDT configuration in MR-DC | 18.9.0 |
| 2024-12 | SA#106 | SP-241634 | 0493 | 2 | A | Rel-18 CR 28.622 Correction of limitation of convex polygons for geographical area | 18.9.0 |
| 2024-12 | SA#106 | SP-241636 | 0497 | 1 | A | Rel-18 CR TS 28.622 Correction to AreaScope (stage 2) | 18.9.0 |
| 2024-12 | SA#106 | SP-241634 | 0500 | 1 | A | Rel-18 CR TS 28.622 Correction to AreaOfInterest | 18.9.0 |
| 2024-12 | SA#106 | SP-241634 | 0503 | - | A | Rel-18 CR TS 28.622 Remove unneeded fileLocation attribute (stage 2) | 18.9.0 |
| 2025-03 | SA#107 | SP-250161 | 0487 | 2 | F | Rel-18 CR TS 28.622 Corrections of measurement type | 18.10.0 |
| 2025-03 | SA#107 | SP-250154 | 0521 | - | A | Rel-18 CR TS 28.622 Correct descriptions for areascope attribute in TraceJob IOC | 18.10.0 |
| 2025-03 | SA#107 | SP-250154 | 0524 | - | A | Rel-18 CR TS 28.622 Correct constraints of Trace-MDT attributes | 18.10.0 |
| 2025-03 | SA#107 | SP-250163 | 0529 | - | F | Rel-18 CR TS 28.622 Fix implementation errors of past CRs | 18.10.0 |
| 2025-03 | SA#107 | SP-250151 | 0533 | 2 | F | Rel-18 CR 28.622 Clarify usage of notifyFileReady for PM | 18.10.0 |
| 2025-06 | SA#108 | SP-250555 | 0556 | 1 | A | Rel-18 CR TS 28.622 Corrections on MDT PLMN List | 18.11.0 |
| 2025-06 | SA#108 | SP-250557 | 0570 | - | F | Rel-18 CR 28.622 Fix missing and duplicate attributes from earlier CR | 18.11.0 |
| 2025-09 | SA#109 | SP-251082 | 0577 | - | A | Rel-18 CR 28.622 Update File IOC | 18.12.0 |
| 2025-09 | SA#109 | SP-251091 | 0583 | 1 | F | Rel-18 CR TS 28.622 Update area scope definition for NPN support (stage 2) | 18.12.0 |
| 2025-12 | SA#110 | SP-251389 | 0594 | - | F | Rel-18 CR TS 28.622 area scope correction | 18.13.0 |
| 2025-12 | SA#110 | SP-251391 | 0604 | 2 | F | Rel-18 CR TS 28.622 Clarify usage of notifyFileReady | 18.13.0 |
| 2026-09 | SA#113 | SP-260828 | 0658 | 1 | A | Rel-18 CR TS 28.622 corrections on listOfMeasurement and reportingTrigger | 18.14.0 |

# History

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